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Efficacy of dezocine on preventing opioids-induced cough during general anesthesia induction: a PRISMA-compliant systematic review and meta-analysis

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Efficacy of dezocine on preventing opioids-induced cough during general anesthesia induction: a PRISMA-compliant systematic review and meta-analysis

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ABSTRACT

Objectives To systematically review the effects of dezocine(DZC) on the incidence and severity of opioid-induced cough(OIC).

Design Systematic review and meta-analysis

Data sources PubMed, Embase, Cochrane Library, Ovid, Web of Science as well as Chinese BioMedical Literature & Retrieval System, China National Knowledge Infrastructure, Wanfang and VIP Data were searched from 1978 to December 31th 2020.

Inclusion criteria All randomized controlled trials(RCTs) comparing DZC with placebo on the incidence and severity of OIC.

Data analysis All data were analyzed by utilizing RevMan 5.3. Each outcome was tested for heterogeneity, and randomized-effects or fixed-effects model was used in the presence or absence of significant heterogeneity.

Results Our search yielded 33 RCTs including 4,442 patients, and 2,521 patients were allocated into the DZC group and 1,921 into the Control(placebo) group. Fentanyl was administrated in 1,880 patients with dosages of 2.0 to 5.0 μg·kg^{-1} and sufentanil in 2,562 patients with dosages of 0.3 to 5.0 μg·kg^{-1} during the induction of general anesthesia. The meta-analysis demonstrated that, DZC significantly reduced the incidence of OIC induced by either fentanyl(8.8% *vs*. 49.7%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001) or sufentanil(5.0% *vs*. 41.5%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001). The meta-analysis also indicated that, the incidence of mild, moderate and severe OIC in the DZC group was remarkably lower than that of the

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Control group(mild: 3.6% *vs.*13.6%, OR=0.19, 95% CI=0.14 to 0.25, *P*<0.00001; moderate: 2.0% *vs.*13.6%, OR=0.12, 95% CI=0.09 to 0.18, *P*<0.00001; severe: 1.0% *vs.*13.9%, OR=0.08, 95% CI=0.05 to 0.12, *P*<0.00001). Additionally, the current meta-analysis indicated that DZC pretreatment was not associated with increased incidence of complications(7.0% *vs.*4.2%, OR=2.34, 95% CI=0.60 to 9.14, *P*=0.22) except for dizziness(11.8% *vs.*0%, OR=8.06, 95% CI=1.40 to 46.35, *P*=0.02).

Conclusion This meta-analysis has demonstrated that, DZC is effective and safe in preventing OIC induced by fentanyl or sufentanil.

PROSPERO registration number CRD42019141255

Keywords: Dezocine, Opioid, Cough, General anesthesia, Meta-analysis

Strengths and limitations of this study

This is the first systematic review to investigate the incidence of opioid-induced cough induced by either fentanyl or sufentanil.

Subgroup analyses were performed on dose-effect of DZC and various kinds of opioids to investigate the optimal dosage of DZC.

The main limitation of this review is that varied quality and heterogeneity of included studies may limit the certainty of the findings of meta-analysis.

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INTRODUCTION

Cough is often observed when administrating a bolus of opioids(*e.g.* fentanyl,[1-4] sufentanil,[5-7] remifentanil,[8-13] alfentanil[14]), with the reported incidence ranging from 7% to 70%.[1-14] The mechanism of opioid-induced cough(OIC) is complex and remains poorly understood, which may involve pulmonary chemoreflex, enhanced activity of parasympathetic nerve, histamine release, opioid receptor dualism and muscular rigidity.[1-3 15-17] OIC is mostly transient, benign and self-limiting, but could be associated with adverse effects such as hypertension, tachycardia, increased intra-cranial, ocular and abdominal pressures, and airway obstruction.[1 2 15-17] OIC could be spasmodic, explosive[18] and life threatening at times.[19] OIC is especially undesirable during the induction of general anesthesia. Numerous pharmacological interventions including lidocaine, atropine, magnesium sulfate($MgSO_4$), dexamethasone, propofol, midazolam, muscular relaxant(rocurounium, vencuronium), ketamine, pentazocine, tramadol, α_2-agonists(clonidine, dexmeditomidine), ß2-agonists(terbutaline, ephedrine), sodium chromoglycate, beclomethasone, salbutamol, dextromethorphan, *etc*, and non-pharmacological interventions such as priming, dilution and slow injection of opioids, have been used to manage OIC.[1 2 4-9 11-13 15 17 19-22] Unfortunately, the efficacy and safety of those anti-tussive interventions remains controversial.

Dezocine(DZC), a mixed opioid agonist/antagnost, was synthesized in 1970s and approved by the Food and Drug Administration of US for perioperative pain management but was discontinued with the closure of its parent company.[23-27]

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Although no longer used clinically in Western countries, DZC has gained popularity in China and been widely used as a peri-operative analgesic for decades.[24 28-32] Recent studies suggested that, pretreatment of intravenous DZC 0.1mg/kg could completely suppress the cough induced by bolus injection of fentanyl or sufentanil during anesthesia induction. For example, Sun and colleagues[4] evaluated the suppressive effect of DZC on fentanyl-induced cough(FIC). One hundred and twenty patients were randomized to receive DZC 0.1 mg/kg or placebo 10 min before fentanyl 5 μg/kg. They demonstrated that, no DZC-pretreated patient had FIC, as compared to 70%(42/60) non-DZC-pretreated patients developing FIC. In another RCT involving 370 patients, Liu and colleagues[6] evaluated the anti-tussive effect of DZC 0.1 mg/kg on sufentanil-induced cough(SIC) during anesthesia induction. They demonstrated the incidence of SIC in the placebo group was 31%(59/185), while no SIC was observed in the DZC group. It is so encouraging that, DZC might be more effective than those above-mentioned anti-tussive interventions, and that DZC could possibly eliminate OIC without causing OIC itself. Therefore, we performed this systemic review and meta-analysis to evaluate the efficacy of DZC on OIC during general anesthesia induction, and possible complications.

METHODS

Patient and public involvement

No patient involved.

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Ethical approval

This study was a meta-analysis of previously published literatures, ethical approval was not necessary according to the Ethical Committee of Fuwai Hospital.

Registration

The protocol of current meta-analysis was published in PROSPERO with the registration number of CRD42019141255 on November 11th 2019.

Search strategy

We conducted a systemic review according to the Preferred Reporting Items for Systemic Reviews and Meta-Analysis Quality of Reporting of Meta-analysis(PRIMSA) Guidelines(**Supplemental Table 1**).[33] Relevant trials were identified by computerized searches of PubMed, Embase, Cochrane Library, Ovid, Web of Science as well as Chinese BioMedical Literature & Retrieval System(SinoMed), China National Knowledge Infrastructure(CNKI), Wanfang Data and VIP Data till December 31th 2019, with an updated databases search on August 25th 2020 prior to submission, using different combination of search words as follows: *(opioid OR fentanyl OR sufentanil OR remifentanil OR alfentanil) AND cough AND dezocine AND (randomized controlled trial OR controlled clinical trial OR randomized OR placebo OR randomly OR trial)*(**Supplemental Table 2**). No language restriction was used. Additionally, we used the bibliography of retrieved articles to further identify relevant studies.

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Criteria for considering studies for this review

We included all RCTs comparing DZC with placebo or blank with respect to their effects on OIC. In studies which also included other comparator drugs, only data of DZC and placebo groups were abstracted. Primary outcomes of interest included the incidence and severity of OIC. The severity of OIC was graded as mild (1-2 coughs), moderate (3-5 coughs), or severe ($>$ 5 coughs).[6] Secondary outcomes of interest include possible complications or adverse effects. Exclusion criteria included (1)studies published as review, case report or abstract; (2)animal or cell studies; (3)duplicate publications; (4) studies lacking information about outcomes of interest. The two authors (LXH and KS) independently reviewed the titles and abstracts of all identified studies for eligibility, excluding obviously ineligible ones. The eligibility of those remaining studies for final inclusion was further determined by reading the full text.

Study quality assessment

Two authors(JM and YYZ) independently assessed the risk of bias, using the tool described in the Cochrane Handbook for Systematic Reviews of Interventions.[34] Each potential source of bias was graded as low, uncertain or high risk of bias and showed as risk of bias summary and graph.

Data abstraction

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The following data were abstracted from the included studies to a data collection form by two authors(LXH and KS) independently: (1)author, year of publication and journal of included studies; (2)total number of patients, number of patients in the DZC and control groups, gender, age; (3)data regarding outcomes of interest in both groups. Disagreements were resolved by discussion among all authors during the process of data abstraction. The authors of the included RCTs were contacted if necessary.

Statistical analysis

All data were analyzed by utilizing RevMan 5.3(Cochrane Collaboration, Oxford, UK). Pooled odds ratio(OR) and 95% confidence interval(CI) were estimated for dichotomous data, and weighted mean difference(WMD) and 95% CI for continuous data, respectively. Each outcome was tested for heterogeneity, and randomized-effects or fixed-effects model was used in the presence or absence of significant heterogeneity(Q-statistical test $P<0.05$). Sensitivity analyses were done by examining the influence of statistical model on estimated treatment effects, and analyses which adopted the fixed-effects model were repeated again by using randomized-effects model and *vice versa*. In addition to that, sensitivity analysis was also performed to evaluate the influence of individual study on the overall effects. The possible effects of opioid type and doses were evaluated by subgroup analysis. Publication bias was explored through visual inspection of funnel plots of the outcomes. All *P*-values were two-sided and statistical significance was defined as

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$P<0.05$.

RESULTS

Characteristics of the included trials

As shown in **Figure 1**, initial literature search generated 70 results. Finally, thirty-three RCTs[4 6 35-65] involving 4,442 patients were included in the meta-analysis. Of the 33 RCTs, thirty[36-65] were written in Chinese, and the other 3[4 6 35] in English(**Table 1**). The 33 RCTs were performed respectively in 2 provincial hospitals,[36 44] 13 affiliated hospitals,[4 6 35 38 41 46 48 49 52 54-56 63] 16 urban hospitals[37 39 40 42 43 45 47 50 51 53 57 59 61 62 64 65] and 2 county hospitals[58 60] from 15 provinces and municipalities in China. All enrolled patients were of ASA(American society of Anesthesiologists) physical status classification Ⅰ-Ⅱ, whose ages ranged from 18 to 85 yr(**Table 1**). No included RCT reported the OIC induced by remifentanil or alfentanil. As shown in **Table 2**, fentanyl was administrated in 1,880 patients during the induction of general anesthesia with dosages of 2.0 to 5.0 μg/kg, and sufentanil in 2,562 patients with dosages of 0.3 to 5.0 μg/kg. The injection duration of fentanyl and sufentanil varied from 2 to 30 s. Out of the 4,442 patients, 2,521 were allocated into the DZC group and 1,921 into the Control(placebo) group. DZC administration protocols differed among the 33 included trials. DZC were administered intravenously with dosages of 0.025 to 0.3 mg/kg(or 2 to 5 mg), one to ten minutes prior to fentanyl or sufentanil injection(**Table 1**).

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Methodological quality

The risk of bias analysis was shown in **Supplemental Figure 1 and 2.** There were no patient withdrawal or dropout, neither selectiveness nor bias in all 33 RCTs.

Effects of interventions

Incidence of OIC

All the 33 included studies reported the incidence of OIC. As shown in **Figure 2**, meta-analysis demonstrated that the incidence of OIC in the DZC group was statistically lower than that of the Control group(6.7% *vs*. 44.5%, OR=0.07, 95% CI=0.05 to 0.11, $P<0.00001$, $I^2=56\%$). To analyze the type effects of opioids(fentanyl and sufentanil), subgroup analysis was performed, which indicated that DZC significantly reduced the incidence of FIC(8.8% *vs*. 49.7%, OR=0.07, 95% CI=0.04 to 0.12, $P<0.00001$, $I^2=61\%$) and SIC(5.0% *vs*. 41.5%, OR=0.07, 95% CI=0.04 to 0.12, $P<0.00001$, $I^2=53\%$). As shown in **Supplemental Figure 3**, subgroup analysis demonstrated that, the FIC incidence increased from 45.0%, 43.1%, 47.5% to 73.1% in the Control group when fentanyl dosage increased from 2, 3, 4 to 5 μg/kg, respectively. Dose-effect of sufentanil dosage on the incidence of SIC was shown in **Supplemental Figure 4**.

Twenty-two RCTs[6 35-37 39 40 43 45-47 50-52 56-64] reported the incidence of mild and moderate OIC. As shown in **Supplemental Figure 5** and **6**, meta-analysis demonstrated that, DZC group showed significantly lower incidence of OIC than Control group both on mild and moderate grades(mild OIC: 3.6% *vs*.13.6%, OR=0.19,

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95% CI=0.14 to 0.25, *P*<0.00001, I^2=22; moderate OIC: 2.0% *vs.*13.6%, OR=0.12, 95% CI=0.09 to 0.18, *P*<0.00001, I^2=0). Subgroup analysis demonstrated that, DZC significantly reduced the occurrence of either FIC(mild FIC: 5.2% *vs.*15.3%, OR=0.25, 95% CI=0.16 to 0.38, *P*<0.00001, I^2=28; moderate FIC: 3.1% *vs.*14.2%, OR=0.17, 95% CI=0.10 to 0.28, *P*<0.00001, I^2=0) or SIC(mild SIC: 2.4% *vs.*12.9%, OR=0.14, 95% CI=0.09 to 0.22, *P*<0.00001, I^2=11; moderate SIC: 1.1% *vs.*13.4%, OR=0.10, 95% CI=0.06 to 0.17, *P*<0.00001, I^2=0) when compared to placebo.

Twenty-five enrolled RCTs[6 35-37 39 40 43-47 50-52 54-64] reported the incidence of severe OIC. As shown in **Supplemental Figure 7**, meta-analysis demonstrated that, the incidence of severe OIC in the DZC group was remarkably lower than that of the Control group(0.9% *vs.*13.7%, OR=0.08, 95% CI=0.05 to 0.12, *P*<0.00001, I^2=0). Subgroup analysis demonstrated that, DZC significantly reduced the occurrence of either severe FIC(1.8% *vs.*13.5%, OR=0.12, 95% CI=0.07 to 0.20, *P*<0.00001, I^2=0) or severe SIC(0.3% *vs.*13.9%, OR=0.05, 95% CI=0.03 to 0.10, *P*<0.00001, I^2=0) when compared to placebo.

Subgroup analyses were also performed to investigate the dose effects of DZC on FIC and SIC incidences. As shown in **Supplemental Figure 8** and **9**, DZC could effectively suppress OIC by fentanyl or sufentanil when administered at dosages ranging from less than 0.1 to 0.3 mg/kg (or 5 mg). The dose of 0.1 mg/kg is mostly investigated and suggested as the optimal dose. Whether the prophylactic effect of DZC on OIC is dose-dependent remains further verification.

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Adverse effects

Six RCTs[48 50 53 58 64 65] reported possible side effects of DZC administration. As shown in **Figure 3**, meta-analysis suggested that, the incidences of drowsiness, truncal rigidity, chill, respiratory inhibition, nausea and emesis of the DZC group were all comparable to those of the Control group, with exception that the DZC-treated patients had higher incidence of dizziness as compared to placebo(11.8% *vs.* 0%, OR=8.06, 95% CI=1.40 to 46.35, P=0.02, I^2=0%).

Sensitivity analyses and publication bias

Sensitivity analysis showed that treatment effects on all the outcomes were not affected by the choice of statistical model (**Table 2**). Sensitivity tests were also performed by exclusion of some studies to analyze the influence of the overall treatment effect on high heterogeneity outcomes(**Table 2**), and no contradictory results were found in pooled OR and 95%CI. For incidence of OIC, heterogeneity changed from 61% to 35% for FIC by exclusion of three studies conducted from Gao et al(female patients only)[39], Wang et al[49] and Yuan et al[58](preoperative medication with phenobarbital), and 53% to 36% for SIC by exclusion of four studies conducted from Shen et al(female patients only),[47] Xu et al(duration of sufentanil injection more than 10s),[57] Wang et al[53] and Zheng et al[63](preoperative medication with phenobarbital). For incidence of complications, heterogeneity changed from 73% to 0% by exclusion of one study from Sheng et al(preoperative medication with phenobarbital).[48] No significant publication bias was detected by funnels plot

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examination for the incidence of OIC(**Supplemental Figure 10A**) and the incidence of mild , moderate and severe OIC(**Supplemental Figure 10B, Supplemental Figure 10C and Supplemental Figure 10D**).

DISCUSSION

Cough suppression is one useful side-effect of opioids, which is the basis of their use in cough suppressants. Opioids depress the cough reflex by directly acting on the medullary cough center.[16] Fentanyl and its derivatives sufentanil are commonly used opioid analgesics in the induction and maintenance of general anesthesia. Intravenous bolus injection of fentanyl or sufentanil often cause cough. The present meta-analysis demonstrated that, the incidences of FIC and SIC were 49.7% and 41.5% respectively, the incidences of severe FIC and severe SIC were 13.5% and 13.9% respectively, which is consistent with previous reports.[2 4 6 7 15] However, significant heterogeneity was found in the results, which may have affected the rigor of those findings. The heterogeneity may be explained by study design. For example, sex of the patients in excluded study in sensitivity analysis were obviously different from others. It was reported by Solanki et al[66] that incidence of FIC was low when studied in female cancer patients(12.7%). However, contradictory results of 57.5% and 28.3% were observed in the two excluded study enrolling female only.[39 47] This may suggest that sex to some extent contributes to heterogeneity. In addition to that, study from Xu et al[57] with significant low SIC incidence(3% in DZC group and 8% in Control group) was excluded owing to prolonged injection time(>30s) in sensitivity analysis, which

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though made no influence on pooled effect, may improve the credibility of current meta-analysis.

Till now, the mechanism of OIC remains poorly understood. Various hypotheses have been proposed, which may involve opioid receptors, C-fiber receptors, rapid adapting pulmonary stretch receptors, histamine release and citrate in fentanyl and sufentanil injection.[1-3 15-17] Additionally, many factors can contribute to the occurrence of OIC, which can be divided into two categories. One is patients' individual physical conditions(age, sex, smoking status, disease history, *etc*). Another is usage of opioids(drug category, dosage, concentration, injection site, injection concentration, injection rate, *etc*).[15] And subgroup analysis suggested possible dose-effects of fentanyl and sufentanil on the incidences of OIC.

OIC is associated with adverse effects and should be avoided. The anti-tussive efficacy of numerous pharmaceutical and non-pharmaceutical interventions have been tested, some proved to be effective, some ineffective, and some have side effects.[15] DZC, a mixed κ and μ opioid receptor agonist-antagonist, is not a well-known drug in Western countries.[24-27] However, DZC is widely applied as peri-operative pain analgesic agent in China for decades.[24-26 28-32] The present meta-analysis demonstrated that DZC could significantly suppress both FIC and SIC, with several trials[4 6 35 41 45 48 50 56 64] reporting that DZC could completely prevent OIC. Furthermore, the subgroup analysis of the present meta-analysis suggested that, the anti-tussive effect of DZC on FIC and SIC may be dose-dependent. The mechanism responsible for the anti-tussive effect of DZC remains unknown. Possible explanation for this phenomenon is that

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DZC suppresses OIC by μ-receptor antagonism or norepinephrine/serotonine reuptake inhibition and reduce cough.[15] Whether a central gating mechanisms via C-fiber receptors or inhibition of histamine release play a role in the cough suppression elicited by DZC needs to be investigated.[4]

Because of its partial μ agonism, DZC exhibits a ceiling effect for common opioids-related complications such as respiratory depression.[24-26] The meta-analysis suggested that, DZC did not increase the incidences of drowsiness, truncal rigidity, chill, respiratory inhibition, nausea and emesis, but was associated with higher incidence of dizziness. Whether DZC pretreatment interferes with opioid analgesia remains to be verified. Initial evidence indicated that, DZC can enhance the analgesic effect of opioids and reduced OIC and opioid-related side effects.[67 68]

This study has some limitations. Firstly, meta-analysis can increase the power of analysis by pooling many small low-quality studies, but different clinical practices, varied quality and heterogeneity of included studies may limit the certainty of the findings of meta-analysis. For example, there were no differences in DZC and Control group on OIC incidence when using preoperative medication of phenobarbital 30 minutes before anesthesia induction.[53 58] One possible explanation is that, sedatives exhibit similar effect on suppressing OIC as well according to previous study.[2] Secondly, all the 33 included RCTs were performed in China. The anti-tussive effectiveness of DZC in other ethnicities remains to be investigated. Thirdly, the doses, injection rates or injection order of fentanyl or sufentanil varied among these included trials. For example, Sun and colleagues[4] reported DZC administered 10 min

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before anesthesia induction could prevent FIC, which may be not a convenient practice in clinical settings. To determine the proper administration protocol of DZC for OIC prevention, a prospective randomized, placebo-controlled, triple-blinded trial is ongoing in our center.

CONCLUSIONS

This meta-analysis has demonstrated that, DZC is effective and safe in preventing OIC induced by fentanyl or sufentanil. DZC is of clinical value, but further studies should be performed on a wider scale to determine more emphasized results.

Contributors LXH and YTY were involved in the study design, data collection, data analysis and drafting the manuscript. KS, YYZ and JM participated in data collection. All authors have read and approved the manuscript.

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Competing interests None declared.

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Table 1. Characteristics of the included RCTs and administration protocols of dezocine

Study	Language	Patient characteristics			Opioids			Group Dezocine		Group Control		Outcomes reported				
		Age(yr)	Sex(M/F)	Type	Dose(μg/kg)	Duration(s)	Timing(min)	n	Dose(mg/kg)	n	Dose	CID	CIC	SCID	SCIC	Complication
Bian 2012[36]	Chinese	23-64	48/53	S	3.0	≤5s	5	50	5mg	50	Equal volume NS	2.00%	30.00%	0	6.00%	NR
Chen 2013[37]	Chinese	20-60	39/41	S	2.0	≤2s	NR	40	5mg	40	2ml NS	5.00%	45.00%	0	25.00%	NR
Fang 2016[38]	Chinese	20-60	66/58	F	5.0	≤3s	10	62	0.1	62	Equal volume NS	9.68%	62.90%	NR	NR	NR
Gao 2013[39]	Chinese	20-50	0/120	F	3.0	NR	10	40	0.05	14	5ml NS	57.50%	64.29%	17.5%	20.00%	NR
					3.0			40	0.1	13		17.50%	53.85%	2.5%		
					3.0			40	0.15	13		15.00%	53.85%	2.5%		
Li 2017[40]	Chinese	23-72	134/106	S	0.3	≤10s	10	80	0.05	40	Equal volume NS	2.50%	32.50%	0	7.50%	NR
					0.3			80	0.1	40		1.25%	35.00%			
Liu 2015[6]	English	18-70	189/181	S	0.5	>3s	2	185	0.1	185	NS	0.00%	31.89%	0	13.51%	NR
Meng 2013[41]	Chinese	28-55	39/41	S	0.4	≤2s	2	40	0.1	40	Equal volume NS	0.00%	72.50%	NR	NR	NR
Pi 2015[42]	Chinese	22-65	51/49	F	4.0	≤3s	10	50	0.1	50	Equal volume NS	12.00%	68.00%	NR	NR	NR
Qiu 2016[43]	Chinese	20-65	119/81	S	0.5	NR	NR	100	5mg	100	2ml NS	5.00%	45.00%	0	39.00%	NR
Qiu L 2016[44]	Chinese	18-45	0/120	F	3.0	≤5s	2	60	0.1	60	5ml NS	1.67%	25.00%	0	3.33%	NR
Qu 2012[45]	Chinese	18-65	40/80	F	4.0	≤5s	2	60	0.05	60	Equal volume NS	0.00%	26.67%	0	11.67%	NR
Ruan 2017[46]	Chinese	24-55	NR	S	0.4	NR	10	35	0.1	35	Equal volume NS	5.71%	57.14%	0	28.57%	NR
Shen 2014[47]	Chinese	20-65	0/120	S	0.3	<5s	5	60	0.05	60	5ml NS	8.33%	28.33%	3.33%	10.00%	NR
Sheng 2017[48]	Chinese	19-70	44/52	S	0.3	≤10s	8	48	0.1	48	5ml NS	0.00%	64.58%	NR	NR	CH, RI, NE
Sun 2011[4]	Chinese	20-60	68/52	F	5.0	≤2s	10	60	0.1	60	NS	0.00%	70.00%	NR	NR	NR
Wang 2015[49]	Chinese	15-60	78/62	F	5.0	≤5s	10	70	0.1	70	10ml NS	1.43%	75.71%	NR	NR	NR
Wang 2016[50]	Chinese	18-70	190/180	S	0.5	NR	Immediately	185	0.1	185	Equal volume NS	0.00%	31.89%	0	29.41%	TR, RI, NE
Wang 2017[51]	Chinese	22-61	67/53	S	NR	NR	NR	60	0.05	60	NS	16.67%	55.00%	0	6.67%	NR
Wang H 2015[52]	Chinese	25-65	42/58	S	0.3	<5s	10	50	0.1	50	2ml NS	2.00%	32.00%	0	8.00%	NR
Wang L 2015[53]	Chinese	60-85	59/41	S	0.3	≤5s	5	25	0.04	9	5ml NS	28.00%	44.44%	NR	NR	DI, DR
					0.3			25	0.08	8		12.00%	37.50%			

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					0.3			25	0.12	8		8.00%	37.50%			
Wang Z 2015[54]	Chinese	18-55	31-29	F	4	≤3s	1	30	0.1	30	2ml NS	13.33%	53.33%	0	23.33%	NR
Wang Z 2016[55]	Chinese	18-55	33-27	F	4	≤3s	1	30	0.1	30	2ml NS	16.67%	50.00%	0	16.67%	NR
					3			40	0.1	14		7.50%	71.43%			
Wu 2014[56]	Chinese	18-60	105/55	F	3	<3s	10	40	0.2	13	Equal volume NS	2.50%	61.54%	0	15.00%	NR
					3			40	0.3	13		0.00%	61.54%			
					0.5	<3s	5	50	0.1	50		6.00%	80.00%	0	7.50%	
Xu 2014[57]	Chinese	20-65	102/98	S							5ml NS					NR
					0.5	<30s	5	50	0.1	50		2.00%	8.00%	0	0	
					3			100	0.025	34		12.00%	41.18%			
Xu 2015[35]	English	20-70	243/157	F	3	<5s	Immediately	100	0.05	33	NS	4.00%	39.39%	0	5.00%	NR
					3			100	0.1	33		0.00%	39.39%			
					3			30	0.05	10		56.67%	60.00%	26.67%		
Yuan 2015[58]	Chinese	25-55	NR	F	3	NR	10	30	0.1	10	NS	13.33%	60.00%	0	33.33%	DI, DR
					3			30	0.2	10		6.67%	60.00%	0		
Zhang 2013[59]	Chinese	25-56	41/39	F	3	<3s	12	40	0.1	40	2ml NS	2.50%	45.00%	0	6.67%	NR
Zhang 2016[60]	Chinese	23-56	72/48	F	4	≤3s	5	30	5mg	30	10ml NS	3.33%	46.67%	0	22.50%	NR
Zhao 2017[61]	Chinese	20-61	19/33	S	5	NR	5-8	26	0.1	26	NS	7.69%	65.38%	0	19.23%	NR
					0.4		1	25	2mg	9		28.00%	66.67%			
Zheng 2018[62]	Chinese	39±5	61/39	S	0.4	≤3s	3	25	2mg	8	Equal volume NS	4.00%	50.00%	0	8.00%	NR
					0.4		8	25	2mg	8		4.00%	50.00%			
Zheng 2019[63]	Chinese	18-56	92/108	S	0.4	≤6s	3	50	5mg	50	Equal volume NS	16.00%	44.00%	0	8.00%	NR
Zhou 2014[64]	Chinese	18-65	23/37	S	0.3	<10s	10	30	0.1	30	5ml NS	0.00%	26.67%	0	3.33%	DI, DR
Zhu 2018[65]	Chinese	22-75	31/29	S	0.4	NR	2	30	0.1	30	Equal volume NS	3.33%	13.33%	NR	NR	NE

CID=Cough incidence of dezocine; CIC=Cough incidence of control; SCID=Severe cough incidence of dezocine; SCIC=Severe cough incidence of control; RI=Respiratory inhibition; NE=Nausea and emesis; TR=Truncal rigidity; NR=Not reported; DI=Dizziness; DR=Drowsiness; CH=Chill.

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Table 2. Reliability of results

Influence of statistical model on estimated treatment effects of primary outcomes			
Statistical model	Cough incidence OR (95% CI)	Severe cough incidence OR (95% CI)	Complications incidence OR (95% CI)
Fixed Effects	0.07(0.05, 0.08)	0.08(0.05,0.12)	1.61(1.09,2.39)
Random Effects	0.07(0.05, 0.10)	0.11(0.07, 0.18)	2.34(0.60,9.14)

Sensitivity analyses of high heterogeneity outcome									
Heterogeneity outcome	Excluded trials	Group DZC (n)	Group C (n)	Heterogeneity		Analysis model	OR	95%CI	Overall effect *P*
				I^2 (%)	*P*				
FIC (%)	39,49,58	280	140	35	0.08	M-H, Fixed	0.06	(0.04,0.08)	<0.00001
SIC（%）	47,53,57,63	285	235	36	0.07	M-H, Fixed	0.04	(0.03,0.06)	<0.00001
Complications (%)	48	48	48	0	0.59	M-H, Fixed	10.75	(4.75,24.33)	<0.00001

FIC=Fentanyl-induced cough; SIC=Sufentanil-induced cough.

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Figure legends

Figure 1. Flow chart

Figure 2. Forest plot of OIC incidence

Figure 3. Possible complications

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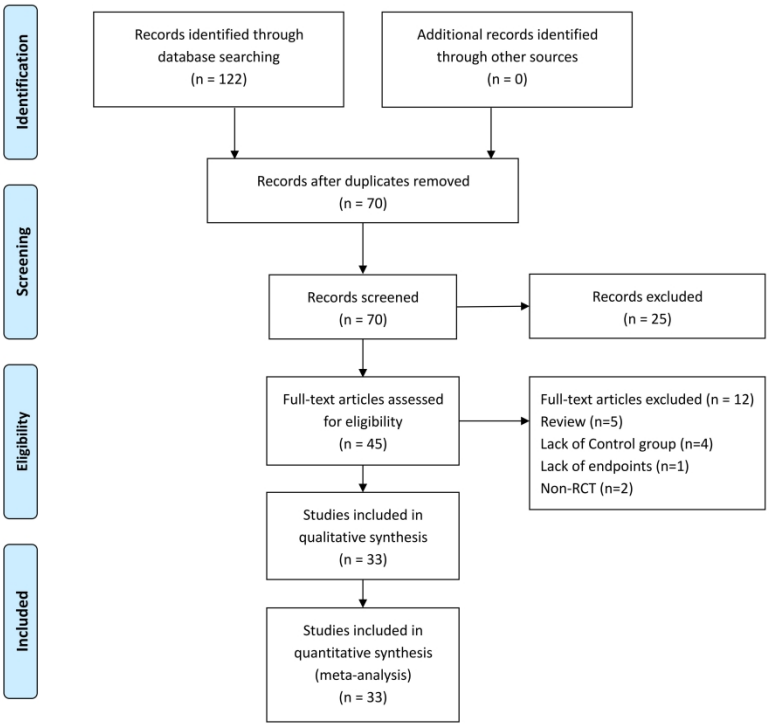


Figure 1. Flow chart

403x570mm (300 x 300 DPI)

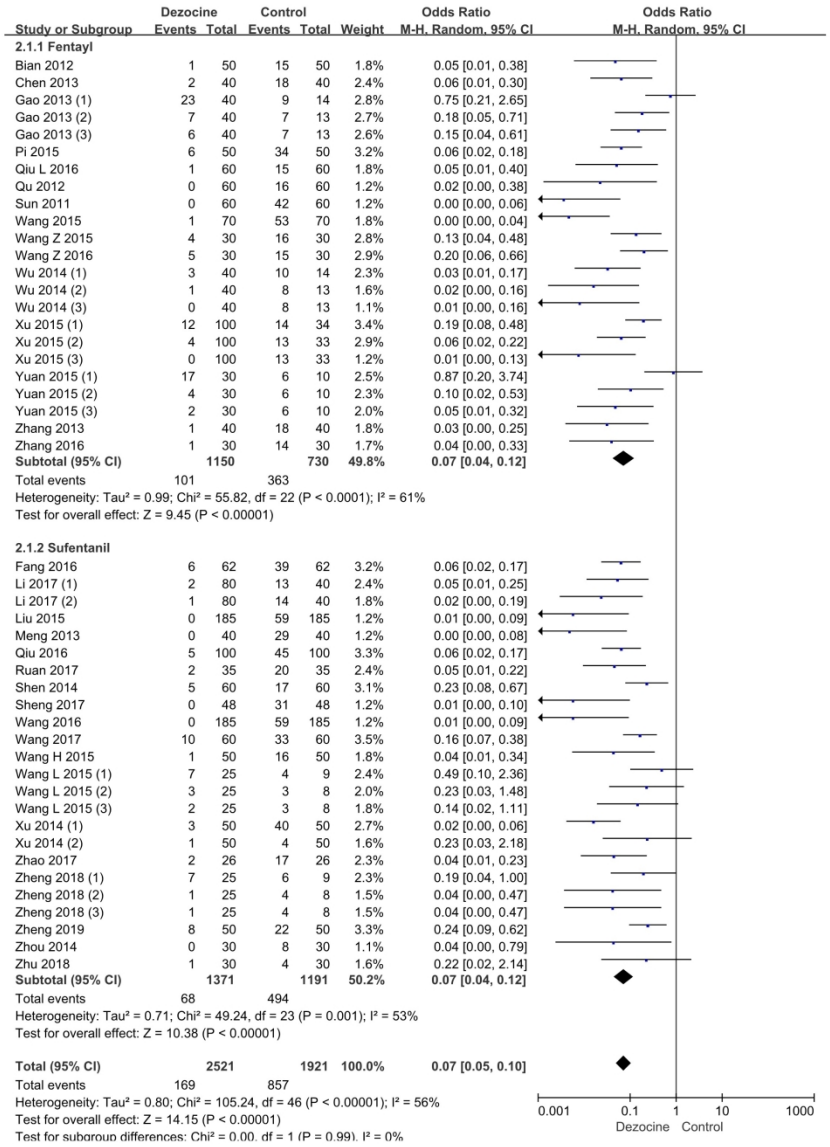


Figure 2. Forest plot of OIC incidence

209x278mm (300 x 300 DPI)

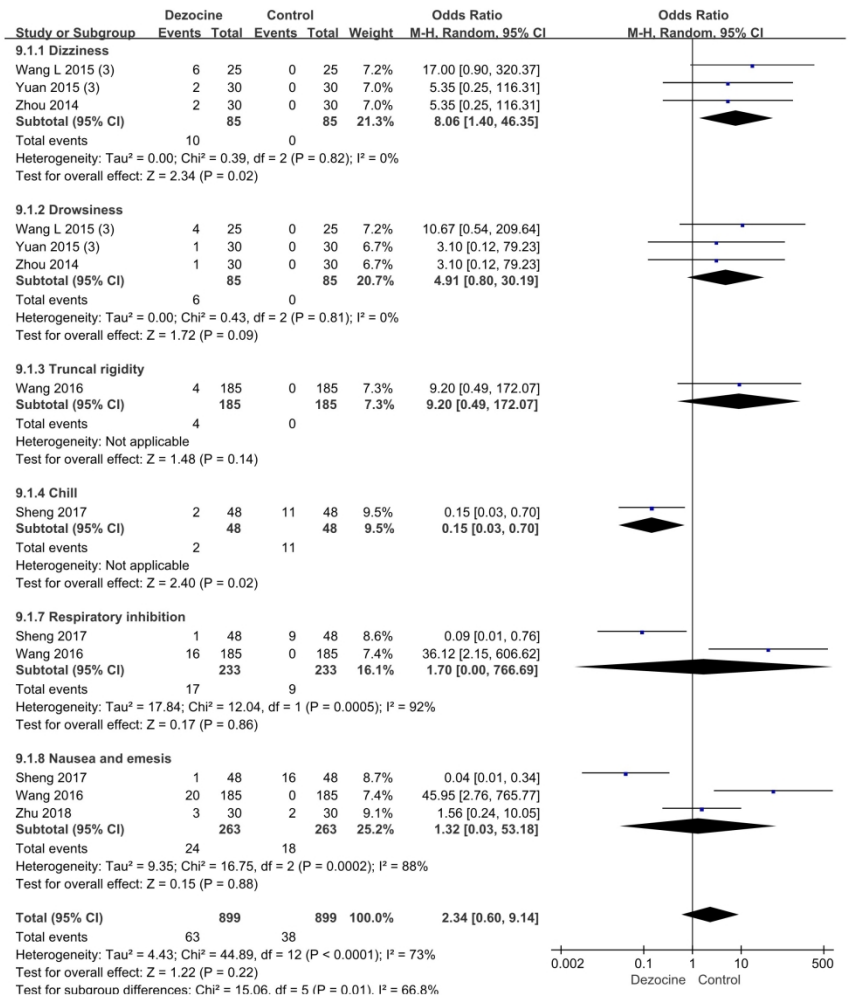


Figure 3. Possible complications

209x278mm (300 x 300 DPI)

Supplemental Table 1. PRISMA checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Methods, page 5-6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Methods—Study inclusion criteria, page 6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Methods—Search strategy, page 6

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Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Methods—Search strategy, page 6, and Supplemental Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Methods—Study inclusion criteria, page 6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Methods—Data abstraction, page 7-8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Methods—Data abstraction, page 7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Methods—Study quality assessmen, page 7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Methods—Statistical analysis, page 8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Methods—Statistical analysis, page 8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Methods—Statistical analysis, page 7-8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Methods—Statistical analysis, page 7-8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Results-Search results page 9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Results-Meta-analysis, page 9-
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Results-Study quality and risk bias, page 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Results-Meta-analysis, page 10-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results-Meta-analysis, page 10-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Results-Meta-analysis, page 12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results- page 12
DISCUSSION			

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Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion- Page 13-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion- Page 15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion- Page 16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding- Page 16

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Supplemental Table 2. Search strategy

PubMed

No.	Search items
#1	"dezocine"[Supplementary Concept] OR dezocine[Title/Abstract]
#2	((((((((((((((((Analgesics, Opioid[MeSH Terms]) OR Opioid[Title/Abstract]) OR Fentanyl[MeSH Terms]) OR Fentanyl[Title/Abstract]) OR Phentanyl[Title/Abstract]) OR Fentanyl Citrate[Title/Abstract]) OR Sufentanil[MeSH Terms]) OR Sufentanil[Title/Abstract]) OR Sulfentanyl[Title/Abstract]) OR Sulfentanil[Title/Abstract]) OR Sufentanil Citrate[Title/Abstract]) OR Remifentanil[MeSH Terms]) OR Remifentanil[Title/Abstract]) OR Remifentanil Hydrochloride[Title/Abstract]) OR Alfentanil[MeSH Terms]) OR Alfentanil[Title/Abstract]) OR Alfentanyl[Title/Abstract]) OR Alfentanil Hydrochloride[Title/Abstract]
#3	((((Cough[MeSH Terms]) OR Cough[Title/Abstract]) OR Coughs[Title/Abstract]) OR Antitussive[Title/Abstract]) OR Anti-tussive[Title/Abstract]
#4	((((((Randomized Controlled Trial[Publication Type]) OR Randomized Controlled Trial) OR Controlled Clinical Trial[Publication Type]) OR Controlled Clinical Trial) OR Randomized) OR Placebo) OR randomly
#5	#1 AND #2 AND #3 AND #4

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Embase

('dezocine'/exp OR dezocine:ab,ti) AND ('opiate agonist'/exp OR opioid:ab,ti OR 'fentanyl derivative'/exp OR fentanyl:ab,ti OR 'fentanyl citrate':ab,ti OR sufentanil:ab,ti OR 'sufentanil citrate':ab,ti OR remifentanil:ab,ti OR alfentanil:ab,ti) AND ('coughing'/exp OR coughing:ab,ti OR cough:ab,ti OR antitussive:ab,ti OR anti-tussive:ab,ti) AND ('randomized controlled trial'/exp OR 'randomized controlled trial':it OR 'randomized controlled trial':ab,ti OR randomized OR placebo OR randomly)

Cochrane Library

No.	Search items
#1	(dezocine): ti, ab, kw
#2	[Analgesics, Opioid] explode all trees OR (opioid): ti, ab, kw OR [Fentanyl] explode all trees OR (fentanyl): ti, ab, kw OR (fentanyl citrate): ti, ab, kw OR (phentanyl): ti, ab, kw OR [Sufentanil] explode all trees OR (sufentanil): ti, ab, kw OR (sufentanil citrate): ti, ab, kw OR (sulfentanyl): ti, ab, kw OR [Remifentanil] explode all trees OR (remifentanil): ti, ab, kw OR (remifentanil monohydrochloride): ti, ab, kw OR (remifentanil hydrochloride): ti, ab, kw OR [Alfentanil] explode all trees OR (alfentanil): ti, ab, kw OR (alfentanil hydrochloride): ti, ab, kw OR (alfentanyl): ti, ab, kw
#3	[Cough] explode all trees OR (cough): ti, ab, kw OR (coughs): ti, ab, kw OR (antitussive):ti, ab, kw OR (anti-tussive):ti, ab, kw
#4	[Randomized Controlled Trial] explode all trees OR (Randomized Controlled Trial): ti, ab, kw OR [Randomized Controlled Trials as Topic] explode all trees OR [Controlled Clinical Trial] explode all trees OR (Controlled Clinical Trial): ti, ab, kw OR [Controlled Clinical Trial as Topic] explode all trees

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#5	#1 AND #2 AND #3 AND #4

Ovid

No.	Search items
#1	dezocine.mp. [mp=title, abstract, full text, caption text]
#2	(opioid or opioid or "Analgesics, Opioid" or fentanyl or phentanyl or "fentanyl citrate" or sufentanil or sulfentanyl or "sufentanil citrate" or remifentanil or "remifentanil hydrochloride" or alfentanil or alfentanyl or "alfentanil hydrochloride").mp. [mp=title, abstract, full text, caption text]
#3	(cough or coughs or coughing or antitussive or anti-tussive).mp. [mp=title, abstract, full text, caption text]
#4	("randomized controlled trial" or "controlled clinical trial" or randomized or placebo or randomly).mp. [mp=title, abstract, full text, caption text]
#5	#1 AND #2 AND #3 AND #4

Web of Science

TS=dezocine AND TS=(opioid OR opioid OR "Analgesics, Opioid" OR fentanyl OR phentanyl OR "fentanyl citrate" OR sufentanil OR sulfentanyl OR "sufentanil citrate" OR remifentanil OR "remifentanil hydrochloride" OR alfentanil OR alfentanyl OR "alfentanil hydrochloride") AND TS=(cough OR coughs OR coughing OR antitussive OR anti-tussive) AND TS=("randomized controlled trial" OR "controlled clinical trial" OR randomized OR placebo OR randomly)

SinoMed

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No.	Search items
#1	"地佐辛"[不加权:扩展] OR "地佐辛"[摘要:智能]
#2	"阿片"[不加权:扩展] OR "阿片"[中文标题:智能] OR "镇痛药,"[不加权:扩展] AND "阿片类"[不加权:扩展] OR "芬太尼"[不加权:扩展] OR "芬太尼"[中文标题:智能] OR "舒芬太尼"[不加权:扩展] OR "舒芬太尼"[中文标题:智能] OR "瑞芬太尼"[中文标题:智能] OR "阿芬太尼"[不加权:扩展] OR "阿芬太尼"[中文标题:智能]
#3	"咳嗽"[不加权:扩展] OR "咳嗽"[中文标题:智能] OR "呛咳"[中文标题:智能] OR "止咳"[不加权:扩展] OR "止咳"[中文标题:智能] OR "镇咳"[不加权:扩展] OR "镇咳"[中文标题:智能]
#4	"随机对照试验"[不加权:扩展] OR "临床对照试验"[不加权:扩展] OR "随机地"[摘要:智能] OR "随机的"[摘要:智能] OR "对照"[摘要:智能] OR "安慰剂"[摘要:智能]
#5	#1 AND #2 AND #3 AND #4

CNKI

(SU='地佐辛' OR AB='地佐辛') AND (SU=('阿片'+'阿片类镇痛药'+'芬太尼'+'舒芬太尼'+'瑞芬太尼'+'阿芬太尼') OR TI=('阿片'+'阿片类镇痛药'+'芬太尼'+'舒芬太尼'+'瑞芬太尼'+'阿芬太尼')) AND (SU=('咳嗽'+'呛咳'+'止咳'+'镇咳') OR TI=('咳嗽'+'呛咳'+'止咳'+'镇咳')) AND (SU=('随机对照试验'+'临床对照试验'+'随机的'+'随机地'+'安慰剂'+'对照') OR AB=('随机对照试验'+'临床对照试验'+'随机的'+'随机地'+'安慰剂'+'对照'))

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Wanfang Data
(主题:地佐辛+摘要:地佐辛)*(主题:(阿片+阿片类镇痛药+芬太尼+舒芬太尼+瑞芬太尼+阿芬太尼)+题名:(阿片+阿片类镇痛药+芬太尼+舒芬太尼+瑞芬太尼+阿芬太尼))*(主题:(咳嗽+呛咳+止咳+镇咳)+题名:(咳嗽+呛咳+止咳+镇咳))*(主题:(随机对照试验+临床对照试验+随机的+随机地+安慰剂+对照)+ 摘要:(随机对照试验+临床对照试验+随机的+随机地+安慰剂+对照))
VIP Data
(M=地佐辛 OR R=地佐辛) AND (M=阿片 OR 阿片类镇痛药 OR 芬太尼 OR 舒芬太尼 OR 瑞芬太尼 OR 阿芬太尼 OR R=阿片 OR 阿片类镇痛药 OR 芬太尼 OR 舒芬太尼 OR 瑞芬太尼 OR 阿芬太尼) AND (M=咳嗽 OR 呛咳 OR 止咳 OR 镇咳 OR R=咳嗽 OR 呛咳 OR 止咳 OR 镇咳) AND (M=随机对照试验 OR 临床对照试验 OR 随机的 OR 随机地 OR 安慰剂 OR 对照 OR R=随机对照试验 OR 临床对照试验 OR 随机的 OR 随机地 OR 安慰剂 OR 对照)

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Supplemental Figure legends

Supplemental Figure 1. Risk of bias graph

Supplemental Figure 2. Risk of bias summary

Supplemental Figure 3. Dose-effect of fentanyl dosage on FIC incidence

Supplemental Figure 4. Dose-effect of sufentanil dosage on SIC incidence

Supplemental Figure 5. Forest plot of mild OIC incidence

Supplemental Figure 6. Forest plot of moderate OIC incidence

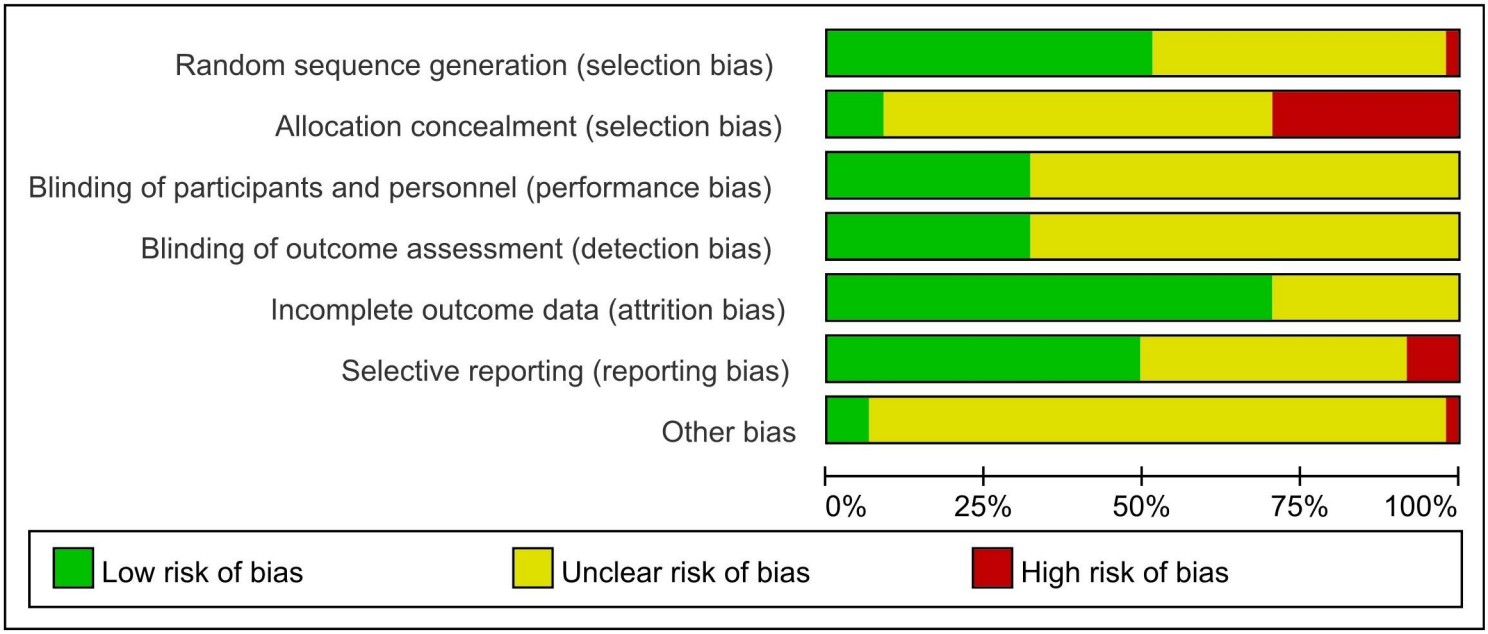
Supplemental Figure 7. Forest plot of severe OIC incidence

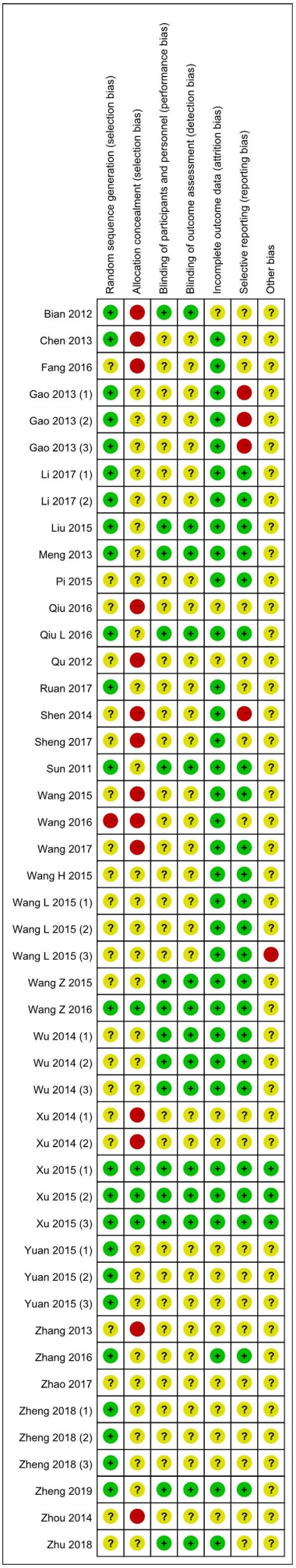
Supplemental Figure 8. Dose-effects of DZC on FIC

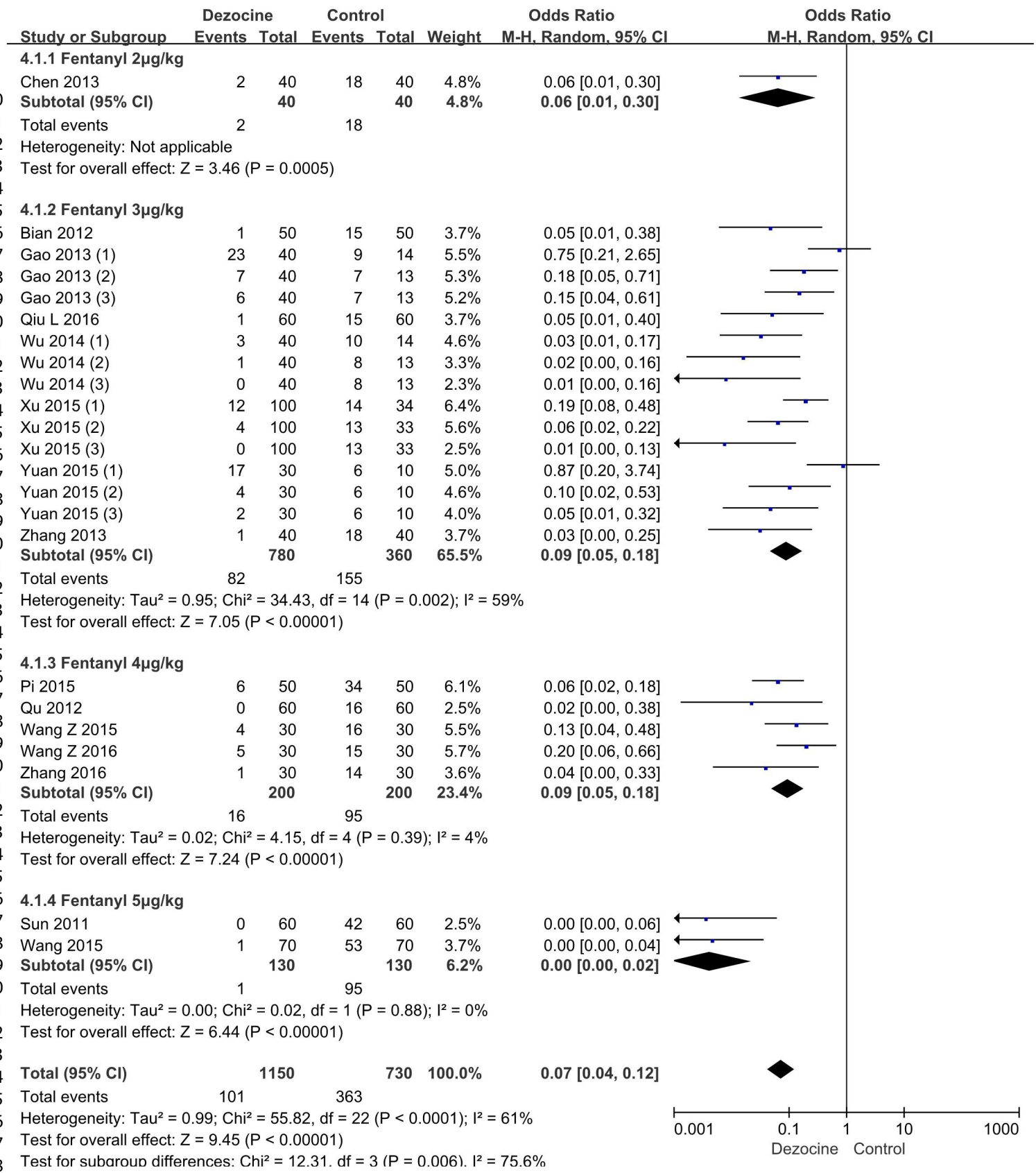
Supplemental Figure 9. Dose-effects of DZC on SIC

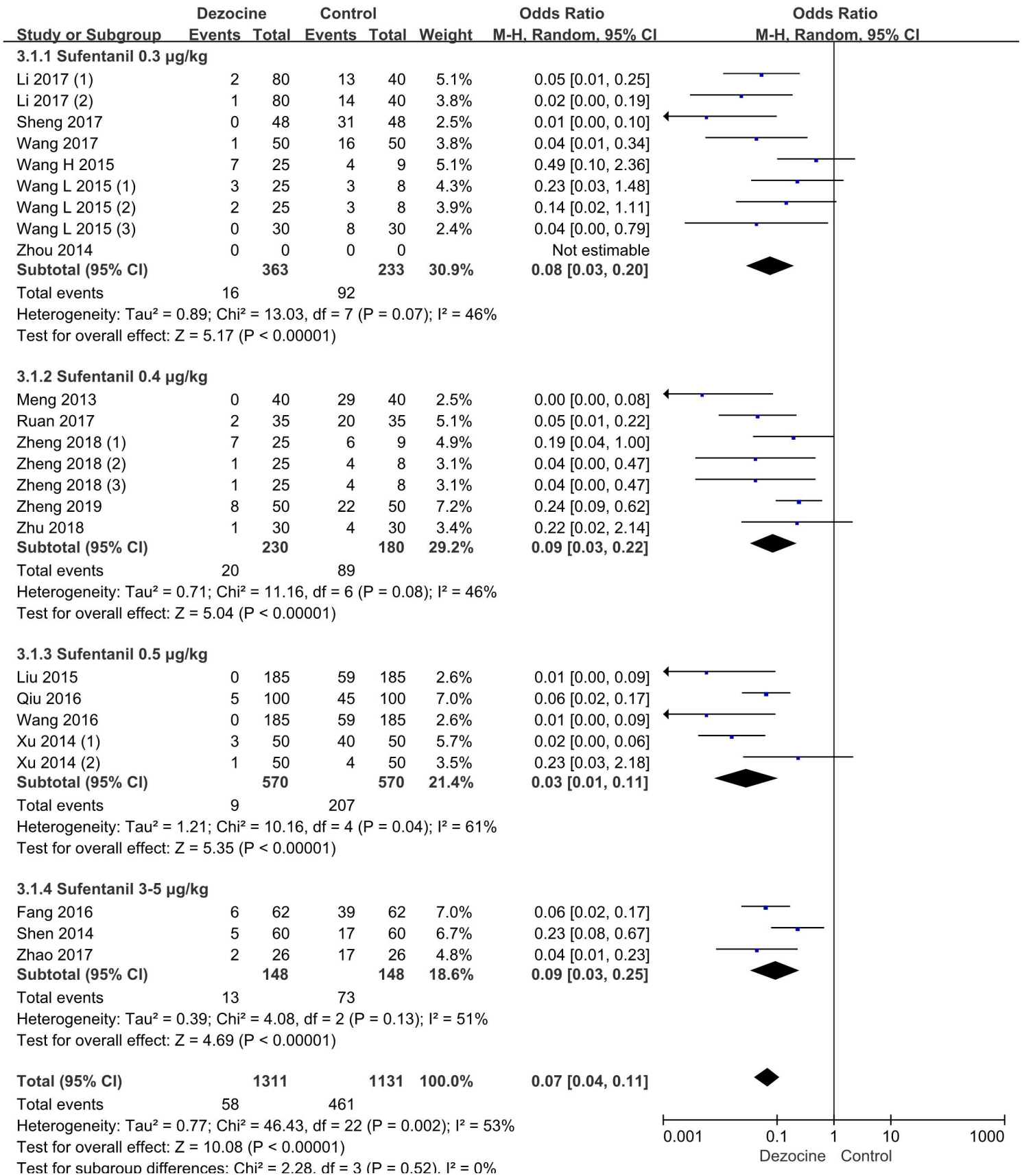
Supplemental Figure 10. Funnel plot of OIC incidence(A), mild OIC incidence(B), moderate OIC incidence(C), severe OIC incidence(D)

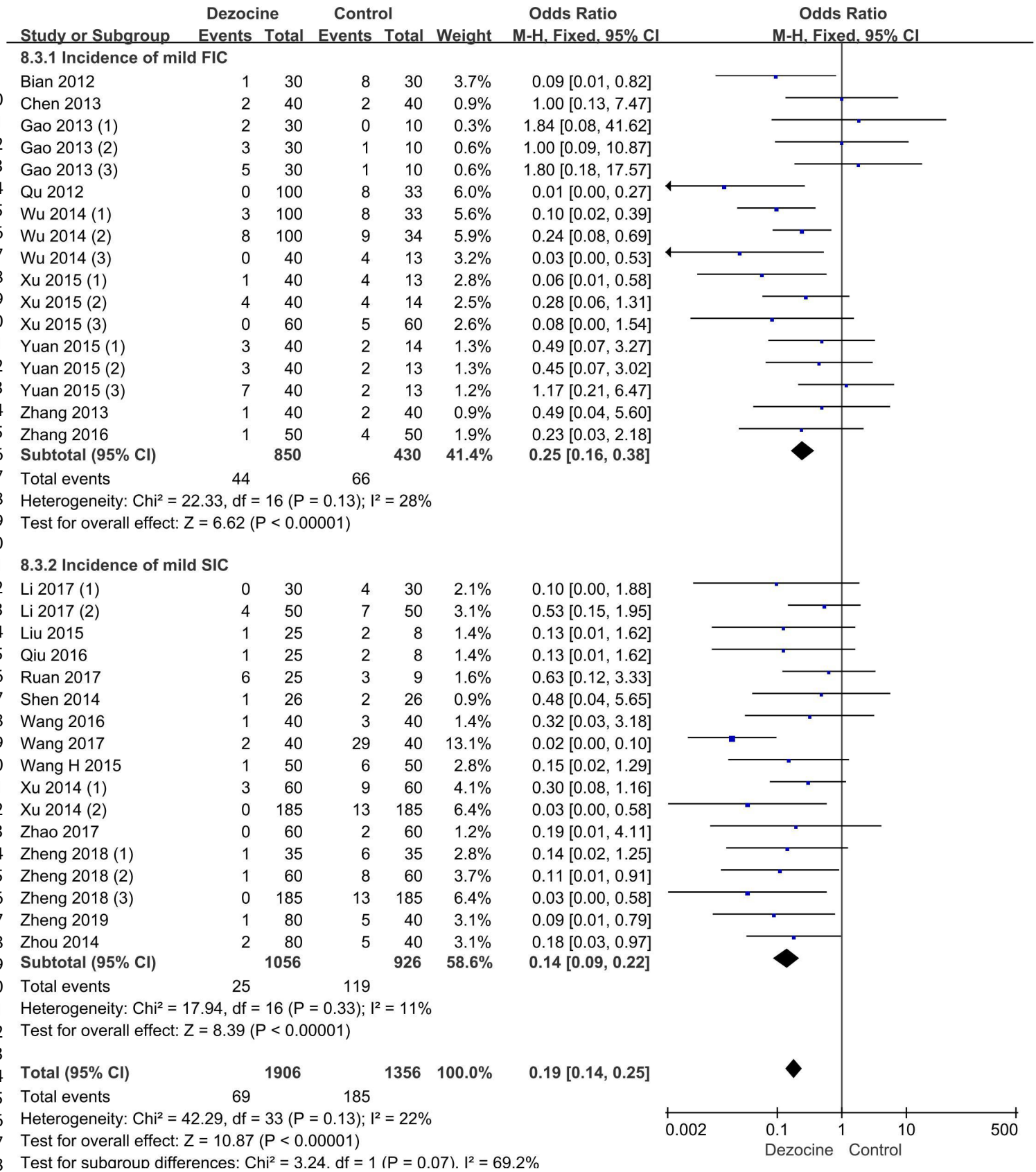
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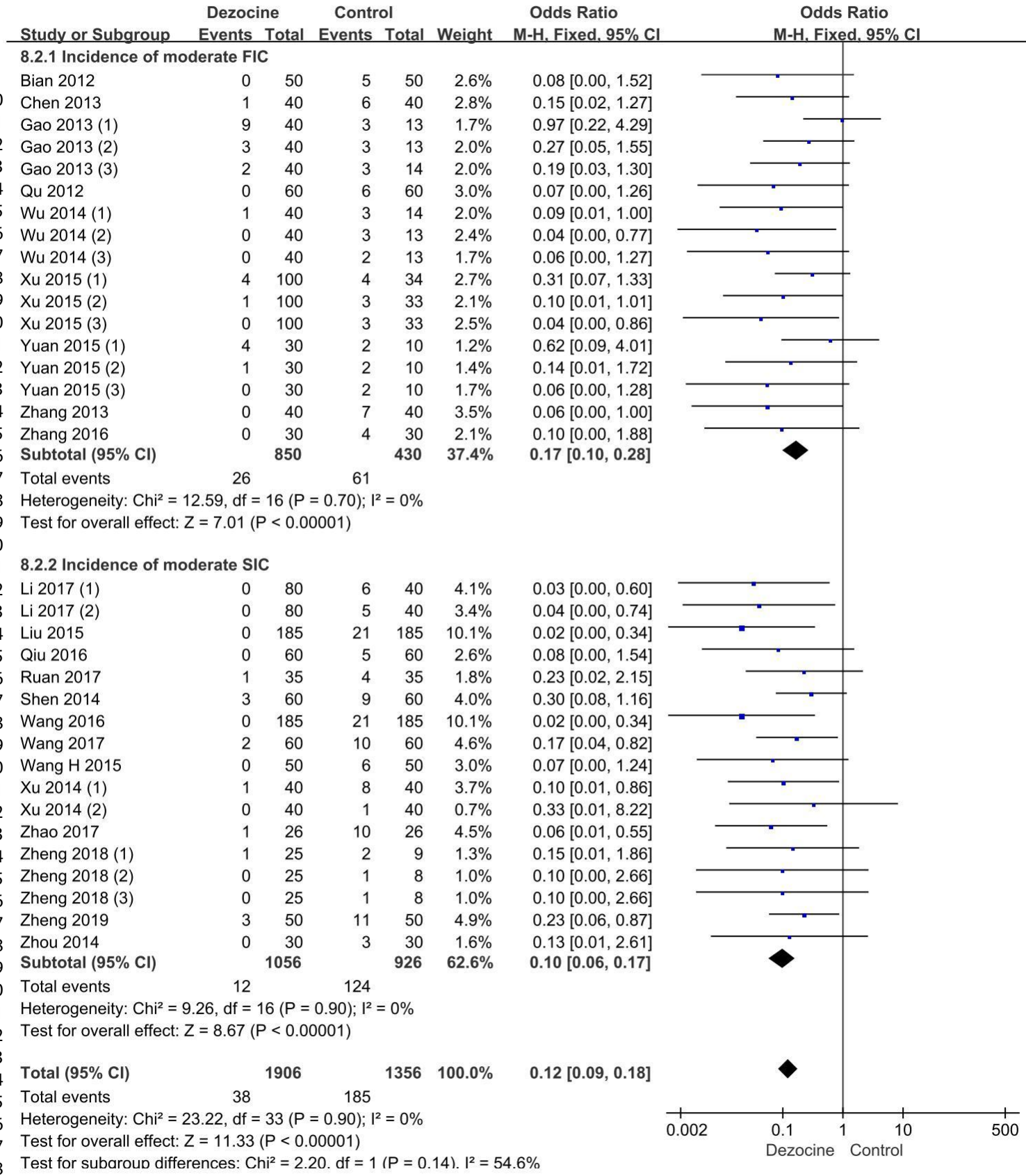

Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participants and personnel (performance bias)
Blinding of outcome assessment (detection bias)
Incomplete outcome data (attrition bias)
Selective reporting (reporting bias)
Other bias
0%
25%
50%
75%
100%
Low risk of bias
Unclear risk of bias
High risk of bias

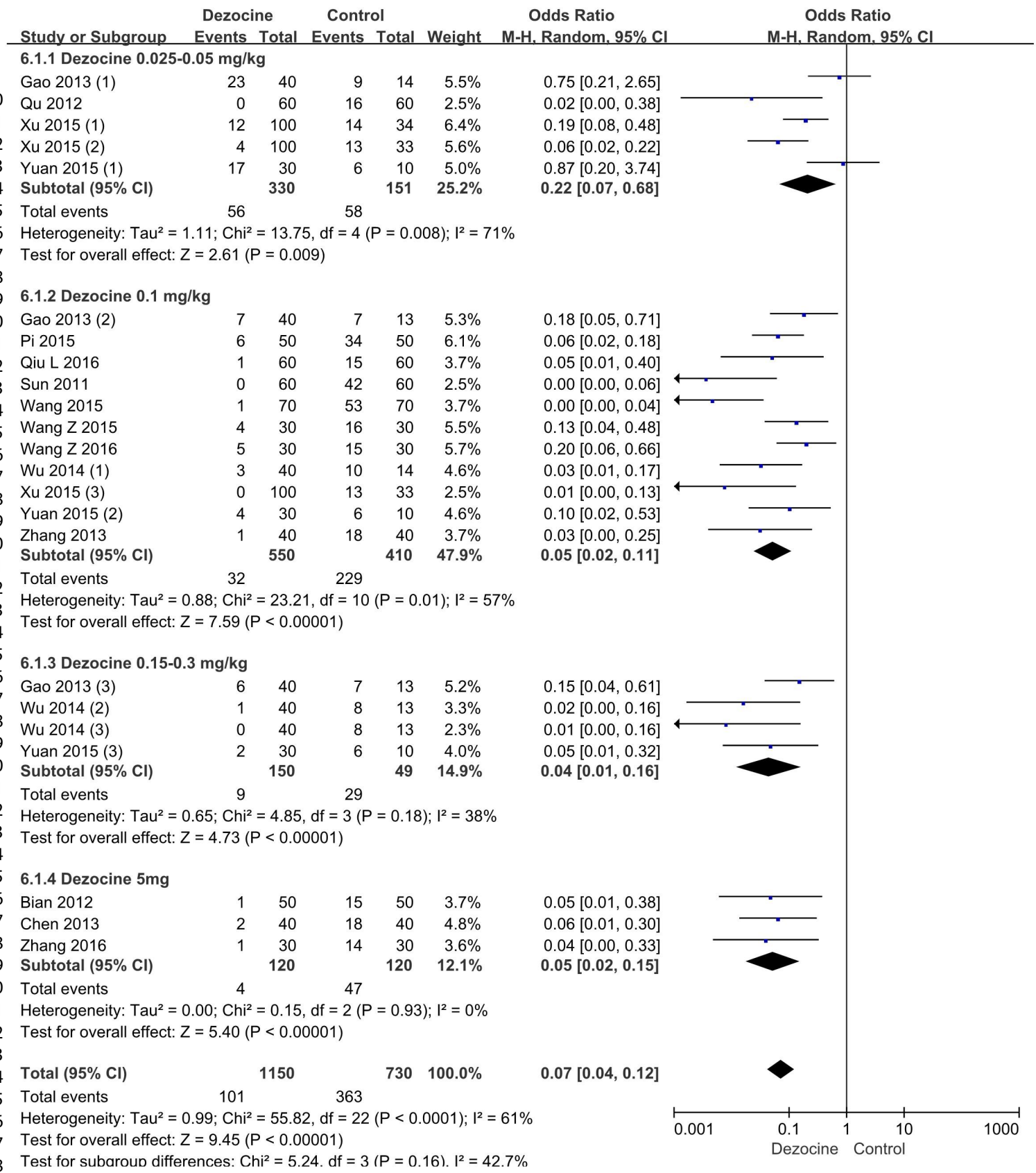

Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participants and personnel (performance bias)
Blinding of outcome assessment (detection bias)
Incomplete outcome data (attrition bias)
Selective reporting (reporting bias)
Other bias
Bian 2012
Chen 2013
Fang 2016
Gao 2013 (1)
Gao 2013 (2)
Gao 2013 (3)
Li 2017 (1)
Li 2017 (2)
Liu 2015
Meng 2013
Pi 2015
Qiu 2016
Qiu L 2016
Qu 2012
Ruan 2017
Shen 2014
Sheng 2017
Sun 2011
Wang 2015
Wang 2016
Wang 2017
Wang H 2015
Wang L 2015 (1)
Wang L 2015 (2)
Wang L 2015 (3)
Wang Z 2015
Wang Z 2016
Wu 2014 (1)
Wu 2014 (2)
Wu 2014 (3)
Xu 2014 (1)
Xu 2014 (2)
Xu 2015 (1)
Xu 2015 (2)
Xu 2015 (3)
Yuan 2015 (1)
Yuan 2015 (2)
Yuan 2015 (3)
Zhang 2013
Zhang 2016
Zhao 2017
Zheng 2018 (1)
Zheng 2018 (2)
Zheng 2018 (3)
Zheng 2019
Zhou 2014
Zhu 2018

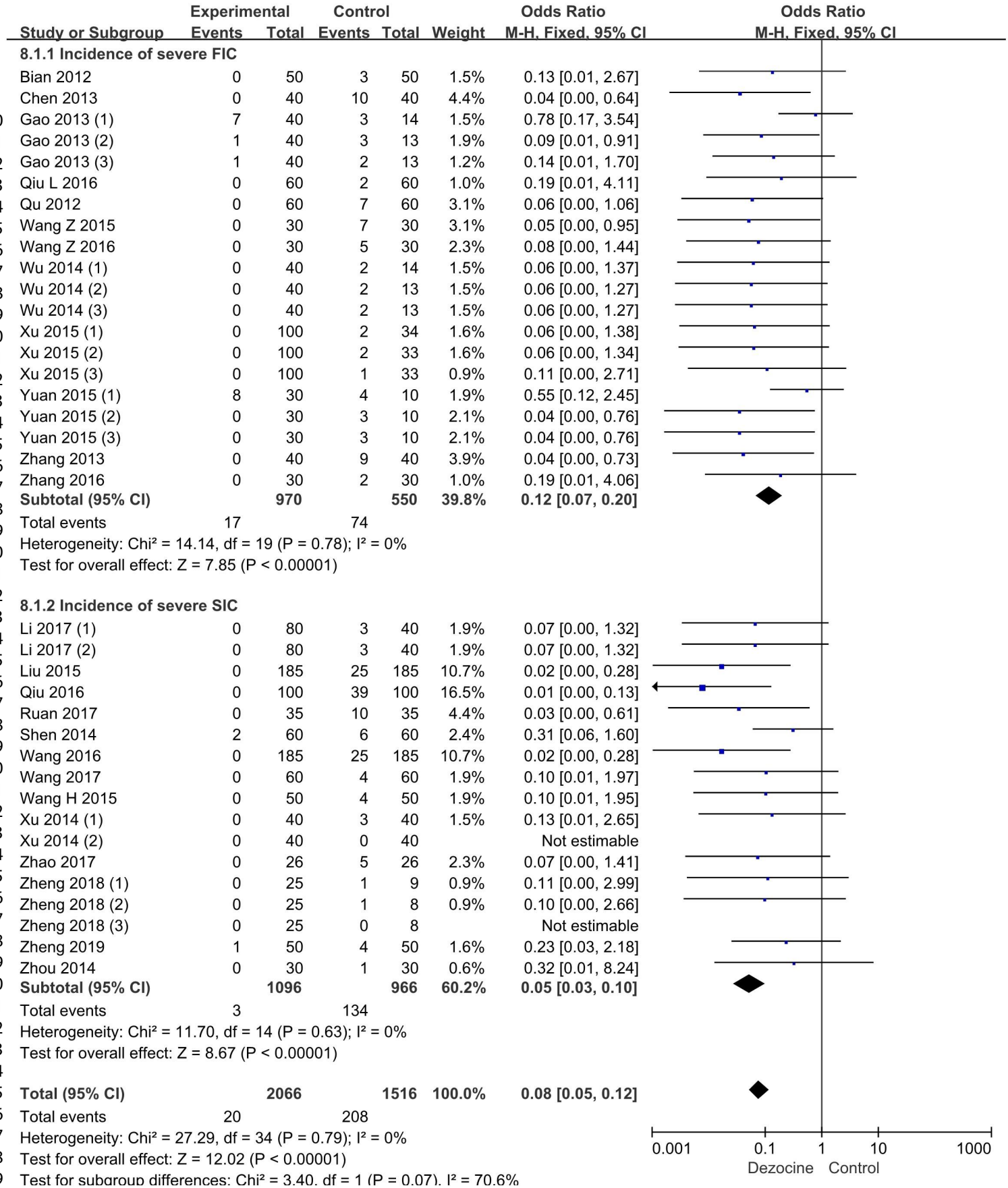

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.1.1 Fentanyl 2μg/kg
Chen 2013 2 40 18 40 4.8% 0.06 [0.01, 0.30]
Subtotal (95% CI) 40 40 4.8% 0.06 [0.01, 0.30]
Total events 2 18
Heterogeneity: Not applicable
Test for overall effect: Z = 3.46 (P = 0.0005)
4.1.2 Fentanyl 3μg/kg
Bian 2012 1 50 15 50 3.7% 0.05 [0.01, 0.38]
Gao 2013 (1) 23 40 9 14 5.5% 0.75 [0.21, 2.65]
Gao 2013 (2) 7 40 7 13 5.3% 0.18 [0.05, 0.71]
Gao 2013 (3) 6 40 7 13 5.2% 0.15 [0.04, 0.61]
Qiu L 2016 1 60 15 60 3.7% 0.05 [0.01, 0.40]
Wu 2014 (1) 3 40 10 14 4.6% 0.03 [0.01, 0.17]
Wu 2014 (2) 1 40 8 13 3.3% 0.02 [0.00, 0.16]
Wu 2014 (3) 0 40 8 13 2.3% 0.01 [0.00, 0.16]
Xu 2015 (1) 12 100 14 34 6.4% 0.19 [0.08, 0.48]
Xu 2015 (2) 4 100 13 33 5.6% 0.06 [0.02, 0.22]
Xu 2015 (3) 0 100 13 33 2.5% 0.01 [0.00, 0.13]
Yuan 2015 (1) 17 30 6 10 5.0% 0.87 [0.20, 3.74]
Yuan 2015 (2) 4 30 6 10 4.6% 0.10 [0.02, 0.53]
Yuan 2015 (3) 2 30 6 10 4.0% 0.05 [0.01, 0.32]
Zhang 2013 1 40 18 40 3.7% 0.03 [0.00, 0.25]
Subtotal (95% CI) 780 360 65.5% 0.09 [0.05, 0.18]
Total events 82 155
Heterogeneity: Tau² = 0.95; Chi² = 34.43, df = 14 (P = 0.002); I² = 59%
Test for overall effect: Z = 7.05 (P < 0.00001)
4.1.3 Fentanyl 4μg/kg
Pi 2015 6 50 34 50 6.1% 0.06 [0.02, 0.18]
Qu 2012 0 60 16 60 2.5% 0.02 [0.00, 0.38]
Wang Z 2015 4 30 16 30 5.5% 0.13 [0.04, 0.48]
Wang Z 2016 5 30 15 30 5.7% 0.20 [0.06, 0.66]
Zhang 2016 1 30 14 30 3.6% 0.04 [0.00, 0.33]
Subtotal (95% CI) 200 200 23.4% 0.09 [0.05, 0.18]
Total events 16 95
Heterogeneity: Tau² = 0.02; Chi² = 4.15, df = 4 (P = 0.39); I² = 4%
Test for overall effect: Z = 7.24 (P < 0.00001)
4.1.4 Fentanyl 5μg/kg
Sun 2011 0 60 42 60 2.5% 0.00 [0.00, 0.06]
Wang 2015 1 70 53 70 3.7% 0.00 [0.00, 0.04]
Subtotal (95% CI) 130 130 6.2% 0.00 [0.00, 0.02]
Total events 1 95
Heterogeneity: Tau² = 0.00; Chi² = 0.02, df = 1 (P = 0.88); I² = 0%
Test for overall effect: Z = 6.44 (P < 0.00001)
Total (95% CI) 1150 730 100.0% 0.07 [0.04, 0.12]
Total events 101 363
Heterogeneity: Tau² = 0.99; Chi² = 55.82, df = 22 (P < 0.0001); I² = 61%
Test for overall effect: Z = 9.45 (P < 0.00001)
Test for subgroup differences: Chi² = 12.31, df = 3 (P = 0.006), I² = 75.6%
0.001
0.1
1
10
1000
Dezocine
Control

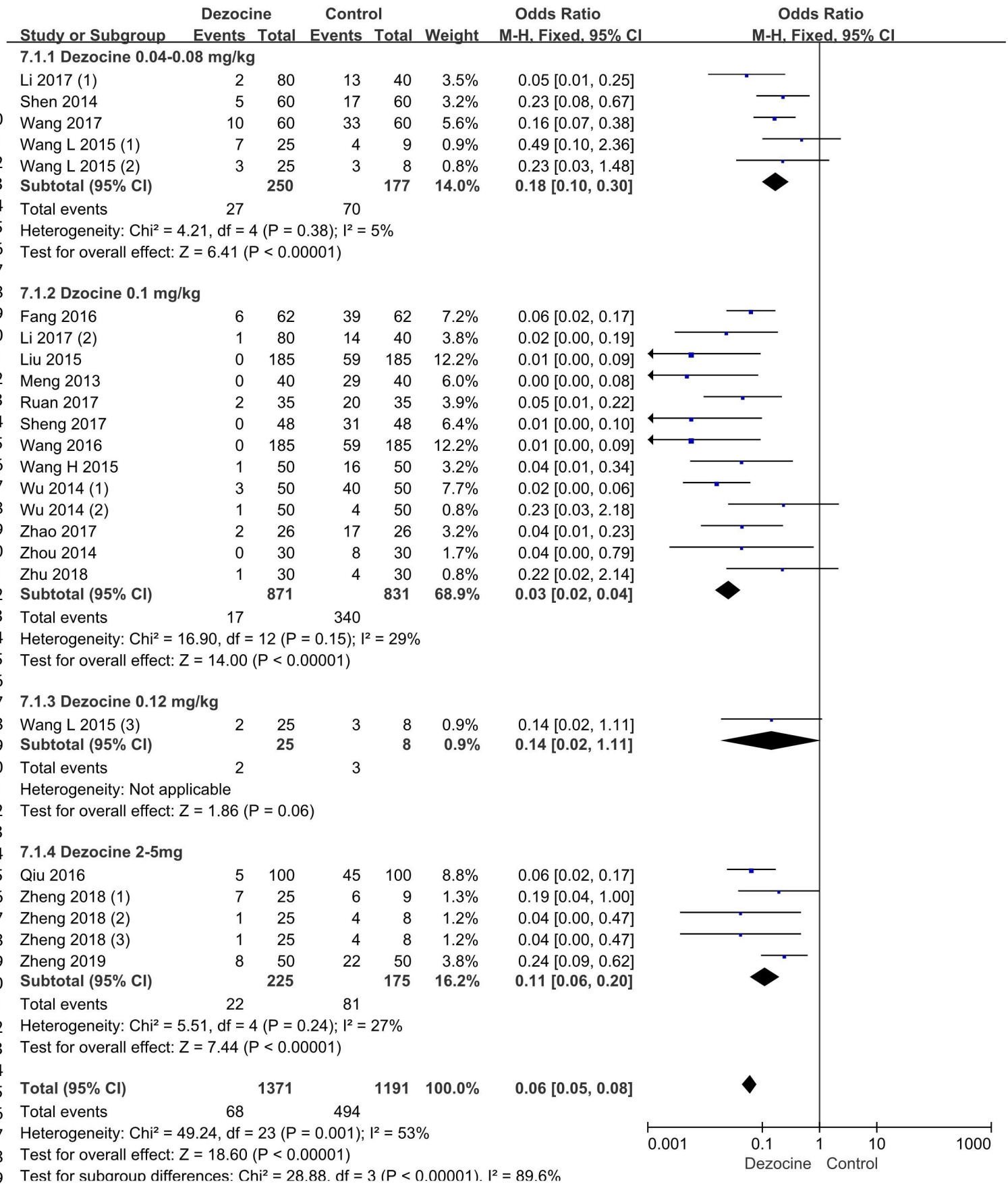

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
3.1.1 Sufentanil 0.3 μg/kg
Li 2017 (1) 2 80 13 40 5.1% 0.05 [0.01, 0.25]
Li 2017 (2) 1 80 14 40 3.8% 0.02 [0.00, 0.19]
Sheng 2017 0 48 31 48 2.5% 0.01 [0.00, 0.10]
Wang 2017 1 50 16 50 3.8% 0.04 [0.01, 0.34]
Wang H 2015 7 25 4 9 5.1% 0.49 [0.10, 2.36]
Wang L 2015 (1) 3 25 3 8 4.3% 0.23 [0.03, 1.48]
Wang L 2015 (2) 2 25 3 8 3.9% 0.14 [0.02, 1.11]
Wang L 2015 (3) 0 30 8 30 2.4% 0.04 [0.00, 0.79]
Zhou 2014 0 0 0 0 Not estimable
Subtotal (95% CI) 363 233 30.9% 0.08 [0.03, 0.20]
Total events 16 92
Heterogeneity: Tau² = 0.89; Chi² = 13.03, df = 7 (P = 0.07); I² = 46%
Test for overall effect: Z = 5.17 (P < 0.00001)
3.1.2 Sufentanil 0.4 μg/kg
Meng 2013 0 40 29 40 2.5% 0.00 [0.00, 0.08]
Ruan 2017 2 35 20 35 5.1% 0.05 [0.01, 0.22]
Zheng 2018 (1) 7 25 6 9 4.9% 0.19 [0.04, 1.00]
Zheng 2018 (2) 1 25 4 8 3.1% 0.04 [0.00, 0.47]
Zheng 2018 (3) 1 25 4 8 3.1% 0.04 [0.00, 0.47]
Zheng 2019 8 50 22 50 7.2% 0.24 [0.09, 0.62]
Zhu 2018 1 30 4 30 3.4% 0.22 [0.02, 2.14]
Subtotal (95% CI) 230 180 29.2% 0.09 [0.03, 0.22]
Total events 20 89
Heterogeneity: Tau² = 0.71; Chi² = 11.16, df = 6 (P = 0.08); I² = 46%
Test for overall effect: Z = 5.04 (P < 0.00001)
3.1.3 Sufentanil 0.5 μg/kg
Liu 2015 0 185 59 185 2.6% 0.01 [0.00, 0.09]
Qiu 2016 5 100 45 100 7.0% 0.06 [0.02, 0.17]
Wang 2016 0 185 59 185 2.6% 0.01 [0.00, 0.09]
Xu 2014 (1) 3 50 40 50 5.7% 0.02 [0.00, 0.06]
Xu 2014 (2) 1 50 4 50 3.5% 0.23 [0.03, 2.18]
Subtotal (95% CI) 570 570 21.4% 0.03 [0.01, 0.11]
Total events 9 207
Heterogeneity: Tau² = 1.21; Chi² = 10.16, df = 4 (P = 0.04); I² = 61%
Test for overall effect: Z = 5.35 (P < 0.00001)
3.1.4 Sufentanil 3-5 μg/kg
Fang 2016 6 62 39 62 7.0% 0.06 [0.02, 0.17]
Shen 2014 5 60 17 60 6.7% 0.23 [0.08, 0.67]
Zhao 2017 2 26 17 26 4.8% 0.04 [0.01, 0.23]
Subtotal (95% CI) 148 148 18.6% 0.09 [0.03, 0.25]
Total events 13 73
Heterogeneity: Tau² = 0.39; Chi² = 4.08, df = 2 (P = 0.13); I² = 51%
Test for overall effect: Z = 4.69 (P < 0.00001)
Total (95% CI) 1311 1131 100.0% 0.07 [0.04, 0.11]
Total events 58 461
Heterogeneity: Tau² = 0.77; Chi² = 46.43, df = 22 (P = 0.002); I² = 53%
Test for overall effect: Z = 10.08 (P < 0.00001)
Test for subgroup differences: Chi² = 2.28, df = 3 (P = 0.52), I² = 0%
0.001
0.1
1
10
1000
Dezocine
Control

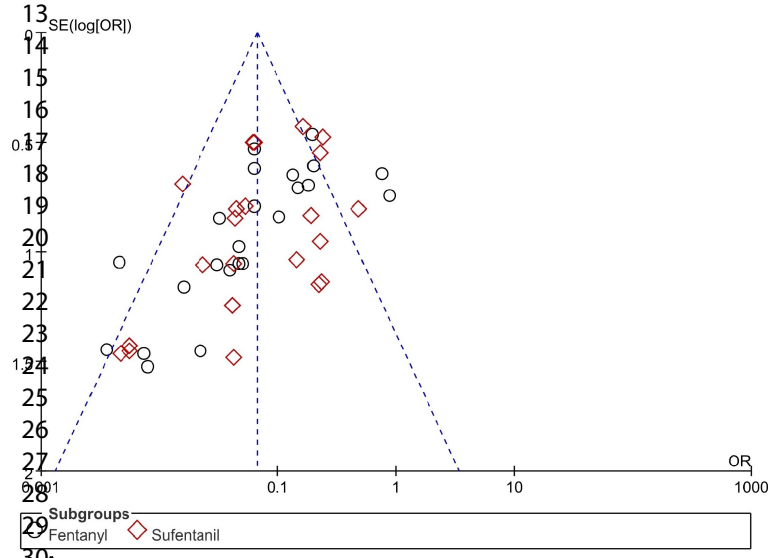

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.3.1 Incidence of mild FIC
Bian 2012 1 30 8 30 3.7% 0.09 [0.01, 0.82]
Chen 2013 2 40 2 40 0.9% 1.00 [0.13, 7.47]
Gao 2013 (1) 2 30 0 10 0.3% 1.84 [0.08, 41.62]
Gao 2013 (2) 3 30 1 10 0.6% 1.00 [0.09, 10.87]
Gao 2013 (3) 5 30 1 10 0.6% 1.80 [0.18, 17.57]
Qu 2012 0 100 8 33 6.0% 0.01 [0.00, 0.27]
Wu 2014 (1) 3 100 8 33 5.6% 0.10 [0.02, 0.39]
Wu 2014 (2) 8 100 9 34 5.9% 0.24 [0.08, 0.69]
Wu 2014 (3) 0 40 4 13 3.2% 0.03 [0.00, 0.53]
Xu 2015 (1) 1 40 4 13 2.8% 0.06 [0.01, 0.58]
Xu 2015 (2) 4 40 4 14 2.5% 0.28 [0.06, 1.31]
Xu 2015 (3) 0 60 5 60 2.6% 0.08 [0.00, 1.54]
Yuan 2015 (1) 3 40 2 14 1.3% 0.49 [0.07, 3.27]
Yuan 2015 (2) 3 40 2 13 1.3% 0.45 [0.07, 3.02]
Yuan 2015 (3) 7 40 2 13 1.2% 1.17 [0.21, 6.47]
Zhang 2013 1 40 2 40 0.9% 0.49 [0.04, 5.60]
Zhang 2016 1 50 4 50 1.9% 0.23 [0.03, 2.18]
Subtotal (95% CI) 850 430 41.4% 0.25 [0.16, 0.38]
Total events 44 66
Heterogeneity: Chi² = 22.33, df = 16 (P = 0.13); I² = 28%
Test for overall effect: Z = 6.62 (P < 0.00001)
8.3.2 Incidence of mild SIC
Li 2017 (1) 0 30 4 30 2.1% 0.10 [0.00, 1.88]
Li 2017 (2) 4 50 7 50 3.1% 0.53 [0.15, 1.95]
Liu 2015 1 25 2 8 1.4% 0.13 [0.01, 1.62]
Qiu 2016 1 25 2 8 1.4% 0.13 [0.01, 1.62]
Ruan 2017 6 25 3 9 1.6% 0.63 [0.12, 3.33]
Shen 2014 1 26 2 26 0.9% 0.48 [0.04, 5.65]
Wang 2016 1 40 3 40 1.4% 0.32 [0.03, 3.18]
Wang 2017 2 40 29 40 13.1% 0.02 [0.00, 0.10]
Wang H 2015 1 50 6 50 2.8% 0.15 [0.02, 1.29]
Xu 2014 (1) 3 60 9 60 4.1% 0.30 [0.08, 1.16]
Xu 2014 (2) 0 185 13 185 6.4% 0.03 [0.00, 0.58]
Zhao 2017 0 60 2 60 1.2% 0.19 [0.01, 4.11]
Zheng 2018 (1) 1 35 6 35 2.8% 0.14 [0.02, 1.25]
Zheng 2018 (2) 1 60 8 60 3.7% 0.11 [0.01, 0.91]
Zheng 2018 (3) 0 185 13 185 6.4% 0.03 [0.00, 0.58]
Zheng 2019 1 80 5 40 3.1% 0.09 [0.01, 0.79]
Zhou 2014 2 80 5 40 3.1% 0.18 [0.03, 0.97]
Subtotal (95% CI) 1056 926 58.6% 0.14 [0.09, 0.22]
Total events 25 119
Heterogeneity: Chi² = 17.94, df = 16 (P = 0.33); I² = 11%
Test for overall effect: Z = 8.39 (P < 0.00001)
Total (95% CI) 1906 1356 100.0% 0.19 [0.14, 0.25]
Total events 69 185
Heterogeneity: Chi² = 42.29, df = 33 (P = 0.13); I² = 22%
Test for overall effect: Z = 10.87 (P < 0.00001)
Test for subgroup differences: Chi² = 3.24, df = 1 (P = 0.07), I² = 69.2%
0.002
0.1
1
10
500
Dezocine
Control


Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup
Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.2.1 Incidence of moderate FIC
Bian 2012 0 50 5 50 2.6% 0.08 [0.00, 1.52]
Chen 2013 1 40 6 40 2.8% 0.15 [0.02, 1.27]
Gao 2013 (1) 9 40 3 13 1.7% 0.97 [0.22, 4.29]
Gao 2013 (2) 3 40 3 13 2.0% 0.27 [0.05, 1.55]
Gao 2013 (3) 2 40 3 14 2.0% 0.19 [0.03, 1.30]
Qu 2012 0 60 6 60 3.0% 0.07 [0.00, 1.26]
Wu 2014 (1) 1 40 3 14 2.0% 0.09 [0.01, 1.00]
Wu 2014 (2) 0 40 3 13 2.4% 0.04 [0.00, 0.77]
Wu 2014 (3) 0 40 2 13 1.7% 0.06 [0.00, 1.27]
Xu 2015 (1) 4 100 4 34 2.7% 0.31 [0.07, 1.33]
Xu 2015 (2) 1 100 3 33 2.1% 0.10 [0.01, 1.01]
Xu 2015 (3) 0 100 3 33 2.5% 0.04 [0.00, 0.86]
Yuan 2015 (1) 4 30 2 10 1.2% 0.62 [0.09, 4.01]
Yuan 2015 (2) 1 30 2 10 1.4% 0.14 [0.01, 1.72]
Yuan 2015 (3) 0 30 2 10 1.7% 0.06 [0.00, 1.28]
Zhang 2013 0 40 7 40 3.5% 0.06 [0.00, 1.00]
Zhang 2016 0 30 4 30 2.1% 0.10 [0.00, 1.88]
Subtotal (95% CI) 850 430 37.4% 0.17 [0.10, 0.28]
Total events 26 61
Heterogeneity: Chi² = 12.59, df = 16 (P = 0.70); I² = 0%
Test for overall effect: Z = 7.01 (P < 0.00001)
8.2.2 Incidence of moderate SIC
Li 2017 (1) 0 80 6 40 4.1% 0.03 [0.00, 0.60]
Li 2017 (2) 0 80 5 40 3.4% 0.04 [0.00, 0.74]
Liu 2015 0 185 21 185 10.1% 0.02 [0.00, 0.34]
Qiu 2016 0 60 5 60 2.6% 0.08 [0.00, 1.54]
Ruan 2017 1 35 4 35 1.8% 0.23 [0.02, 2.15]
Shen 2014 3 60 9 60 4.0% 0.30 [0.08, 1.16]
Wang 2016 0 185 21 185 10.1% 0.02 [0.00, 0.34]
Wang 2017 2 60 10 60 4.6% 0.17 [0.04, 0.82]
Wang H 2015 0 50 6 50 3.0% 0.07 [0.00, 1.24]
Xu 2014 (1) 1 40 8 40 3.7% 0.10 [0.01, 0.86]
Xu 2014 (2) 0 40 1 40 0.7% 0.33 [0.01, 8.22]
Zhao 2017 1 26 10 26 4.5% 0.06 [0.01, 0.55]
Zheng 2018 (1) 1 25 2 9 1.3% 0.15 [0.01, 1.86]
Zheng 2018 (2) 0 25 1 8 1.0% 0.10 [0.00, 2.66]
Zheng 2018 (3) 0 25 1 8 1.0% 0.10 [0.00, 2.66]
Zheng 2019 3 50 11 50 4.9% 0.23 [0.06, 0.87]
Zhou 2014 0 30 3 30 1.6% 0.13 [0.01, 2.61]
Subtotal (95% CI) 1056 926 62.6% 0.10 [0.06, 0.17]
Total events 12 124
Heterogeneity: Chi² = 9.26, df = 16 (P = 0.90); I² = 0%
Test for overall effect: Z = 8.67 (P < 0.00001)
Total (95% CI) 1906 1356 100.0% 0.12 [0.09, 0.18]
Total events 38 185
Heterogeneity: Chi² = 23.22, df = 33 (P = 0.90); I² = 0%
Test for overall effect: Z = 11.33 (P < 0.00001)
Test for subgroup differences: Chi² = 2.20, df = 1 (P = 0.14), I² = 54.6%
0.002
0.1
1
10
500
Dezocine
Control

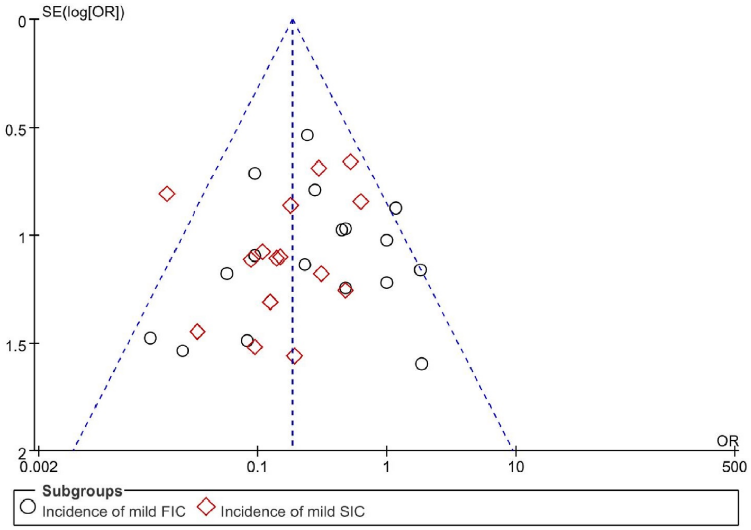

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
6.1.1 Dezocine 0.025-0.05 mg/kg
Gao 2013 (1) 23 40 9 14 5.5% 0.75 [0.21, 2.65]
Qu 2012 0 60 16 60 2.5% 0.02 [0.00, 0.38]
Xu 2015 (1) 12 100 14 34 6.4% 0.19 [0.08, 0.48]
Xu 2015 (2) 4 100 13 33 5.6% 0.06 [0.02, 0.22]
Yuan 2015 (1) 17 30 6 10 5.0% 0.87 [0.20, 3.74]
Subtotal (95% CI) 330 151 25.2% 0.22 [0.07, 0.68]
Total events 56 58
Heterogeneity: Tau² = 1.11; Chi² = 13.75, df = 4 (P = 0.008); I² = 71%
Test for overall effect: Z = 2.61 (P = 0.009)
6.1.2 Dezocine 0.1 mg/kg
Gao 2013 (2) 7 40 7 13 5.3% 0.18 [0.05, 0.71]
Pi 2015 6 50 34 50 6.1% 0.06 [0.02, 0.18]
Qiu L 2016 1 60 15 60 3.7% 0.05 [0.01, 0.40]
Sun 2011 0 60 42 60 2.5% 0.00 [0.00, 0.06]
Wang 2015 1 70 53 70 3.7% 0.00 [0.00, 0.04]
Wang Z 2015 4 30 16 30 5.5% 0.13 [0.04, 0.48]
Wang Z 2016 5 30 15 30 5.7% 0.20 [0.06, 0.66]
Wu 2014 (1) 3 40 10 14 4.6% 0.03 [0.01, 0.17]
Xu 2015 (3) 0 100 13 33 2.5% 0.01 [0.00, 0.13]
Yuan 2015 (2) 4 30 6 10 4.6% 0.10 [0.02, 0.53]
Zhang 2013 1 40 18 40 3.7% 0.03 [0.00, 0.25]
Subtotal (95% CI) 550 410 47.9% 0.05 [0.02, 0.11]
Total events 32 229
Heterogeneity: Tau² = 0.88; Chi² = 23.21, df = 10 (P = 0.01); I² = 57%
Test for overall effect: Z = 7.59 (P < 0.00001)
6.1.3 Dezocine 0.15-0.3 mg/kg
Gao 2013 (3) 6 40 7 13 5.2% 0.15 [0.04, 0.61]
Wu 2014 (2) 1 40 8 13 3.3% 0.02 [0.00, 0.16]
Wu 2014 (3) 0 40 8 13 2.3% 0.01 [0.00, 0.16]
Yuan 2015 (3) 2 30 6 10 4.0% 0.05 [0.01, 0.32]
Subtotal (95% CI) 150 49 14.9% 0.04 [0.01, 0.16]
Total events 9 29
Heterogeneity: Tau² = 0.65; Chi² = 4.85, df = 3 (P = 0.18); I² = 38%
Test for overall effect: Z = 4.73 (P < 0.00001)
6.1.4 Dezocine 5mg
Bian 2012 1 50 15 50 3.7% 0.05 [0.01, 0.38]
Chen 2013 2 40 18 40 4.8% 0.06 [0.01, 0.30]
Zhang 2016 1 30 14 30 3.6% 0.04 [0.00, 0.33]
Subtotal (95% CI) 120 120 12.1% 0.05 [0.02, 0.15]
Total events 4 47
Heterogeneity: Tau² = 0.00; Chi² = 0.15, df = 2 (P = 0.93); I² = 0%
Test for overall effect: Z = 5.40 (P < 0.00001)
Total (95% CI) 1150 730 100.0% 0.07 [0.04, 0.12]
Total events 101 363
Heterogeneity: Tau² = 0.99; Chi² = 55.82, df = 22 (P < 0.0001); I² = 61%
Test for overall effect: Z = 9.45 (P < 0.00001)
Test for subgroup differences: Chi² = 5.24, df = 3 (P = 0.16), I² = 42.7%
0.001
0.1
1
10
1000
Dezocine
Control


Experimental
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.1.1 Incidence of severe FIC
Bian 2012 0 50 3 50 1.5% 0.13 [0.01, 2.67]
Chen 2013 0 40 10 40 4.4% 0.04 [0.00, 0.64]
Gao 2013 (1) 7 40 3 14 1.5% 0.78 [0.17, 3.54]
Gao 2013 (2) 1 40 3 13 1.9% 0.09 [0.01, 0.91]
Gao 2013 (3) 1 40 2 13 1.2% 0.14 [0.01, 1.70]
Qiu L 2016 0 60 2 60 1.0% 0.19 [0.01, 4.11]
Qu 2012 0 60 7 60 3.1% 0.06 [0.00, 1.06]
Wang Z 2015 0 30 7 30 3.1% 0.05 [0.00, 0.95]
Wang Z 2016 0 30 5 30 2.3% 0.08 [0.00, 1.44]
Wu 2014 (1) 0 40 2 14 1.5% 0.06 [0.00, 1.37]
Wu 2014 (2) 0 40 2 13 1.5% 0.06 [0.00, 1.27]
Wu 2014 (3) 0 40 2 13 1.5% 0.06 [0.00, 1.27]
Xu 2015 (1) 0 100 2 34 1.6% 0.06 [0.00, 1.38]
Xu 2015 (2) 0 100 2 33 1.6% 0.06 [0.00, 1.34]
Xu 2015 (3) 0 100 1 33 0.9% 0.11 [0.00, 2.71]
Yuan 2015 (1) 8 30 4 10 1.9% 0.55 [0.12, 2.45]
Yuan 2015 (2) 0 30 3 10 2.1% 0.04 [0.00, 0.76]
Yuan 2015 (3) 0 30 3 10 2.1% 0.04 [0.00, 0.76]
Zhang 2013 0 40 9 40 3.9% 0.04 [0.00, 0.73]
Zhang 2016 0 30 2 30 1.0% 0.19 [0.01, 4.06]
Subtotal (95% CI) 970 550 39.8% 0.12 [0.07, 0.20]
Total events 17 74
Heterogeneity: Chi² = 14.14, df = 19 (P = 0.78); I² = 0%
Test for overall effect: Z = 7.85 (P < 0.00001)
8.1.2 Incidence of severe SIC
Li 2017 (1) 0 80 3 40 1.9% 0.07 [0.00, 1.32]
Li 2017 (2) 0 80 3 40 1.9% 0.07 [0.00, 1.32]
Liu 2015 0 185 25 185 10.7% 0.02 [0.00, 0.28]
Qiu 2016 0 100 39 100 16.5% 0.01 [0.00, 0.13]
Ruan 2017 0 35 10 35 4.4% 0.03 [0.00, 0.61]
Shen 2014 2 60 6 60 2.4% 0.31 [0.06, 1.60]
Wang 2016 0 185 25 185 10.7% 0.02 [0.00, 0.28]
Wang 2017 0 60 4 60 1.9% 0.10 [0.01, 1.97]
Wang H 2015 0 50 4 50 1.9% 0.10 [0.01, 1.95]
Xu 2014 (1) 0 40 3 40 1.5% 0.13 [0.01, 2.65]
Xu 2014 (2) 0 40 0 40 Not estimable
Zhao 2017 0 26 5 26 2.3% 0.07 [0.00, 1.41]
Zheng 2018 (1) 0 25 1 9 0.9% 0.11 [0.00, 2.99]
Zheng 2018 (2) 0 25 1 8 0.9% 0.10 [0.00, 2.66]
Zheng 2018 (3) 0 25 0 8 Not estimable
Zheng 2019 1 50 4 50 1.6% 0.23 [0.03, 2.18]
Zhou 2014 0 30 1 30 0.6% 0.32 [0.01, 8.24]
Subtotal (95% CI) 1096 966 60.2% 0.05 [0.03, 0.10]
Total events 3 134
Heterogeneity: Chi² = 11.70, df = 14 (P = 0.63); I² = 0%
Test for overall effect: Z = 8.67 (P < 0.00001)
Total (95% CI) 2066 1516 100.0% 0.08 [0.05, 0.12]
Total events 20 208
Heterogeneity: Chi² = 27.29, df = 34 (P = 0.79); I² = 0%
Test for overall effect: Z = 12.02 (P < 0.00001)
Test for subgroup differences: Chi² = 3.40, df = 1 (P = 0.07), I² = 70.6%
0.001
0.1
1
10
1000
Dezocine
Control

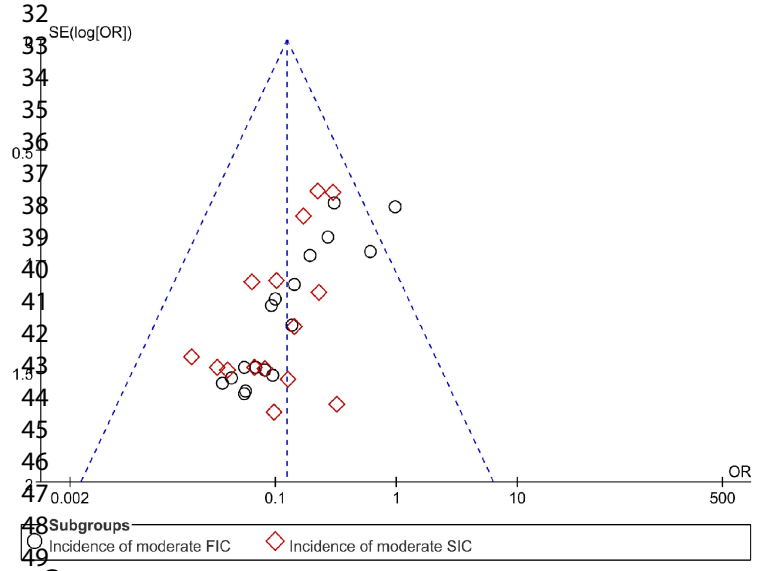

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
7.1.1 Dezocine 0.04-0.08 mg/kg
Li 2017 (1) 2 80 13 40 3.5% 0.05 [0.01, 0.25]
Shen 2014 5 60 17 60 3.2% 0.23 [0.08, 0.67]
Wang 2017 10 60 33 60 5.6% 0.16 [0.07, 0.38]
Wang L 2015 (1) 7 25 4 9 0.9% 0.49 [0.10, 2.36]
Wang L 2015 (2) 3 25 3 8 0.8% 0.23 [0.03, 1.48]
Subtotal (95% CI) 250 177 14.0% 0.18 [0.10, 0.30]
Total events 27 70
Heterogeneity: Chi² = 4.21, df = 4 (P = 0.38); I² = 5%
Test for overall effect: Z = 6.41 (P < 0.00001)
7.1.2 Dzocine 0.1 mg/kg
Fang 2016 6 62 39 62 7.2% 0.06 [0.02, 0.17]
Li 2017 (2) 1 80 14 40 3.8% 0.02 [0.00, 0.19]
Liu 2015 0 185 59 185 12.2% 0.01 [0.00, 0.09]
Meng 2013 0 40 29 40 6.0% 0.00 [0.00, 0.08]
Ruan 2017 2 35 20 35 3.9% 0.05 [0.01, 0.22]
Sheng 2017 0 48 31 48 6.4% 0.01 [0.00, 0.10]
Wang 2016 0 185 59 185 12.2% 0.01 [0.00, 0.09]
Wang H 2015 1 50 16 50 3.2% 0.04 [0.01, 0.34]
Wu 2014 (1) 3 50 40 50 7.7% 0.02 [0.00, 0.06]
Wu 2014 (2) 1 50 4 50 0.8% 0.23 [0.03, 2.18]
Zhao 2017 2 26 17 26 3.2% 0.04 [0.01, 0.23]
Zhou 2014 0 30 8 30 1.7% 0.04 [0.00, 0.79]
Zhu 2018 1 30 4 30 0.8% 0.22 [0.02, 2.14]
Subtotal (95% CI) 871 831 68.9% 0.03 [0.02, 0.04]
Total events 17 340
Heterogeneity: Chi² = 16.90, df = 12 (P = 0.15); I² = 29%
Test for overall effect: Z = 14.00 (P < 0.00001)
7.1.3 Dezocine 0.12 mg/kg
Wang L 2015 (3) 2 25 3 8 0.9% 0.14 [0.02, 1.11]
Subtotal (95% CI) 25 8 0.9% 0.14 [0.02, 1.11]
Total events 2 3
Heterogeneity: Not applicable
Test for overall effect: Z = 1.86 (P = 0.06)
7.1.4 Dezocine 2-5mg
Qiu 2016 5 100 45 100 8.8% 0.06 [0.02, 0.17]
Zheng 2018 (1) 7 25 6 9 1.3% 0.19 [0.04, 1.00]
Zheng 2018 (2) 1 25 4 8 1.2% 0.04 [0.00, 0.47]
Zheng 2018 (3) 1 25 4 8 1.2% 0.04 [0.00, 0.47]
Zheng 2019 8 50 22 50 3.8% 0.24 [0.09, 0.62]
Subtotal (95% CI) 225 175 16.2% 0.11 [0.06, 0.20]
Total events 22 81
Heterogeneity: Chi² = 5.51, df = 4 (P = 0.24); I² = 27%
Test for overall effect: Z = 7.44 (P < 0.00001)
Total (95% CI) 1371 1191 100.0% 0.06 [0.05, 0.08]
Total events 68 494
Heterogeneity: Chi² = 49.24, df = 23 (P = 0.001); I² = 53%
Test for overall effect: Z = 18.60 (P < 0.00001)
Test for subgroup differences: Chi² = 28.88, df = 3 (P < 0.00001), I² = 89.6%
0.001
0.1
1
10
1000
Dezocine
Control


SE(log[OR])
OR
0.001
0.1
1
10
1000
Subgroups
Fentanyl
Sufentanil

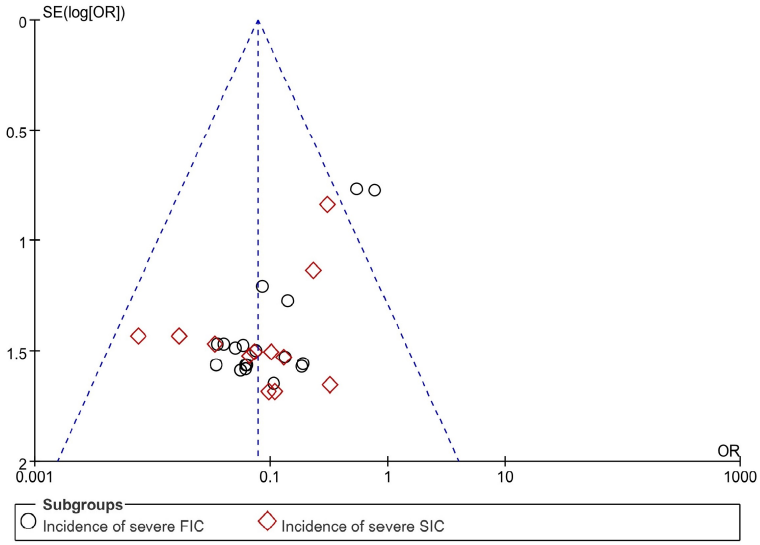
A


SE(log[OR])
0
0.5
1
1.5
2
OR
0.002
0.1
1
10
500
Subgroups
Incidence of mild FIC
Incidence of mild SIC

B


SE(log[OR])
OR
0.002
0.1
1
10
500
Subgroups
Incidence of moderate FIC
Incidence of moderate SIC

C


SE(log[OR])
0
0.5
1
1.5
2
OR
0.001
0.1
1
10
1000
Subgroups
Incidence of severe FIC
Incidence of severe SIC

D

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Efficacy of dezocine on preventing opioids-induced cough during general anesthesia induction: a PRISMA-compliant systematic review and meta-analysis

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Efficacy of dezocine on preventing opioids-induced cough during general anesthesia induction: a PRISMA-compliant systematic review and meta-analysis

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ABSTRACT

Objectives To systematically review the effects of dezocine on the occurrence rate and severity of opioid-induced cough(OIC).

Design Systematic review and meta-analysis

Data sources PubMed, Embase, Cochrane Library, Ovid, Web of Science as well as Chinese BioMedical Literature & Retrieval System, China National Knowledge Infrastructure, Wanfang and VIP Data were searched from 1978 to December 31th 2020.

Inclusion criteria All randomized controlled trials(RCTs) comparing dezocine with placebo on the occurrence rate and severity of OIC.

Data analysis All data were analyzed by utilizing RevMan 5.3. Each outcome was tested for heterogeneity, and randomized-effects or fixed-effects model was used in the presence or absence of significant heterogeneity.

Results Our search yielded 33 RCTs including 4,442 patients, and 2,521 patients were allocated into the dezocine group and 1,921 into the Control group. Fentanyl was administrated in 1,880 patients and sufentanil in 2,562 patients during the induction of general anesthesia. The meta-analysis demonstrated that, dezocine significantly reduced the occurrence rate of OIC induced by either fentanyl(8.8% *vs*. 49.7%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001) or sufentanil(5.0% *vs*. 41.5%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001). The meta-analysis also indicated that, the occurrence rate of mild, moderate and severe OIC in the dezocine group was remarkably lower than that of the Control group(mild: 3.6% *vs*.13.6%, OR=0.19, 95%

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CI=0.14 to 0.25, *P*<0.00001; moderate: 2.0% *vs.*13.6%, OR=0.12, 95% CI=0.09 to 0.18, *P*<0.00001; severe: 1.0% *vs.*13.9%, OR=0.08, 95% CI=0.05 to 0.12, *P*<0.00001). Additionally, the current meta-analysis indicated that DZC pretreatment was not associated with increased occurrence rate of adverse effects (7.0% *vs.*4.2%, OR=2.34, 95% CI=0.60 to 9.14, *P*=0.22) except for dizziness(11.8% *vs.*0%, OR=8.06, 95% CI=1.40 to 46.35, *P*=0.02).

Conclusion This meta-analysis demonstrated that, dezocine significantly inhibited OIC and may be used to manage OIC. More high-quality RCTs are needed to complement the safety of DZC.

PROSPERO registration number CRD42019141255

Keywords: Dezocine, Opioid, Cough, General anesthesia, Meta-analysis

Strengths and limitations of this study

This is the first systematic review to investigate the occurrence rate of opioid-induced cough induced by either fentanyl or sufentanil.

Subgroup analyses were performed on dose-effect of DZC and various kinds of opioids to investigate the optimal dosage of DZC.

The main limitation of this review is that varied quality and heterogeneity of included studies may limit the certainty of the findings of meta-analysis.

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INTRODUCTION

Cough is often observed when administrating a bolus of opioids(*e.g.* fentanyl,[1-4] sufentanil,[5-7] remifentanil,[8-13] alfentanil[14]), with the reported occurrence rate ranging from 7% to 70%.[1-14] The mechanism of opioid-induced cough(OIC) is complex and remains poorly understood, which may involve pulmonary chemoreflex, enhanced activity of parasympathetic nerve, histamine release, opioid receptor dualism and muscular rigidity.[1-3 15-17] OIC is mostly transient, benign and self-limiting, but could be associated with adverse effects such as hypertension, tachycardia, increased intra-cranial, ocular and abdominal pressures, and airway obstruction.[1 2 15-17] OIC could be spasmodic, explosive[18] and life threatening at times.[19] OIC is especially undesirable during the induction of general anesthesia. Numerous pharmacological interventions including lidocaine, atropine, magnesium sulfate($MgSO_4$), dexamethasone, propofol, midazolam, muscular relaxant(rocurounium, vencuronium), ketamine, pentazocine, tramadol, α_2-agonists(clonidine, dexmeditomidine), ß2-agonists(terbutaline, ephedrine), sodium chromoglycate, beclomethasone, salbutamol, dextromethorphan, *etc*, and non-pharmacological interventions such as priming, dilution and slow injection of opioids, have been used to manage OIC.[1 2 4-9 11-13 15 17 19-22] Unfortunately, the efficacy and safety of those anti-tussive interventions remains controversial.

Dezocine(DZC), a mixed opioid agonist/antagnost, was synthesized in 1970s and approved by the Food and Drug Administration of US for perioperative pain management but was discontinued with the closure of its parent company.[23-27]

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Although no longer used clinically in Western countries, DZC has gained popularity in China and been widely used as a peri-operative analgesic for decades.[24 28-32] Recent studies suggested that, pretreatment of intravenous DZC 0.1mg/kg could completely suppress the cough induced by bolus injection of fentanyl or sufentanil during anesthesia induction. For example, Sun and colleagues[4] evaluated the suppressive effect of DZC on fentanyl-induced cough(FIC). One hundred and twenty patients were randomized to receive DZC 0.1 mg/kg or placebo 10 min before fentanyl 5 μg/kg. They demonstrated that, no DZC-pretreated patient had FIC, as compared to 70%(42/60) non-DZC-pretreated patients developing FIC. In another RCT involving 370 patients, Liu and colleagues[6] evaluated the anti-tussive effect of DZC 0.1 mg/kg on sufentanil-induced cough(SIC) during anesthesia induction. They demonstrated the occurrence rate of SIC in the placebo group was 31%(59/185), while no SIC was observed in the DZC group. It is so encouraging that, DZC might be more effective than those above-mentioned anti-tussive interventions, and that DZC could possibly eliminate OIC without causing OIC itself. Therefore, we performed this systemic review and meta-analysis to evaluate the efficacy of DZC on OIC during general anesthesia induction, and possible adverse effects.

METHODS

Patient and public involvement

No patient involved.

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Ethical approval

This study was a meta-analysis of previously published literatures, ethical approval was not necessary according to the Ethical Committee of Fuwai Hospital.

Registration

The protocol of current meta-analysis was published in PROSPERO with the registration number of CRD42019141255 on November 11th 2019.

Search strategy

We conducted a systemic review according to the Preferred Reporting Items for Systemic Reviews and Meta-Analysis Quality of Reporting of Meta-analysis(PRIMSA) Guidelines(**Supplemental Table 1**).[33] Relevant trials were identified by computerized searches of PubMed, Embase, Cochrane Library, Ovid, Web of Science as well as Chinese BioMedical Literature & Retrieval System(SinoMed), China National Knowledge Infrastructure(CNKI), Wanfang Data and VIP Data till December 31th 2019, with an updated databases search on December 31th 2020 prior to submission, using different combination of search words as follows: *(opioid OR fentanyl OR sufentanil OR remifentanil OR alfentanil) AND cough AND dezocine AND (randomized controlled trial OR controlled clinical trial OR randomized OR placebo OR randomly OR trial)*(**Supplemental Table 2**). No language restriction was used. Additionally, we used the bibliography of retrieved articles to further identify relevant studies.

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Criteria for considering studies for this review

We included all RCTs comparing DZC with placebo or blank with respect to their effects on OIC. In studies which also included other comparator drugs, only data of DZC and placebo groups were abstracted. Primary outcomes of interest included the occurrence rate and severity of OIC. The severity of OIC was graded as mild (1-2 coughs), moderate (3-5 coughs), or severe (> 5 coughs).[6] Secondary outcomes of interest include possible adverse effects. Exclusion criteria included (1)studies published as review, case report or abstract; (2)animal or cell studies; (3)duplicate publications; (4) studies lacking information about outcomes of interest. The two authors (LXH and KS) independently reviewed the titles and abstracts of all identified studies for eligibility, excluding obviously ineligible ones. The eligibility of those remaining studies for final inclusion was further determined by reading the full text.

Study quality assessment

Two authors(JM and YYZ) independently assessed the risk of bias, using the tool described in the Cochrane Handbook for Systematic Reviews of Interventions[34] and GRADE scoring. Each potential source of bias was graded as low, uncertain or high risk of bias and showed as risk of bias summary and graph. The quality of each outcome was assigned a score of High Quality, Moderate Quality, Low Quality and Very Low Quality.

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Data abstraction

The following data were abstracted from the included studies to a data collection form by two authors(LXH and KS) independently: (1)author, year of publication and journal of included studies; (2)total number of patients, number of patients in the DZC and control groups, gender, age; (3)data regarding outcomes of interest in both groups. Disagreements were resolved by discussion among all authors during the process of data abstraction. The authors of the included RCTs were contacted if necessary.

Statistical analysis

All data were analyzed by utilizing RevMan 5.3(Cochrane Collaboration, Oxford, UK). Pooled odds ratio(OR) and 95% confidence interval(CI) were estimated for dichotomous data, and weighted mean difference(WMD) and 95% CI for continuous data, respectively. Each outcome was tested for heterogeneity, and randomized-effects or fixed-effects model was used in the presence or absence of significant heterogeneity(Q-statistical test $P<0.05$). Sensitivity analyses were done by examining the influence of statistical model on estimated treatment effects, and analyses which adopted the fixed-effects model were repeated again by using randomized-effects model and *vice versa*. In addition to that, sensitivity analysis was also performed to evaluate the influence of individual study on the overall effects. The possible effects of opioid type and doses were evaluated by subgroup analysis. Publication bias was explored through visual inspection of funnel plots of the

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outcomes. All *P*-values were two-sided and statistical significance was defined as $P<0.05$.

RESULTS

Characteristics of the included trials

As shown in **Figure 1**, initial literature search generated 70 results. Finally, thirty-three RCTs[4 6 35-65] involving 4,442 patients were included in the meta-analysis. Of the 33 RCTs, thirty[36-65] were written in Chinese, and the other 3[4 6 35] in English(**Table 1**). The 33 RCTs were performed respectively in 2 provincial hospitals,[36 44] 13 affiliated hospitals,[4 6 35 38 41 46 48 49 52 54-56 63] 16 urban hospitals[37 39 40 42 43 45 47 50 51 53 57 59 61 62 64 65] and 2 county hospitals[58 60] from 15 provinces and municipalities in China. All enrolled patients were of ASA(American society of Anesthesiologists) physical status classification Ⅰ-Ⅱ, whose ages ranged from 18 to 85 yr(**Table 1**). No included RCT reported the OIC induced by remifentanil or alfentanil. As shown in **Table 1**, fentanyl was administrated in 1,880 patients during the induction of general anesthesia with dosages of 2.0 to 5.0 μg/kg, and sufentanil in 2,562 patients with dosages of 0.3 to 5.0 μg/kg. The injection duration of fentanyl and sufentanil varied from 2 to 30 s. Out of the 4,442 patients, 2,521 were allocated into the DZC group and 1,921 into the Control(placebo) group. DZC administration protocols differed among the 33 included trials. DZC were administered intravenously with dosages of 0.025 to 0.3 mg/kg(or 2 to 5 mg), one to ten minutes prior to fentanyl or sufentanil injection(**Table 1**).

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Methodological quality

The risk of bias analysis was shown in **Figure 2 and 3.** There were no patient withdrawal or dropout, neither selectiveness nor bias in all 33 RCTs.

Quality of evidence

For primary outcome, GRADE scoring shows high quality of evidence on DZC preventing OIC(**Table 2**). While for secondary out comes, high quality of evidence appeared in drowsiness, moderate quality of evidence in dizziness and nausea, very low quality of evidence in truncal rigidity, chill and respiratory inhibition(**Table 3**).

Effects of interventions

Occurrence rate of OIC

All the 33 included studies reported the occurrence rate of OIC. As shown in **Figure 4**, meta-analysis demonstrated that the occurrence rate of OIC in the DZC group was statistically lower than that of the Control group(6.7% *vs*. 44.5%, OR=0.07, 95% CI=0.05 to 0.11, P<0.00001, I^2=56%). To analyze the type effects of opioids(fentanyl and sufentanil), subgroup analysis was performed, which indicated that DZC significantly reduced the occurrence rate of FIC(8.8% *vs*. 49.7%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001, I^2=61%) and SIC(5.0% *vs*. 41.5%, OR=0.07, 95% CI=0.04 to 0.12, P<0.00001, I^2=53%). As shown in **Supplemental Figure 1**, subgroup analysis demonstrated that, the FIC occurrence rate increased from 45.0%,

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43.1%, 47.5% to 73.1% in the Control group when fentanyl dosage increased from 2, 3, 4 to 5 μg/kg, respectively. Dose-effect of sufentanil dosage on the occurrence rate of SIC was shown in **Supplemental Figure 2**.

Twenty-two RCTs[6 35-37 39 40 43 45-47 50-52 56-64] reported the occurrence rate of mild and moderate OIC. As shown in **Supplemental Figure 3** and **4**, meta-analysis demonstrated that, DZC group showed significantly lower occurrence rate of OIC than Control group both on mild and moderate grades(mild OIC: 3.6% *vs.*13.6%, OR=0.19, 95% CI=0.14 to 0.25, $P<0.00001$, $I^2=22$; moderate OIC: 2.0% *vs.*13.6%, OR=0.12, 95% CI=0.09 to 0.18, $P<0.00001$, $I^2=0$). Subgroup analysis demonstrated that, DZC significantly reduced the occurrence of either FIC(mild FIC: 5.2% *vs.*15.3%, OR=0.25, 95% CI=0.16 to 0.38, $P<0.00001$, $I^2=28$; moderate FIC: 3.1% *vs.*14.2%, OR=0.17, 95% CI=0.10 to 0.28, $P<0.00001$, $I^2=0$) or SIC(mild SIC: 2.4% *vs.*12.9%, OR=0.14, 95% CI=0.09 to 0.22, $P<0.00001$, $I^2=11$; moderate SIC: 1.1% *vs.*13.4%, OR=0.10, 95% CI=0.06 to 0.17, $P<0.00001$, $I^2=0$) when compared to placebo.

Twenty-five enrolled RCTs[6 35-37 39 40 43-47 50-52 54-64] reported the occurrence rate of severe OIC. As shown in **Supplemental Figure 5**, meta-analysis demonstrated that, the occurrence rate of severe OIC in the DZC group was remarkably lower than that of the Control group(0.9% *vs.*13.7%, OR=0.08, 95% CI=0.05 to 0.12, $P<0.00001$, $I^2=0$). Subgroup analysis demonstrated that, DZC significantly reduced the occurrence of either severe FIC(1.8% *vs.*13.5%, OR=0.12, 95% CI=0.07 to 0.20, $P<0.00001$, $I^2=0$) or severe SIC(0.3% *vs.*13.9%, OR=0.05, 95% CI=0.03 to 0.10, $P<0.00001$, $I^2=0$)

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when compared to placebo.

Subgroup analyses were also performed to investigate the dose effects of DZC on FIC and SIC occurrence rates. As shown in **Supplemental Figure 6** and **7**, DZC could effectively suppress OIC by fentanyl or sufentanil when administered at dosages ranging from less than 0.1 to 0.3 mg/kg (or 5 mg). The dose of 0.1 mg/kg is mostly investigated and suggested as the optimal dose. Whether the prophylactic effect of DZC on OIC is dose-dependent remains further verification.

Adverse effects

Six RCTs[48 50 53 58 64 65] reported possible side effects of DZC administration. As shown in **Figure 5**, meta-analysis suggested that, the occurrence rates of drowsiness, truncal rigidity, chill, respiratory inhibition, nausea and emesis of the DZC group were all comparable to those of the Control group, with exception that the DZC-treated patients had higher occurrence rate of dizziness as compared to placebo(11.8% *vs.* 0%, OR=8.06, 95% CI=1.40 to 46.35, P=0.02, I^2=0%).

Sensitivity analyses and publication bias

Sensitivity analysis showed that treatment effects on all the outcomes were not affected by the choice of statistical model (**Table 4**). Sensitivity tests were also performed by exclusion of some studies to analyze the influence of the overall treatment effect on high heterogeneity outcomes(**Table 4**), and no contradictory results were found in pooled OR and 95%CI. For occurrence rate of OIC,

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heterogeneity changed from 61% to 35% for FIC by exclusion of three studies conducted from Gao et al(female patients only)[39], Wang et al[49] and Yuan et al[58](preoperative medication with phenobarbital), and 53% to 36% for SIC by exclusion of four studies conducted from Shen et al(female patients only),[47] Xu et al(duration of sufentanil injection more than 10s),[57] Wang et al[53] and Zheng et al[63](preoperative medication with phenobarbital). For occurrence rate of adverse effects, heterogeneity changed from 73% to 0% by exclusion of one study from Sheng et al(preoperative medication with phenobarbital).[48] No significant publication bias was detected by funnels plot examination for the occurrence rate of OIC(**Supplemental Figure 8A**) and the occurrence rate of mild， moderate and severe OIC(**Supplemental Figure 8B, Supplemental Figure 8C and Supplemental Figure 8D**).

DISCUSSION

Cough suppression is one useful side-effect of opioids, which is the basis of their use in cough suppressants. Opioids depress the cough reflex by directly acting on the medullary cough center.[16] Fentanyl and its derivatives sufentanil are commonly used opioid analgesics in the induction and maintenance of general anesthesia. Intravenous bolus injection of fentanyl or sufentanil often cause cough. The present meta-analysis demonstrated that, the occurrence rates of FIC and SIC were 49.7% and 41.5% respectively, the occurrence rates of severe FIC and severe SIC were 13.5% and 13.9% respectively, which is consistent with previous reports.[2 4 6 7 15] However,

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significant heterogeneity was found in the results, which may have affected the rigor of those findings. The heterogeneity may be explained by study design. For example, sex of the patients in excluded study in sensitivity analysis were obviously different from others. It was reported by Solanki et al[66] that occurrence rate of FIC was low when studied in female cancer patients(12.7%). However, contradictory results of 57.5% and 28.3% were observed in the two excluded study enrolling female only.[39 47] This may suggest that sex to some extent contributes to heterogeneity. In addition to that, study from Xu et al[57] with significant low SIC occurrence rate(3% in DZC group and 8% in Control group) was excluded owing to prolonged injection time(>30s) in sensitivity analysis, which though made no influence on pooled effect, may improve the credibility of current meta-analysis.

Till now, the mechanism of OIC remains poorly understood. Various hypotheses have been proposed, which may involve opioid receptors, C-fiber receptors, rapid adapting pulmonary stretch receptors, histamine release and citrate in fentanyl and sufentanil injection.[1-3 15-17] Additionally, many factors can contribute to the occurrence of OIC, which can be divided into two categories. One is patients' individual physical conditions(age, sex, smoking status, disease history, *etc*). Another is usage of opioids(drug category, dosage, concentration, injection site, injection concentration, injection rate, *etc*).[15] And subgroup analysis suggested possible dose-effects of fentanyl and sufentanil on the occurrence rates of OIC.

OIC is associated with adverse effects and should be avoided. The anti-tussive efficacy of numerous pharmaceutical and non-pharmaceutical interventions have been

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tested, some proved to be effective, some ineffective, and some have side effects.[15] DZC, a mixed κ and μ opioid receptor agonist-antagonist, is not a well-known drug in Western countries.[24-27] However, DZC is widely applied as peri-operative pain analgesic agent in China for decades.[24-26 28-32] The present meta-analysis demonstrated that DZC could significantly suppress both FIC and SIC, with several trials[4 6 35 41 45 48 50 56 64] reporting that DZC could completely prevent OIC. Furthermore, the subgroup analysis of the present meta-analysis suggested that, the anti-tussive effect of DZC on FIC and SIC may be dose-dependent. The mechanism responsible for the anti-tussive effect of DZC remains unknown. Possible explanation for this phenomenon is that DZC suppresses OIC by μ-receptor antagonism or norepinephrine/serotonine reuptake inhibition and reduce cough.[15] Whether a central gating mechanisms via C-fiber receptors or inhibition of histamine release play a role in the cough suppression elicited by DZC needs to be investigated.[4]

Because of its partial μ agonism, DZC exhibits a ceiling effect for common opioids-related adverse effects such as respiratory depression.[24-26] The meta-analysis suggested that, DZC did not increase the occurrence rates of drowsiness, truncal rigidity, chill, respiratory inhibition, nausea and emesis, but was associated with higher occurrence rate of dizziness. Whether DZC pretreatment interferes with opioid analgesia remains to be verified. Initial evidence indicated that, DZC can enhance the analgesic effect of opioids and reduced OIC and opioid-related side effects.[67 68]

This study has some limitations. Firstly, meta-analysis can increase the power of analysis by pooling many small low-quality studies, but different clinical practices,

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varied quality and heterogeneity of included studies may limit the certainty of the findings of meta-analysis. For example, there were no differences in DZC and Control group on OIC occurrence rate when using preoperative medication of phenobarbital 30 minutes before anesthesia induction.[53 58] One possible explanation is that, sedatives exhibit similar effect on suppressing OIC as well according to previous study.[2] Secondly, all the 33 included RCTs were performed in China. The anti-tussive effectiveness of DZC may not be generalised to the whole world and remains to be investigated in other ethnicities. Thirdly, the doses, injection rates or injection order of fentanyl or sufentanil varied among these included trials. For example, Sun and colleagues[4] reported DZC administered 10 min before anesthesia induction could prevent FIC, which may be not a convenient practice in clinical settings. To determine the proper administration protocol of DZC for OIC prevention, a prospective randomized, placebo-controlled, triple-blinded trial is ongoing in our center.

CONCLUSIONS

This meta-analysis has demonstrated that, DZC significantly inhibited OIC and may be used to manage OIC induced by fentanyl or sufentanil. More high-quality RCTs are needed to complement the safety of DZC.

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Data sharing All data relevant to the study are included in the article or uploaded as supplementary information.

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Table 1. Characteristics of the included RCTs and administration protocols of dezocine

Study	Language	Patient characteristics			Opioids			Group Dezocine		Group Control		Outcomes reported				
		Age(yr)	Sex(M/F)	Type	Dose(μg/kg)	Duration(s)	Timing(min)	n	Dose(mg/kg)	n	Dose	CID	CIC	SCID	SCIC	Adverse effect
Bian 2012[36]	Chinese	23-64	48/53	S	3.0	≤5s	5	50	5mg	50	Equal volume NS	2.00%	30.00%	0	6.00%	NR
Chen 2013[37]	Chinese	20-60	39/41	S	2.0	≤2s	NR	40	5mg	40	2ml NS	5.00%	45.00%	0	25.00%	NR
Fang 2016[38]	Chinese	20-60	66/58	F	5.0	≤3s	10	62	0.1	62	Equal volume NS	9.68%	62.90%	NR	NR	NR
Gao 2013[39]	Chinese	20-50	0/120	F	3.0	NR	10	40	0.05	14	5ml NS	57.50%	64.29%	17.5%	20.00%	NR
					3.0			40	0.1	13		17.50%	53.85%	2.5%		
					3.0			40	0.15	13		15.00%	53.85%	2.5%		
Li 2017[40]	Chinese	23-72	134/106	S	0.3	≤10s	10	80	0.05	40	Equal volume NS	2.50%	32.50%	0	7.50%	NR
					0.3			80	0.1	40		1.25%	35.00%			
Liu 2015[6]	English	18-70	189/181	S	0.5	>3s	2	185	0.1	185	NS	0.00%	31.89%	0	13.51%	NR
Meng 2013[41]	Chinese	28-55	39/41	S	0.4	≤2s	2	40	0.1	40	Equal volume NS	0.00%	72.50%	NR	NR	NR
Pi 2015[42]	Chinese	22-65	51/49	F	4.0	≤3s	10	50	0.1	50	Equal volume NS	12.00%	68.00%	NR	NR	NR
Qiu 2016[43]	Chinese	20-65	119/81	S	0.5	NR	NR	100	5mg	100	2ml NS	5.00%	45.00%	0	39.00%	NR
Qiu L 2016[44]	Chinese	18-45	0/120	F	3.0	≤5s	2	60	0.1	60	5ml NS	1.67%	25.00%	0	3.33%	NR
Qu 2012[45]	Chinese	18-65	40/80	F	4.0	≤5s	2	60	0.05	60	Equal volume NS	0.00%	26.67%	0	11.67%	NR
Ruan 2017[46]	Chinese	24-55	NR	S	0.4	NR	10	35	0.1	35	Equal volume NS	5.71%	57.14%	0	28.57%	NR
Shen 2014[47]	Chinese	20-65	0/120	S	0.3	<5s	5	60	0.05	60	5ml NS	8.33%	28.33%	3.33%	10.00%	NR
Sheng 2017[48]	Chinese	19-70	44/52	S	0.3	≤10s	8	48	0.1	48	5ml NS	0.00%	64.58%	NR	NR	CH, RI, NE
Sun 2011[4]	Chinese	20-60	68/52	F	5.0	≤2s	10	60	0.1	60	NS	0.00%	70.00%	NR	NR	NR
Wang 2015[49]	Chinese	15-60	78/62	F	5.0	≤5s	10	70	0.1	70	10ml NS	1.43%	75.71%	NR	NR	NR
Wang 2016[50]	Chinese	18-70	190/180	S	0.5	NR	Immediately	185	0.1	185	Equal volume NS	0.00%	31.89%	0	29.41%	TR, RI, NE
Wang 2017[51]	Chinese	22-61	67/53	S	NR	NR	NR	60	0.05	60	NS	16.67%	55.00%	0	6.67%	NR
Wang H 2015[52]	Chinese	25-65	42/58	S	0.3	<5s	10	50	0.1	50	2ml NS	2.00%	32.00%	0	8.00%	NR
Wang L 2015[53]	Chinese	60-85	59/41	S	0.3	≤5s	5	25	0.04	9	5ml NS	28.00%	44.44%	NR	NR	DI, DR
					0.3			25	0.08	8		12.00%	37.50%			

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					0.3			25	0.12	8		8.00%	37.50%			
Wang Z 2015[54]	Chinese	18-55	31-29	F	4	≤3s	1	30	0.1	30	2ml NS	13.33%	53.33%	0	23.33%	NR
Wang Z 2016[55]	Chinese	18-55	33-27	F	4	≤3s	1	30	0.1	30	2ml NS	16.67%	50.00%	0	16.67%	NR
					3			40	0.1	14		7.50%	71.43%			
Wu 2014[56]	Chinese	18-60	105/55	F	3	<3s	10	40	0.2	13	Equal volume NS	2.50%	61.54%	0	15.00%	NR
					3			40	0.3	13		0.00%	61.54%			
					0.5	<3s	5	50	0.1	50		6.00%	80.00%	0	7.50%	
Xu 2014[57]	Chinese	20-65	102/98	S	0.5	<30s	5	50	0.1	50	5ml NS	2.00%	8.00%	0	0	NR
					3			100	0.025	34		12.00%	41.18%			
Xu 2015[35]	English	20-70	243/157	F	3	<5s	Immediately	100	0.05	33	NS	4.00%	39.39%	0	5.00%	NR
					3			100	0.1	33		0.00%	39.39%			
					3			30	0.05	10		56.67%	60.00%	26.67%		
Yuan 2015[58]	Chinese	25-55	NR	F	3	NR	10	30	0.1	10	NS	13.33%	60.00%	0	33.33%	DI, DR
					3			30	0.2	10		6.67%	60.00%	0		
Zhang 2013[59]	Chinese	25-56	41/39	F	3	<3s	12	40	0.1	40	2ml NS	2.50%	45.00%	0	6.67%	NR
Zhang 2016[60]	Chinese	23-56	72/48	F	4	≤3s	5	30	5mg	30	10ml NS	3.33%	46.67%	0	22.50%	NR
Zhao 2017[61]	Chinese	20-61	19/33	S	5	NR	5-8	26	0.1	26	NS	7.69%	65.38%	0	19.23%	NR
					0.4		1	25	2mg	9		28.00%	66.67%			
Zheng 2018[62]	Chinese	39±5	61/39	S	0.4	≤3s	3	25	2mg	8	Equal volume NS	4.00%	50.00%	0	8.00%	NR
					0.4		8	25	2mg	8		4.00%	50.00%			
Zheng 2019[63]	Chinese	18-56	92/108	S	0.4	≤6s	3	50	5mg	50	Equal volume NS	16.00%	44.00%	0	8.00%	NR
Zhou 2014[64]	Chinese	18-65	23/37	S	0.3	<10s	10	30	0.1	30	5ml NS	0.00%	26.67%	0	3.33%	DI, DR
Zhu 2018[65]	Chinese	22-75	31/29	S	0.4	NR	2	30	0.1	30	Equal volume NS	3.33%	13.33%	NR	NR	NE

CID=Cough occurrence rate of dezocine; CIC=Cough occurrence rate of control; SCID=Severe cough occurrence rate of dezocine; SCIC=Severe cough occurrence rate of control;

RI=Respiratory inhibition; NE=Nausea and emesis; TR=Truncal rigidity; NR=Not reported; DI=Dizziness; DR=Drowsiness; CH=Chill.

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Table 2. Quality assessment for primary outcomes

Quality assessment							No of patients		Effect		Quality	Importance
No of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	DZC	Control	Relative (95% CI)	Absolute		
Effect of DZC on OIC occurrence rate												
47	randomised trials	serious	serious[1]	no serious indirectness	serious	strong association reduced effect for RR >> 1 or RR << 1 dose response gradient	169/2521 (6.7%)	857/1921 (44.6%)	OR 0.07 (0.05 to 0.1)	393 fewer per 1000 (from 372 fewer to 407 fewer)	⊕⊕⊕⊕ HIGH	CRITICAL
								50%		435 fewer per 1000 (from 409 fewer to 452 fewer)		
Effect of DZC on FIC occurrence rate												
23	randomised trials	serious	serious	no serious indirectness	serious	strong association[1] reduced effect for RR >> 1 or RR << 1 dose response gradient	101/1150 (8.8%)	363/730 (49.7%)	OR 0.07 (0.04 to 0.12)	433 fewer per 1000 (from 391 fewer to 459 fewer)	⊕⊕⊕⊕ HIGH	CRITICAL
								53.9%		463 fewer per 1000 (from 416 fewer to 494 fewer)		
Effect of DZC on SIC occurrence rate												
24	randomised trials	serious	serious	no serious indirectness	serious	strong association reduced effect for RR >> 1 or RR << 1 dose response gradient	68/1371 (5%)	494/1191 (41.5%)	OR 0.07 (0.04 to 0.12)	368 fewer per 1000 (from 336 fewer to 387 fewer)	⊕⊕⊕⊕ HIGH	CRITICAL
								44.2%		389 fewer per 1000 (from 355 fewer to 411 fewer)		

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Table 3. Quality assessment for secondary outcomes

Quality assessment							No of patients		Effect		Quality	Importance
No of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	DZC	Control	Relative (95% CI)	Absolute		
Dizziness												
3	randomised trials	serious[1]	serious	no serious indirectness	serious	strong association reduced effect for RR >> 1 or RR << 1	10/85 (11.8%)	0/85 (0%)	OR 8.06 (1.40 to 46.35)	-	⊕⊕⊕O MODERATE	IMPORTANT
								0%		-		
Drowsiness												
3	randomised trials	serious[1]	no serious inconsistency	no serious indirectness	serious	strong association reduced effect for RR >> 1 or RR << 1	6/85 (7.1%)	0/85 (0%)	OR 4.91 (0.80 to 30.19)	-	⊕⊕⊕⊕ HIGH	IMPORTANT
								0%		-		
Truncal rigidity												
1	randomised trials	very serious	no serious inconsistency	no serious indirectness	serious	none	4/185 (2.2%)	0/185 (0%)	OR 9.2 (0.49 to 172.07)	-	⊕OOO VERY LOW	CRITICAL
								0%		-		
Chill												
1	randomised trials	serious	serious	no serious indirectness	very serious	none	2/48 (4.2%)	11/48 (22.9%)	not pooled	not pooled	⊕OOO VERY LOW	IMPORTANT
								22.9%		not pooled		
Respiratory inhibition												
2	randomised trials	serious	very serious	no serious indirectness	serious	none	17/233 (7.3%)	9/233 (3.9%)	OR 1.7 (0.00 to 766.69)	25 more per 1000 (from 39 fewer to 930 more)	⊕OOO VERY LOW	CRITICAL
								9.4%		56 more per 1000 (from 94 fewer to 894 more)		

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Nausea and emesis												
3	randomised trials	serious	serious	no serious indirectness	no serious imprecision	reduced effect for RR >> 1 or RR << 1	24/263 (9.1%)	18/263 (6.8%)	OR 1.32 (0.03 to 53.18)	20 more per 1000 (from 66 fewer to 728 more)	⊕⊕⊕O MODERATE	IMPORTANT
								6.7%		20 more per 1000 (from 65 fewer to 725 more)		

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Table 4. Reliability of results

Influence of statistical model on estimated treatment effects of primary outcomes			
Statistical model	Cough occurrence rate OR (95% CI)	Severe cough occurrence rate OR (95% CI)	Adverse effects occurrence rate OR (95% CI)
Fixed Effects	0.07(0.05, 0.08)	0.08(0.05,0.12)	1.61(1.09,2.39)
Random Effects	0.07(0.05, 0.10)	0.11(0.07, 0.18)	2.34(0.60,9.14)

Sensitivity analyses of high heterogeneity outcome									
Heterogeneity outcome	Excluded trials	Group DZC (n)	Group C (n)	Heterogeneity		Analysis model	OR	95%CI	Overall effect *P*
				I^2 (%)	*P*				
FIC (%)	39,49,58	280	140	35	0.08	M-H, Fixed	0.06	(0.04,0.08)	<0.00001
SIC（%）	47,53,57,63	285	235	36	0.07	M-H, Fixed	0.04	(0.03,0.06)	<0.00001
Adverse effects (%)	48	48	48	0	0.59	M-H, Fixed	10.75	(4.75,24.33)	<0.00001

FIC=Fentanyl-induced cough; SIC=Sufentanil-induced cough.

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Figure legends

Figure 1. Flow chart

Figure 2. Quality assessment for primary outcomes

Figure 3. Quality assessment for secondary outcomes

Figure 4. Forest plot of OIC occurrence rate

Figure 5. Possible adverse effects

Supplemental Figure 1. Dose-effect of fentanyl dosage on FIC incidence

Supplemental Figure 2. Dose-effect of sufentanil dosage on SIC incidence

Supplemental Figure 3. Forest plot of mild OIC incidence

Supplemental Figure 4. Forest plot of moderate OIC incidence

Supplemental Figure 5. Forest plot of severe OIC incidence

Supplemental Figure 6. Dose-effects of DZC on FIC

Supplemental Figure 7. Dose-effects of DZC on SIC

Supplemental Figure 8. Funnel plot of OIC incidence(A), mild OIC incidence(B), moderate OIC incidence(C), severe OIC incidence(D)

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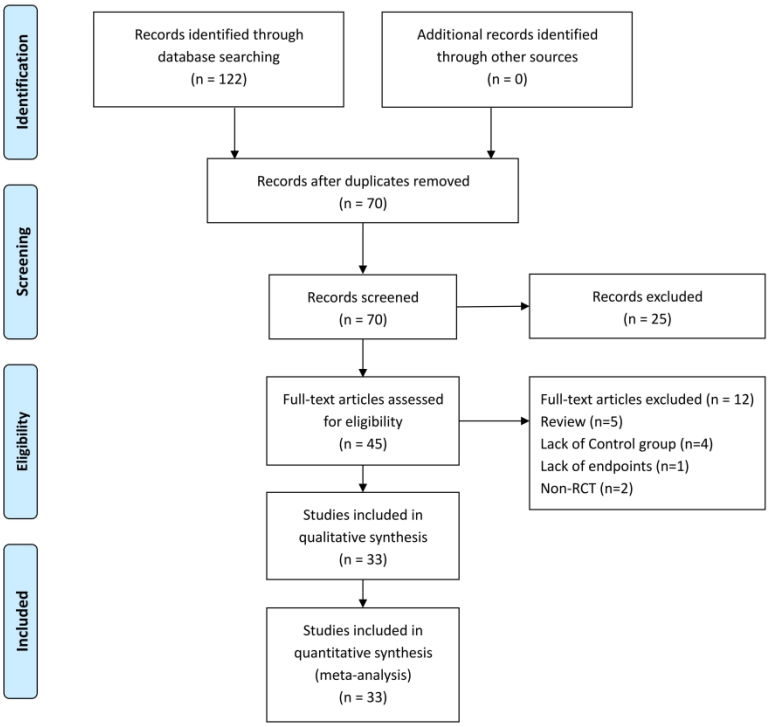


Figure 1. Flow chart

403x570mm (300 x 300 DPI)

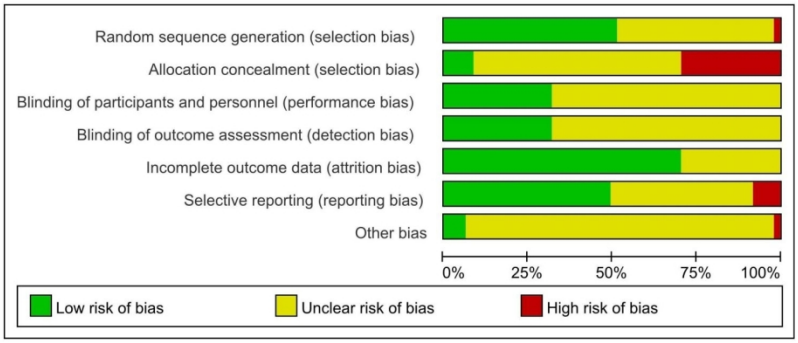


Figure 2. Quality assessment for primary outcomes

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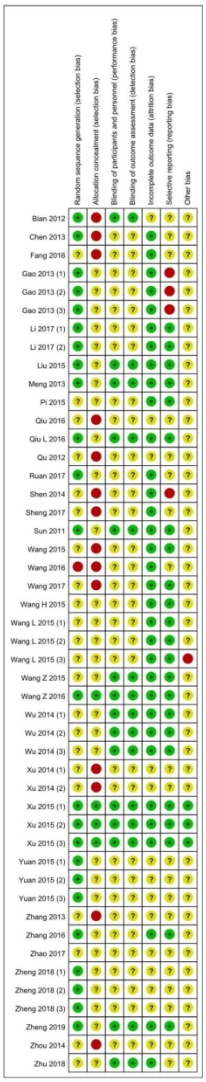


Figure 3. Quality assessment for secondary outcomes

210x296mm (192 x 192 DPI)

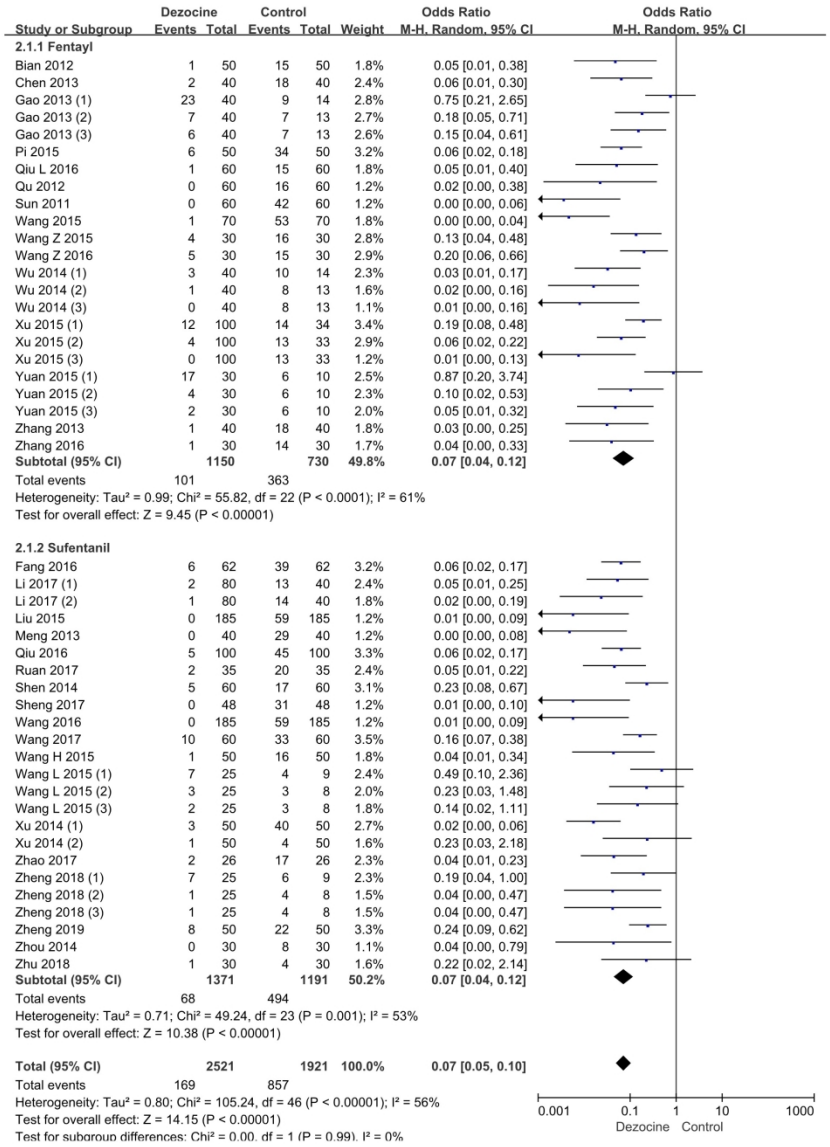


Figure 4. Forest plot of OIC occurrence rate

209x278mm (300 x 300 DPI)

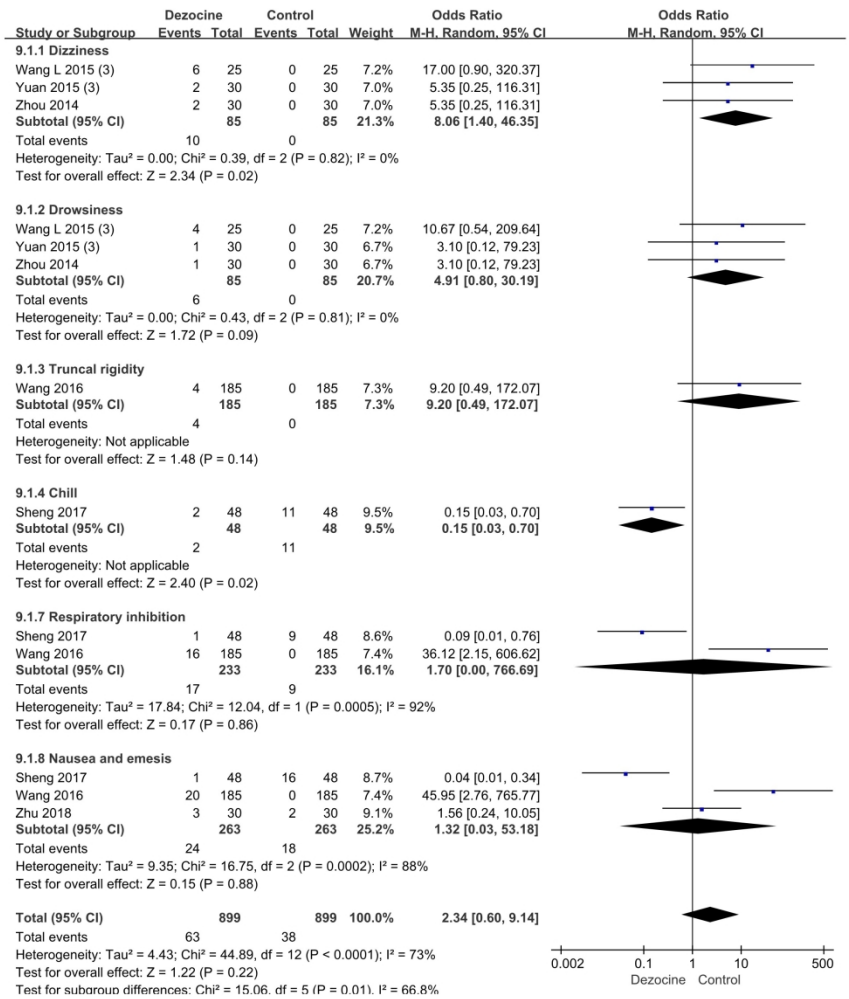


Figure 5. Possible adverse effects

209x278mm (300 x 300 DPI)

Supplemental Table 1. PRISMA checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract, page 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Introduction, page 4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Introduction, page 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Methods, page 5-6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Methods—Study inclusion criteria, page 6-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Methods—Search strategy, page 6

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Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Methods—Search strategy, page 6, and Supplemental Table 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Methods—Study inclusion criteria, page 6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Methods—Data abstraction, page 7-8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Methods—Data abstraction, page 7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Methods—Study quality assessmen, page 7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Methods—Statistical analysis, page 8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	Methods—Statistical analysis, page 8

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Methods—Statistical analysis, page 7-8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Methods—Statistical analysis, page 7-8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Results-Search results page 9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Results-Meta-analysis, page 9-
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Results-Study quality and risk bias, page 9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Results-Meta-analysis, page 10-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Results-Meta-analysis, page 10-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Results-Meta-analysis, page 12
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Results- page 12
DISCUSSION			

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Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Discussion- Page 13-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Discussion- Page 15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Discussion- Page 16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Funding- Page 16

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Supplemental Table 2. Search strategy

PubMed

No.	Search items
#1	"dezocine"[Supplementary Concept] OR dezocine[Title/Abstract]
#2	((((((((((((((((Analgesics, Opioid[MeSH Terms]) OR Opioid[Title/Abstract]) OR Fentanyl[MeSH Terms]) OR Fentanyl[Title/Abstract]) OR Phentanyl[Title/Abstract]) OR Fentanyl Citrate[Title/Abstract]) OR Sufentanil[MeSH Terms]) OR Sufentanil[Title/Abstract]) OR Sulfentanyl[Title/Abstract]) OR Sulfentanil[Title/Abstract]) OR Sufentanil Citrate[Title/Abstract]) OR Remifentanil[MeSH Terms]) OR Remifentanil[Title/Abstract]) OR Remifentanil Hydrochloride[Title/Abstract]) OR Alfentanil[MeSH Terms]) OR Alfentanil[Title/Abstract]) OR Alfentanyl[Title/Abstract]) OR Alfentanil Hydrochloride[Title/Abstract]
#3	((((Cough[MeSH Terms]) OR Cough[Title/Abstract]) OR Coughs[Title/Abstract]) OR Antitussive[Title/Abstract]) OR Anti-tussive[Title/Abstract]
#4	((((((Randomized Controlled Trial[Publication Type]) OR Randomized Controlled Trial) OR Controlled Clinical Trial[Publication Type]) OR Controlled Clinical Trial) OR Randomized) OR Placebo) OR randomly
#5	#1 AND #2 AND #3 AND #4

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Embase

('dezocine'/exp OR dezocine:ab,ti) AND ('opiate agonist'/exp OR opioid:ab,ti OR 'fentanyl derivative'/exp OR fentanyl:ab,ti OR 'fentanyl citrate':ab,ti OR sufentanil:ab,ti OR 'sufentanil citrate':ab,ti OR remifentanil:ab,ti OR alfentanil:ab,ti) AND ('coughing'/exp OR coughing:ab,ti OR cough:ab,ti OR antitussive:ab,ti OR anti-tussive:ab,ti) AND ('randomized controlled trial'/exp OR 'randomized controlled trial':it OR 'randomized controlled trial':ab,ti OR randomized OR placebo OR randomly)

Cochrane Library

No.	Search items
#1	(dezocine): ti, ab, kw
#2	[Analgesics, Opioid] explode all trees OR (opioid): ti, ab, kw OR [Fentanyl] explode all trees OR (fentanyl): ti, ab, kw OR (fentanyl citrate): ti, ab, kw OR (phentanyl): ti, ab, kw OR [Sufentanil] explode all trees OR (sufentanil): ti, ab, kw OR (sufentanil citrate): ti, ab, kw OR (sulfentanyl): ti, ab, kw OR [Remifentanil] explode all trees OR (remifentanil): ti, ab, kw OR (remifentanil monohydrochloride): ti, ab, kw OR (remifentanil hydrochloride): ti, ab, kw OR [Alfentanil] explode all trees OR (alfentanil): ti, ab, kw OR (alfentanil hydrochloride): ti, ab, kw OR (alfentanyl): ti, ab, kw
#3	[Cough] explode all trees OR (cough): ti, ab, kw OR (coughs): ti, ab, kw OR (antitussive):ti, ab, kw OR (anti-tussive):ti, ab, kw
#4	[Randomized Controlled Trial] explode all trees OR (Randomized Controlled Trial): ti, ab, kw OR [Randomized Controlled Trials as Topic] explode all trees OR [Controlled Clinical Trial] explode all trees OR (Controlled Clinical Trial): ti, ab, kw OR [Controlled Clinical Trial as Topic] explode all trees

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#5	#1 AND #2 AND #3 AND #4

Ovid

No.	Search items
#1	dezocine.mp. [mp=title, abstract, full text, caption text]
#2	(opioid or opioid or "Analgesics, Opioid" or fentanyl or phentanyl or "fentanyl citrate" or sufentanil or sulfentanyl or "sufentanil citrate" or remifentanil or "remifentanil hydrochloride" or alfentanil or alfentanyl or "alfentanil hydrochloride").mp. [mp=title, abstract, full text, caption text]
#3	(cough or coughs or coughing or antitussive or anti-tussive).mp. [mp=title, abstract, full text, caption text]
#4	("randomized controlled trial" or "controlled clinical trial" or randomized or placebo or randomly).mp. [mp=title, abstract, full text, caption text]
#5	#1 AND #2 AND #3 AND #4

Web of Science

TS=dezocine AND TS=(opioid OR opioid OR "Analgesics, Opioid" OR fentanyl OR phentanyl OR "fentanyl citrate" OR sufentanil OR sulfentanyl OR "sufentanil citrate" OR remifentanil OR "remifentanil hydrochloride" OR alfentanil OR alfentanyl OR "alfentanil hydrochloride") AND TS=(cough OR coughs OR coughing OR antitussive OR anti-tussive) AND TS=("randomized controlled trial" OR "controlled clinical trial" OR randomized OR placebo OR randomly)

SinoMed

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No.	Search items
#1	"地佐辛"[不加权:扩展] OR "地佐辛"[摘要:智能]
#2	"阿片"[不加权:扩展] OR "阿片"[中文标题:智能] OR "镇痛药,"[不加权:扩展] AND "阿片类"[不加权:扩展] OR "芬太尼"[不加权:扩展] OR "芬太尼"[中文标题:智能] OR "舒芬太尼"[不加权:扩展] OR "舒芬太尼"[中文标题:智能] OR "瑞芬太尼"[中文标题:智能] OR "阿芬太尼"[不加权:扩展] OR "阿芬太尼"[中文标题:智能]
#3	"咳嗽"[不加权:扩展] OR "咳嗽"[中文标题:智能] OR "呛咳"[中文标题:智能] OR "止咳"[不加权:扩展] OR "止咳"[中文标题:智能] OR "镇咳"[不加权:扩展] OR "镇咳"[中文标题:智能]
#4	"随机对照试验"[不加权:扩展] OR "临床对照试验"[不加权:扩展] OR "随机地"[摘要:智能] OR "随机的"[摘要:智能] OR "对照"[摘要:智能] OR "安慰剂"[摘要:智能]
#5	#1 AND #2 AND #3 AND #4

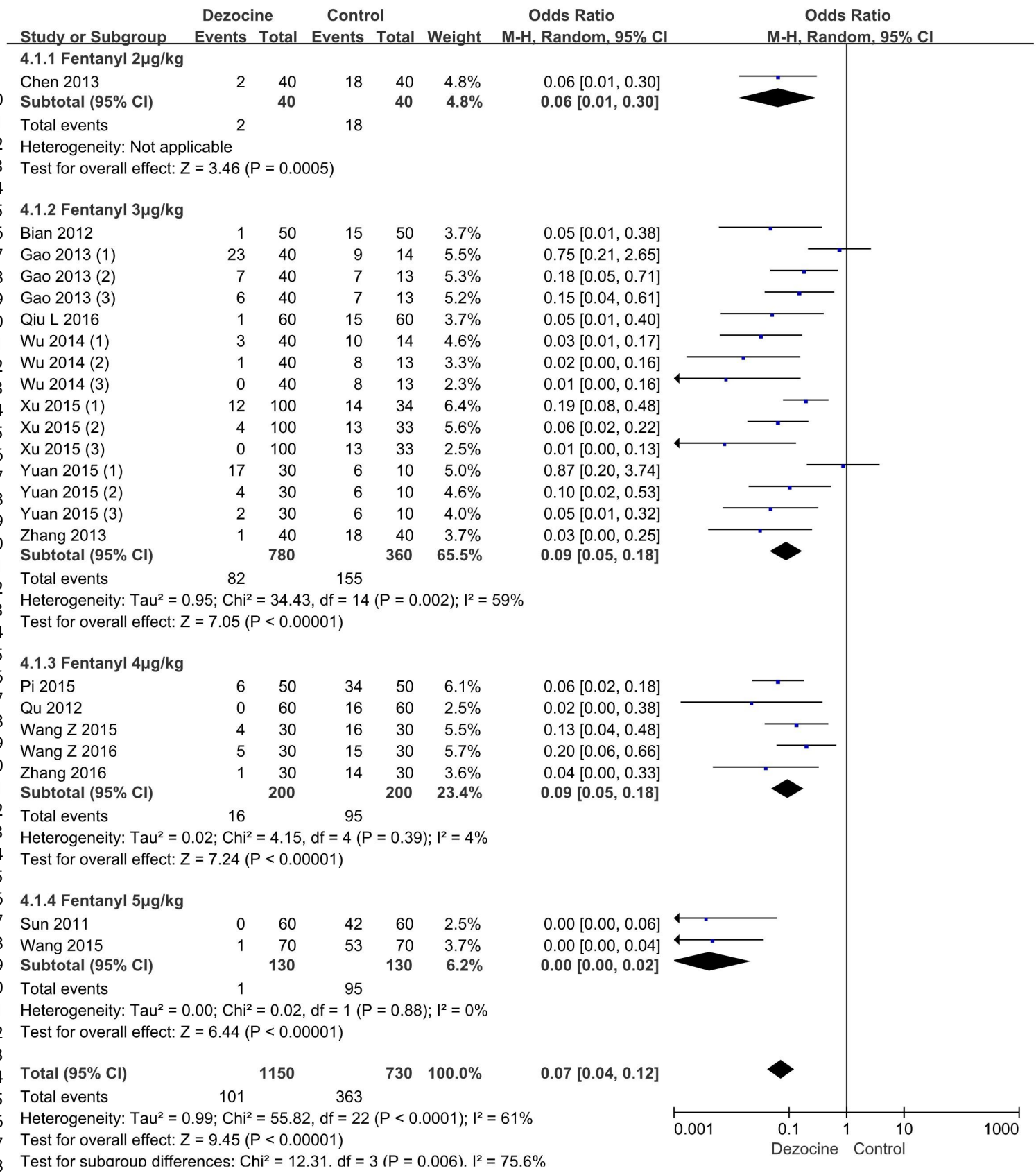
CNKI

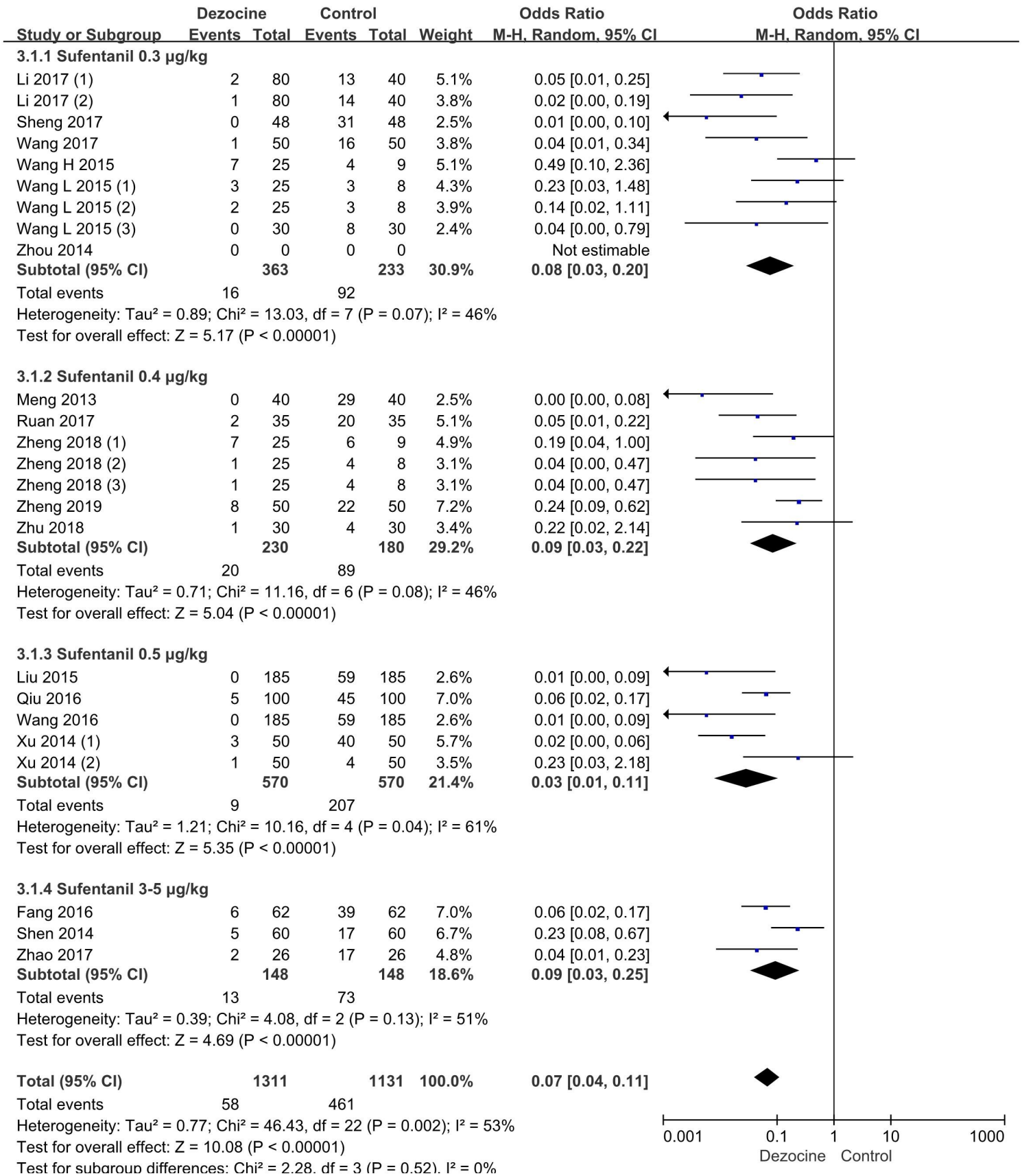
(SU='地佐辛' OR AB='地佐辛') AND (SU=('阿片'+'阿片类镇痛药'+'芬太尼'+'舒芬太尼'+'瑞芬太尼'+'阿芬太尼') OR TI=('阿片'+'阿片类镇痛药'+'芬太尼'+'舒芬太尼'+'瑞芬太尼'+'阿芬太尼')) AND (SU=('咳嗽'+'呛咳'+'止咳'+'镇咳') OR TI=('咳嗽'+'呛咳'+'止咳'+'镇咳')) AND (SU=('随机对照试验'+'临床对照试验'+'随机的'+'随机地'+'安慰剂'+'对照') OR AB=('随机对照试验'+'临床对照试验'+'随机的'+'随机地'+'安慰剂'+'对照'))

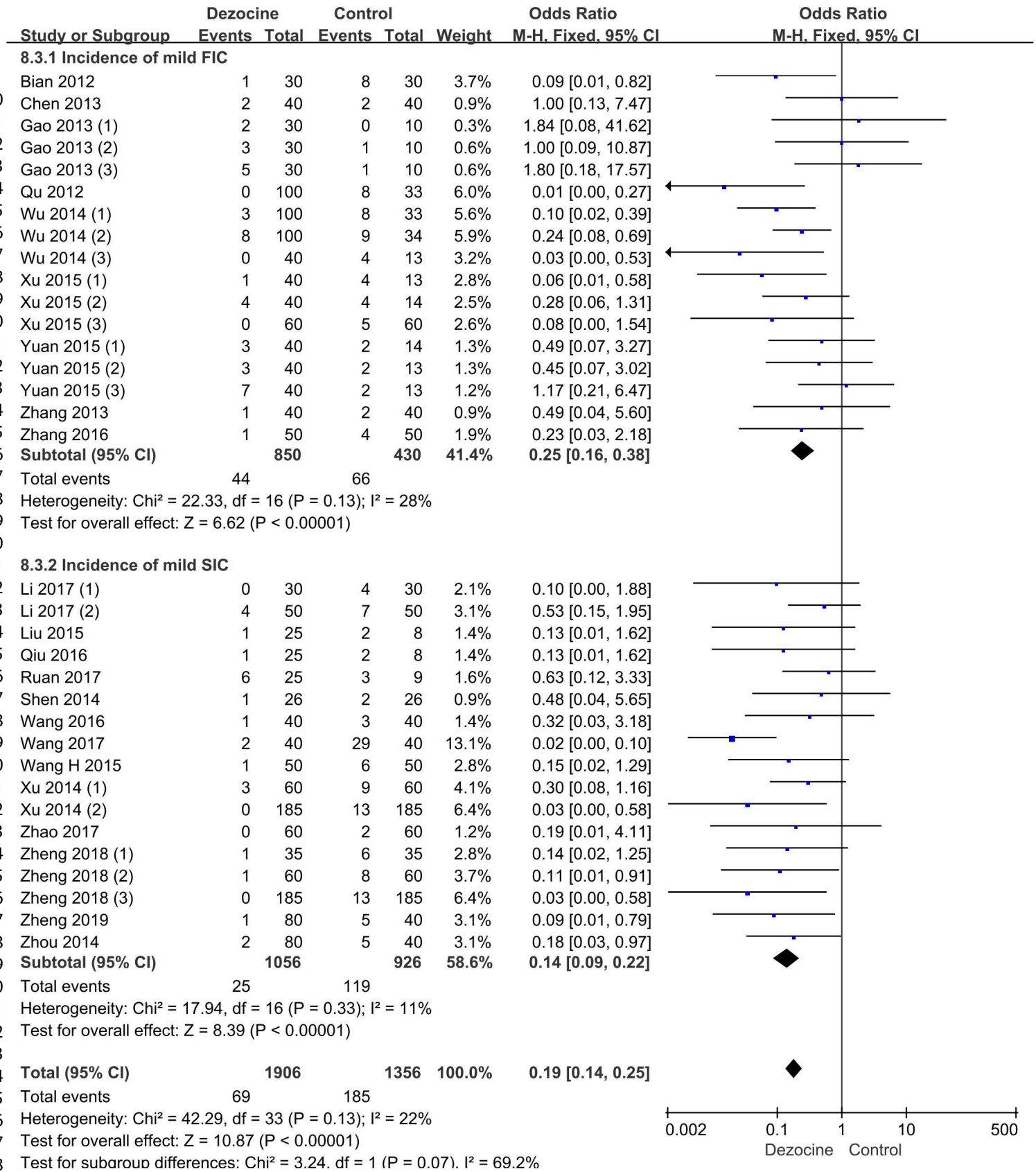
For peer review only

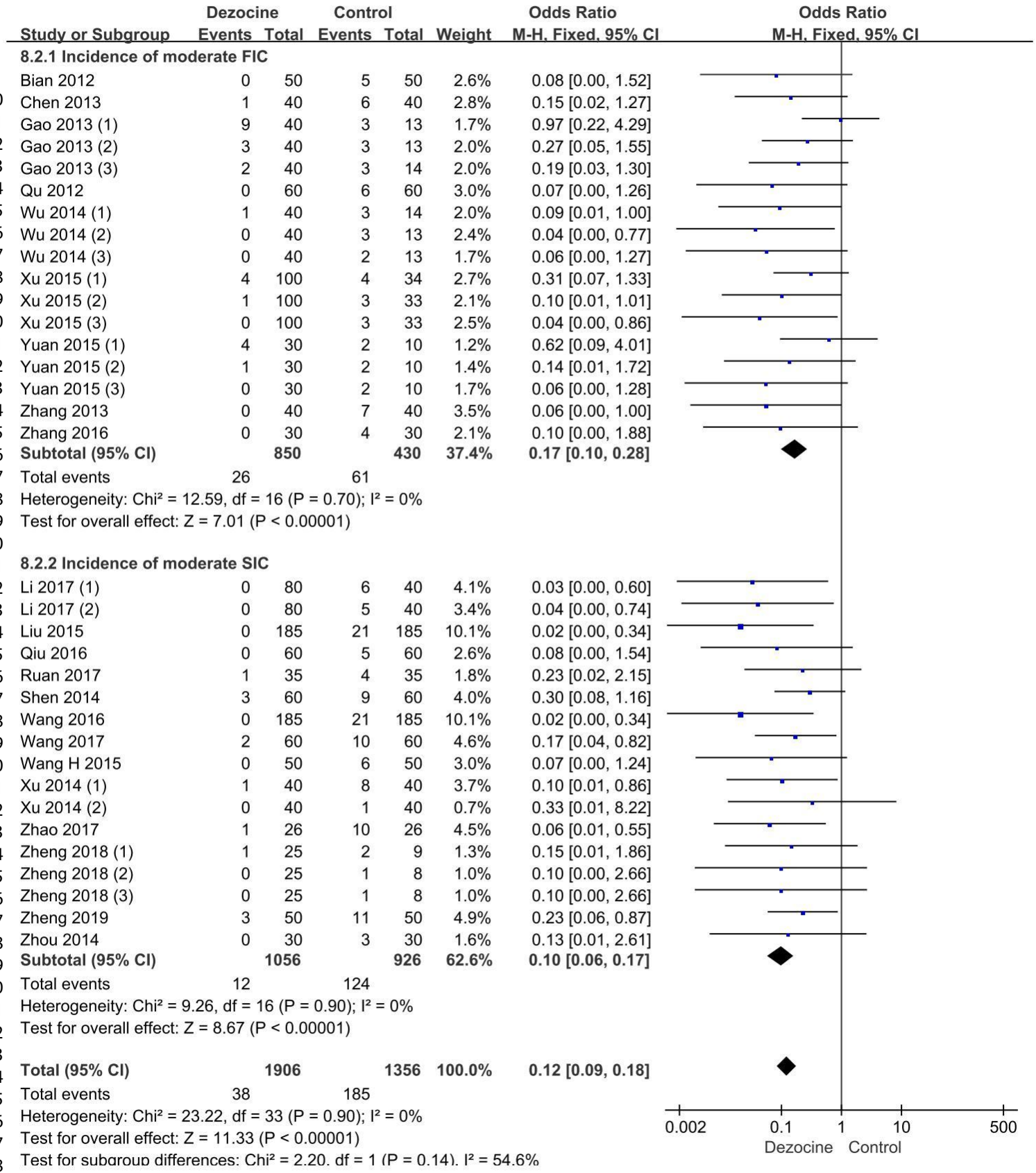
Wanfang Data
(主题:地佐辛+摘要:地佐辛)*(主题:(阿片+阿片类镇痛药+芬太尼+舒芬太尼+瑞芬太尼+阿芬太尼)+题名:(阿片+阿片类镇痛药+芬太尼+舒芬太尼+瑞芬太尼+阿芬太尼))*(主题:(咳嗽+呛咳+止咳+镇咳)+题名:(咳嗽+呛咳+止咳+镇咳))*(主题:(随机对照试验+临床对照试验+随机的+随机地+安慰剂+对照)+ 摘要:(随机对照试验+临床对照试验+随机的+随机地+安慰剂+对照))
VIP Data
(M=地佐辛 OR R=地佐辛) AND (M=阿片 OR 阿片类镇痛药 OR 芬太尼 OR 舒芬太尼 OR 瑞芬太尼 OR 阿芬太尼 OR R=阿片 OR 阿片类镇痛药 OR 芬太尼 OR 舒芬太尼 OR 瑞芬太尼 OR 阿芬太尼) AND (M=咳嗽 OR 呛咳 OR 止咳 OR 镇咳 OR R=咳嗽 OR 呛咳 OR 止咳 OR 镇咳) AND (M=随机对照试验 OR 临床对照试验 OR 随机的 OR 随机地 OR 安慰剂 OR 对照 OR R=随机对照试验 OR 临床对照试验 OR 随机的 OR 随机地 OR 安慰剂 OR 对照)

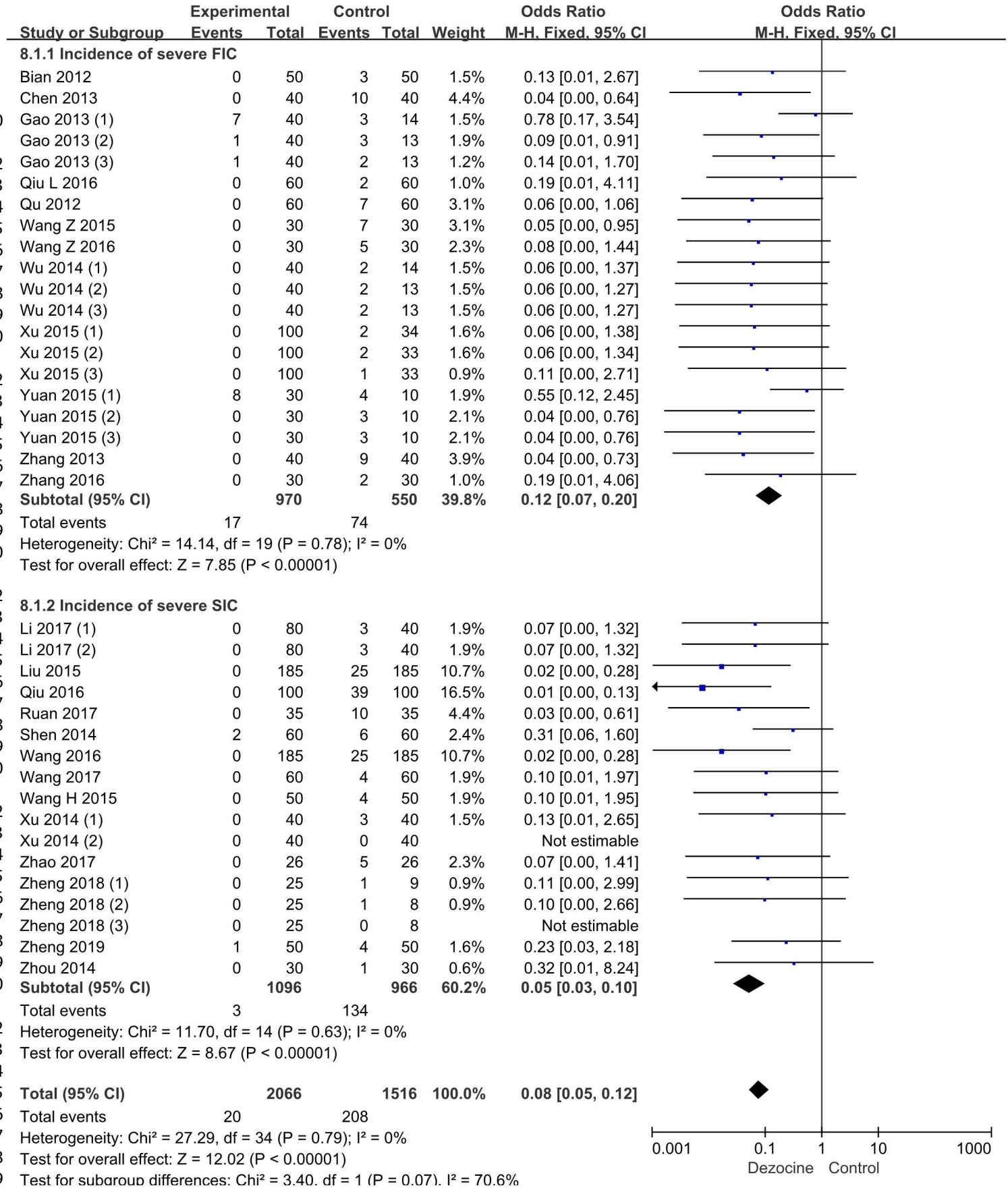
For peer review only

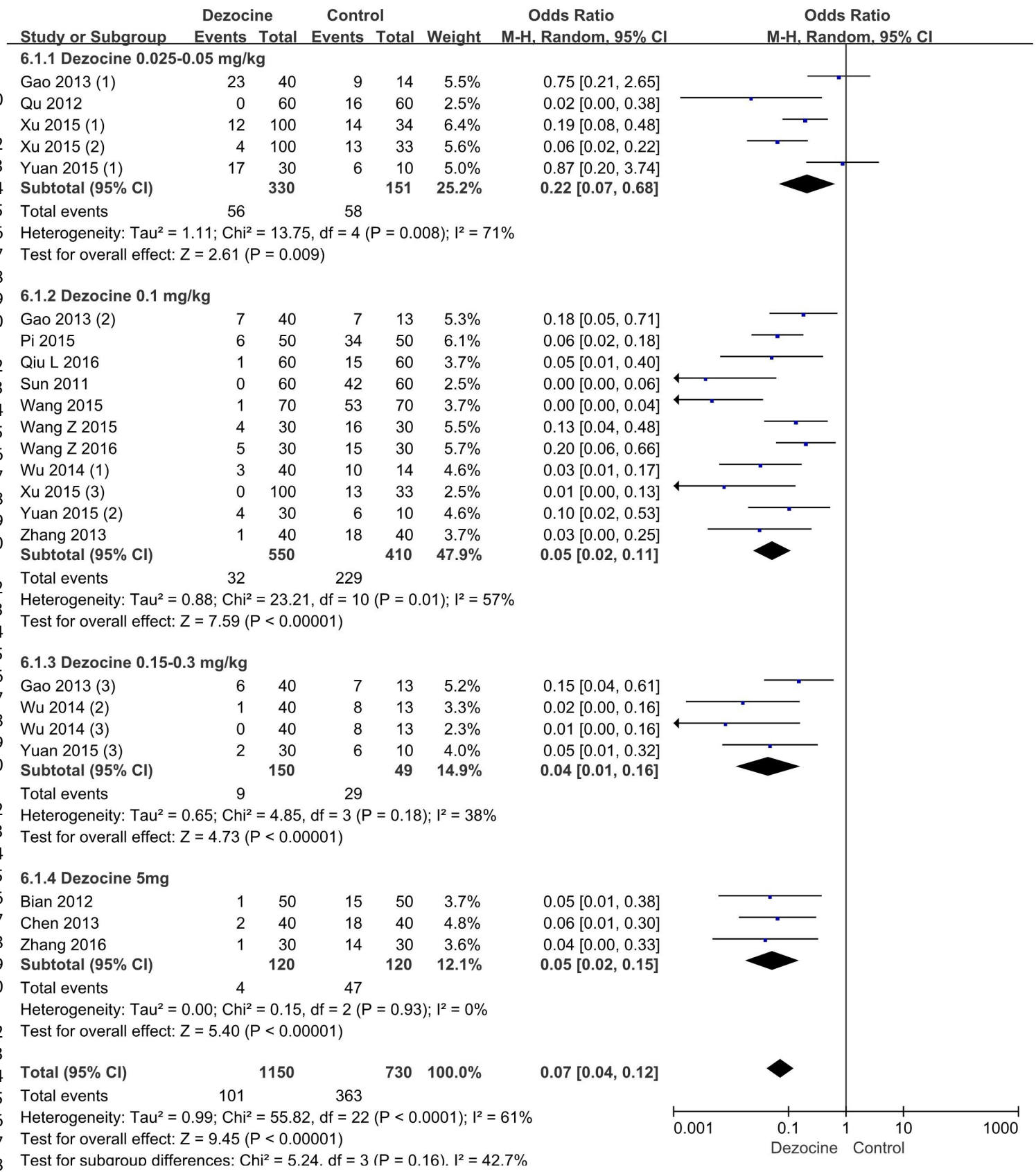

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
4.1.1 Fentanyl 2μg/kg
Chen 2013 2 40 18 40 4.8% 0.06 [0.01, 0.30]
Subtotal (95% CI) 40 40 4.8% 0.06 [0.01, 0.30]
Total events 2 18
Heterogeneity: Not applicable
Test for overall effect: Z = 3.46 (P = 0.0005)
4.1.2 Fentanyl 3μg/kg
Bian 2012 1 50 15 50 3.7% 0.05 [0.01, 0.38]
Gao 2013 (1) 23 40 9 14 5.5% 0.75 [0.21, 2.65]
Gao 2013 (2) 7 40 7 13 5.3% 0.18 [0.05, 0.71]
Gao 2013 (3) 6 40 7 13 5.2% 0.15 [0.04, 0.61]
Qiu L 2016 1 60 15 60 3.7% 0.05 [0.01, 0.40]
Wu 2014 (1) 3 40 10 14 4.6% 0.03 [0.01, 0.17]
Wu 2014 (2) 1 40 8 13 3.3% 0.02 [0.00, 0.16]
Wu 2014 (3) 0 40 8 13 2.3% 0.01 [0.00, 0.16]
Xu 2015 (1) 12 100 14 34 6.4% 0.19 [0.08, 0.48]
Xu 2015 (2) 4 100 13 33 5.6% 0.06 [0.02, 0.22]
Xu 2015 (3) 0 100 13 33 2.5% 0.01 [0.00, 0.13]
Yuan 2015 (1) 17 30 6 10 5.0% 0.87 [0.20, 3.74]
Yuan 2015 (2) 4 30 6 10 4.6% 0.10 [0.02, 0.53]
Yuan 2015 (3) 2 30 6 10 4.0% 0.05 [0.01, 0.32]
Zhang 2013 1 40 18 40 3.7% 0.03 [0.00, 0.25]
Subtotal (95% CI) 780 360 65.5% 0.09 [0.05, 0.18]
Total events 82 155
Heterogeneity: Tau² = 0.95; Chi² = 34.43, df = 14 (P = 0.002); I² = 59%
Test for overall effect: Z = 7.05 (P < 0.00001)
4.1.3 Fentanyl 4μg/kg
Pi 2015 6 50 34 50 6.1% 0.06 [0.02, 0.18]
Qu 2012 0 60 16 60 2.5% 0.02 [0.00, 0.38]
Wang Z 2015 4 30 16 30 5.5% 0.13 [0.04, 0.48]
Wang Z 2016 5 30 15 30 5.7% 0.20 [0.06, 0.66]
Zhang 2016 1 30 14 30 3.6% 0.04 [0.00, 0.33]
Subtotal (95% CI) 200 200 23.4% 0.09 [0.05, 0.18]
Total events 16 95
Heterogeneity: Tau² = 0.02; Chi² = 4.15, df = 4 (P = 0.39); I² = 4%
Test for overall effect: Z = 7.24 (P < 0.00001)
4.1.4 Fentanyl 5μg/kg
Sun 2011 0 60 42 60 2.5% 0.00 [0.00, 0.06]
Wang 2015 1 70 53 70 3.7% 0.00 [0.00, 0.04]
Subtotal (95% CI) 130 130 6.2% 0.00 [0.00, 0.02]
Total events 1 95
Heterogeneity: Tau² = 0.00; Chi² = 0.02, df = 1 (P = 0.88); I² = 0%
Test for overall effect: Z = 6.44 (P < 0.00001)
Total (95% CI) 1150 730 100.0% 0.07 [0.04, 0.12]
Total events 101 363
Heterogeneity: Tau² = 0.99; Chi² = 55.82, df = 22 (P < 0.0001); I² = 61%
Test for overall effect: Z = 9.45 (P < 0.00001)
Test for subgroup differences: Chi² = 12.31, df = 3 (P = 0.006), I² = 75.6%
0.001
0.1
1
10
1000
Dezocine
Control

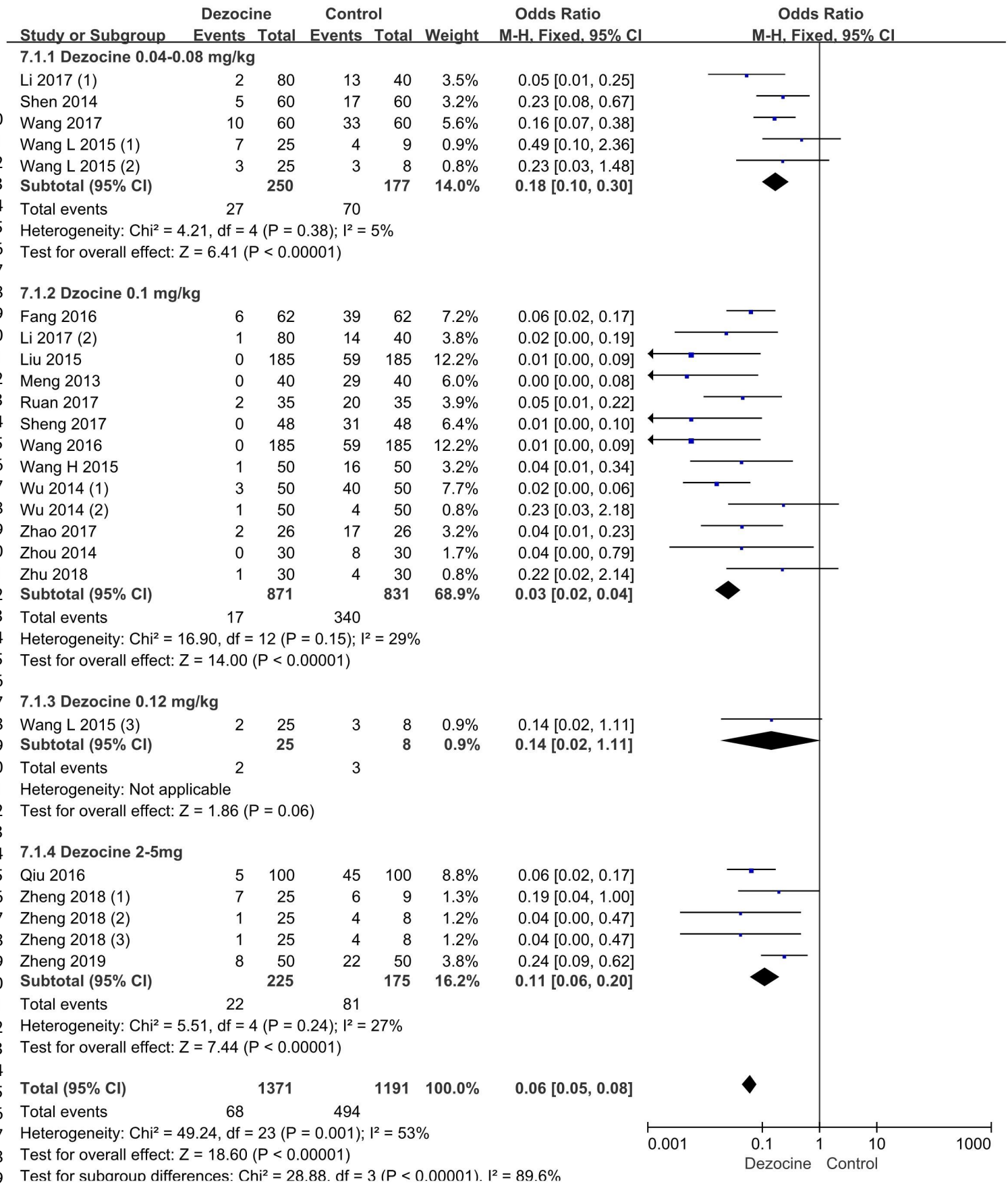

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
3.1.1 Sufentanil 0.3 μg/kg
Li 2017 (1) 2 80 13 40 5.1% 0.05 [0.01, 0.25]
Li 2017 (2) 1 80 14 40 3.8% 0.02 [0.00, 0.19]
Sheng 2017 0 48 31 48 2.5% 0.01 [0.00, 0.10]
Wang 2017 1 50 16 50 3.8% 0.04 [0.01, 0.34]
Wang H 2015 7 25 4 9 5.1% 0.49 [0.10, 2.36]
Wang L 2015 (1) 3 25 3 8 4.3% 0.23 [0.03, 1.48]
Wang L 2015 (2) 2 25 3 8 3.9% 0.14 [0.02, 1.11]
Wang L 2015 (3) 0 30 8 30 2.4% 0.04 [0.00, 0.79]
Zhou 2014 0 0 0 0 Not estimable
Subtotal (95% CI) 363 233 30.9% 0.08 [0.03, 0.20]
Total events 16 92
Heterogeneity: Tau² = 0.89; Chi² = 13.03, df = 7 (P = 0.07); I² = 46%
Test for overall effect: Z = 5.17 (P < 0.00001)
3.1.2 Sufentanil 0.4 μg/kg
Meng 2013 0 40 29 40 2.5% 0.00 [0.00, 0.08]
Ruan 2017 2 35 20 35 5.1% 0.05 [0.01, 0.22]
Zheng 2018 (1) 7 25 6 9 4.9% 0.19 [0.04, 1.00]
Zheng 2018 (2) 1 25 4 8 3.1% 0.04 [0.00, 0.47]
Zheng 2018 (3) 1 25 4 8 3.1% 0.04 [0.00, 0.47]
Zheng 2019 8 50 22 50 7.2% 0.24 [0.09, 0.62]
Zhu 2018 1 30 4 30 3.4% 0.22 [0.02, 2.14]
Subtotal (95% CI) 230 180 29.2% 0.09 [0.03, 0.22]
Total events 20 89
Heterogeneity: Tau² = 0.71; Chi² = 11.16, df = 6 (P = 0.08); I² = 46%
Test for overall effect: Z = 5.04 (P < 0.00001)
3.1.3 Sufentanil 0.5 μg/kg
Liu 2015 0 185 59 185 2.6% 0.01 [0.00, 0.09]
Qiu 2016 5 100 45 100 7.0% 0.06 [0.02, 0.17]
Wang 2016 0 185 59 185 2.6% 0.01 [0.00, 0.09]
Xu 2014 (1) 3 50 40 50 5.7% 0.02 [0.00, 0.06]
Xu 2014 (2) 1 50 4 50 3.5% 0.23 [0.03, 2.18]
Subtotal (95% CI) 570 570 21.4% 0.03 [0.01, 0.11]
Total events 9 207
Heterogeneity: Tau² = 1.21; Chi² = 10.16, df = 4 (P = 0.04); I² = 61%
Test for overall effect: Z = 5.35 (P < 0.00001)
3.1.4 Sufentanil 3-5 μg/kg
Fang 2016 6 62 39 62 7.0% 0.06 [0.02, 0.17]
Shen 2014 5 60 17 60 6.7% 0.23 [0.08, 0.67]
Zhao 2017 2 26 17 26 4.8% 0.04 [0.01, 0.23]
Subtotal (95% CI) 148 148 18.6% 0.09 [0.03, 0.25]
Total events 13 73
Heterogeneity: Tau² = 0.39; Chi² = 4.08, df = 2 (P = 0.13); I² = 51%
Test for overall effect: Z = 4.69 (P < 0.00001)
Total (95% CI) 1311 1131 100.0% 0.07 [0.04, 0.11]
Total events 58 461
Heterogeneity: Tau² = 0.77; Chi² = 46.43, df = 22 (P = 0.002); I² = 53%
Test for overall effect: Z = 10.08 (P < 0.00001)
Test for subgroup differences: Chi² = 2.28, df = 3 (P = 0.52), I² = 0%
0.001
0.1
1
10
1000
Dezocine
Control

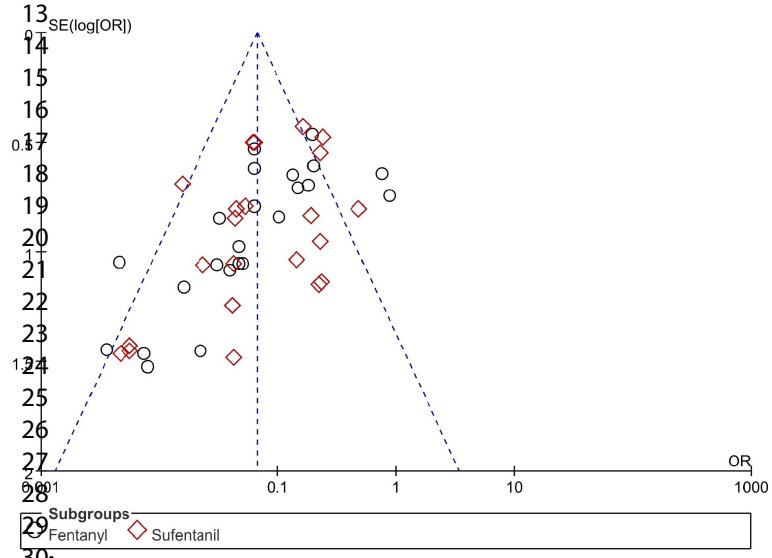

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.3.1 Incidence of mild FIC
Bian 2012 1 30 8 30 3.7% 0.09 [0.01, 0.82]
Chen 2013 2 40 2 40 0.9% 1.00 [0.13, 7.47]
Gao 2013 (1) 2 30 0 10 0.3% 1.84 [0.08, 41.62]
Gao 2013 (2) 3 30 1 10 0.6% 1.00 [0.09, 10.87]
Gao 2013 (3) 5 30 1 10 0.6% 1.80 [0.18, 17.57]
Qu 2012 0 100 8 33 6.0% 0.01 [0.00, 0.27]
Wu 2014 (1) 3 100 8 33 5.6% 0.10 [0.02, 0.39]
Wu 2014 (2) 8 100 9 34 5.9% 0.24 [0.08, 0.69]
Wu 2014 (3) 0 40 4 13 3.2% 0.03 [0.00, 0.53]
Xu 2015 (1) 1 40 4 13 2.8% 0.06 [0.01, 0.58]
Xu 2015 (2) 4 40 4 14 2.5% 0.28 [0.06, 1.31]
Xu 2015 (3) 0 60 5 60 2.6% 0.08 [0.00, 1.54]
Yuan 2015 (1) 3 40 2 14 1.3% 0.49 [0.07, 3.27]
Yuan 2015 (2) 3 40 2 13 1.3% 0.45 [0.07, 3.02]
Yuan 2015 (3) 7 40 2 13 1.2% 1.17 [0.21, 6.47]
Zhang 2013 1 40 2 40 0.9% 0.49 [0.04, 5.60]
Zhang 2016 1 50 4 50 1.9% 0.23 [0.03, 2.18]
Subtotal (95% CI) 850 430 41.4% 0.25 [0.16, 0.38]
Total events 44 66
Heterogeneity: Chi² = 22.33, df = 16 (P = 0.13); I² = 28%
Test for overall effect: Z = 6.62 (P < 0.00001)
8.3.2 Incidence of mild SIC
Li 2017 (1) 0 30 4 30 2.1% 0.10 [0.00, 1.88]
Li 2017 (2) 4 50 7 50 3.1% 0.53 [0.15, 1.95]
Liu 2015 1 25 2 8 1.4% 0.13 [0.01, 1.62]
Qiu 2016 1 25 2 8 1.4% 0.13 [0.01, 1.62]
Ruan 2017 6 25 3 9 1.6% 0.63 [0.12, 3.33]
Shen 2014 1 26 2 26 0.9% 0.48 [0.04, 5.65]
Wang 2016 1 40 3 40 1.4% 0.32 [0.03, 3.18]
Wang 2017 2 40 29 40 13.1% 0.02 [0.00, 0.10]
Wang H 2015 1 50 6 50 2.8% 0.15 [0.02, 1.29]
Xu 2014 (1) 3 60 9 60 4.1% 0.30 [0.08, 1.16]
Xu 2014 (2) 0 185 13 185 6.4% 0.03 [0.00, 0.58]
Zhao 2017 0 60 2 60 1.2% 0.19 [0.01, 4.11]
Zheng 2018 (1) 1 35 6 35 2.8% 0.14 [0.02, 1.25]
Zheng 2018 (2) 1 60 8 60 3.7% 0.11 [0.01, 0.91]
Zheng 2018 (3) 0 185 13 185 6.4% 0.03 [0.00, 0.58]
Zheng 2019 1 80 5 40 3.1% 0.09 [0.01, 0.79]
Zhou 2014 2 80 5 40 3.1% 0.18 [0.03, 0.97]
Subtotal (95% CI) 1056 926 58.6% 0.14 [0.09, 0.22]
Total events 25 119
Heterogeneity: Chi² = 17.94, df = 16 (P = 0.33); I² = 11%
Test for overall effect: Z = 8.39 (P < 0.00001)
Total (95% CI) 1906 1356 100.0% 0.19 [0.14, 0.25]
Total events 69 185
Heterogeneity: Chi² = 42.29, df = 33 (P = 0.13); I² = 22%
Test for overall effect: Z = 10.87 (P < 0.00001)
Test for subgroup differences: Chi² = 3.24, df = 1 (P = 0.07), I² = 69.2%
0.002
0.1
1
10
500
Dezocine
Control


Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.2.1 Incidence of moderate FIC
Bian 2012 0 50 5 50 2.6% 0.08 [0.00, 1.52]
Chen 2013 1 40 6 40 2.8% 0.15 [0.02, 1.27]
Gao 2013 (1) 9 40 3 13 1.7% 0.97 [0.22, 4.29]
Gao 2013 (2) 3 40 3 13 2.0% 0.27 [0.05, 1.55]
Gao 2013 (3) 2 40 3 14 2.0% 0.19 [0.03, 1.30]
Qu 2012 0 60 6 60 3.0% 0.07 [0.00, 1.26]
Wu 2014 (1) 1 40 3 14 2.0% 0.09 [0.01, 1.00]
Wu 2014 (2) 0 40 3 13 2.4% 0.04 [0.00, 0.77]
Wu 2014 (3) 0 40 2 13 1.7% 0.06 [0.00, 1.27]
Xu 2015 (1) 4 100 4 34 2.7% 0.31 [0.07, 1.33]
Xu 2015 (2) 1 100 3 33 2.1% 0.10 [0.01, 1.01]
Xu 2015 (3) 0 100 3 33 2.5% 0.04 [0.00, 0.86]
Yuan 2015 (1) 4 30 2 10 1.2% 0.62 [0.09, 4.01]
Yuan 2015 (2) 1 30 2 10 1.4% 0.14 [0.01, 1.72]
Yuan 2015 (3) 0 30 2 10 1.7% 0.06 [0.00, 1.28]
Zhang 2013 0 40 7 40 3.5% 0.06 [0.00, 1.00]
Zhang 2016 0 30 4 30 2.1% 0.10 [0.00, 1.88]
Subtotal (95% CI) 850 430 37.4% 0.17 [0.10, 0.28]
Total events 26 61
Heterogeneity: Chi² = 12.59, df = 16 (P = 0.70); I² = 0%
Test for overall effect: Z = 7.01 (P < 0.00001)
8.2.2 Incidence of moderate SIC
Li 2017 (1) 0 80 6 40 4.1% 0.03 [0.00, 0.60]
Li 2017 (2) 0 80 5 40 3.4% 0.04 [0.00, 0.74]
Liu 2015 0 185 21 185 10.1% 0.02 [0.00, 0.34]
Qiu 2016 0 60 5 60 2.6% 0.08 [0.00, 1.54]
Ruan 2017 1 35 4 35 1.8% 0.23 [0.02, 2.15]
Shen 2014 3 60 9 60 4.0% 0.30 [0.08, 1.16]
Wang 2016 0 185 21 185 10.1% 0.02 [0.00, 0.34]
Wang 2017 2 60 10 60 4.6% 0.17 [0.04, 0.82]
Wang H 2015 0 50 6 50 3.0% 0.07 [0.00, 1.24]
Xu 2014 (1) 1 40 8 40 3.7% 0.10 [0.01, 0.86]
Xu 2014 (2) 0 40 1 40 0.7% 0.33 [0.01, 8.22]
Zhao 2017 1 26 10 26 4.5% 0.06 [0.01, 0.55]
Zheng 2018 (1) 1 25 2 9 1.3% 0.15 [0.01, 1.86]
Zheng 2018 (2) 0 25 1 8 1.0% 0.10 [0.00, 2.66]
Zheng 2018 (3) 0 25 1 8 1.0% 0.10 [0.00, 2.66]
Zheng 2019 3 50 11 50 4.9% 0.23 [0.06, 0.87]
Zhou 2014 0 30 3 30 1.6% 0.13 [0.01, 2.61]
Subtotal (95% CI) 1056 926 62.6% 0.10 [0.06, 0.17]
Total events 12 124
Heterogeneity: Chi² = 9.26, df = 16 (P = 0.90); I² = 0%
Test for overall effect: Z = 8.67 (P < 0.00001)
Total (95% CI) 1906 1356 100.0% 0.12 [0.09, 0.18]
Total events 38 185
Heterogeneity: Chi² = 23.22, df = 33 (P = 0.90); I² = 0%
Test for overall effect: Z = 11.33 (P < 0.00001)
Test for subgroup differences: Chi² = 2.20, df = 1 (P = 0.14), I² = 54.6%
0.002
0.1
1
10
500
Dezocine
Control

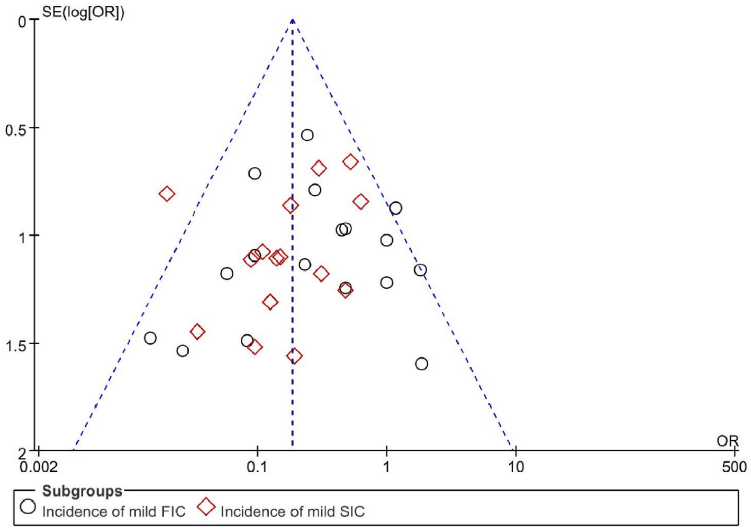

Experimental
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
8.1.1 Incidence of severe FIC
Bian 2012 0 50 3 50 1.5% 0.13 [0.01, 2.67]
Chen 2013 0 40 10 40 4.4% 0.04 [0.00, 0.64]
Gao 2013 (1) 7 40 3 14 1.5% 0.78 [0.17, 3.54]
Gao 2013 (2) 1 40 3 13 1.9% 0.09 [0.01, 0.91]
Gao 2013 (3) 1 40 2 13 1.2% 0.14 [0.01, 1.70]
Qiu L 2016 0 60 2 60 1.0% 0.19 [0.01, 4.11]
Qu 2012 0 60 7 60 3.1% 0.06 [0.00, 1.06]
Wang Z 2015 0 30 7 30 3.1% 0.05 [0.00, 0.95]
Wang Z 2016 0 30 5 30 2.3% 0.08 [0.00, 1.44]
Wu 2014 (1) 0 40 2 14 1.5% 0.06 [0.00, 1.37]
Wu 2014 (2) 0 40 2 13 1.5% 0.06 [0.00, 1.27]
Wu 2014 (3) 0 40 2 13 1.5% 0.06 [0.00, 1.27]
Xu 2015 (1) 0 100 2 34 1.6% 0.06 [0.00, 1.38]
Xu 2015 (2) 0 100 2 33 1.6% 0.06 [0.00, 1.34]
Xu 2015 (3) 0 100 1 33 0.9% 0.11 [0.00, 2.71]
Yuan 2015 (1) 8 30 4 10 1.9% 0.55 [0.12, 2.45]
Yuan 2015 (2) 0 30 3 10 2.1% 0.04 [0.00, 0.76]
Yuan 2015 (3) 0 30 3 10 2.1% 0.04 [0.00, 0.76]
Zhang 2013 0 40 9 40 3.9% 0.04 [0.00, 0.73]
Zhang 2016 0 30 2 30 1.0% 0.19 [0.01, 4.06]
Subtotal (95% CI) 970 550 39.8% 0.12 [0.07, 0.20]
Total events 17 74
Heterogeneity: Chi² = 14.14, df = 19 (P = 0.78); I² = 0%
Test for overall effect: Z = 7.85 (P < 0.00001)
8.1.2 Incidence of severe SIC
Li 2017 (1) 0 80 3 40 1.9% 0.07 [0.00, 1.32]
Li 2017 (2) 0 80 3 40 1.9% 0.07 [0.00, 1.32]
Liu 2015 0 185 25 185 10.7% 0.02 [0.00, 0.28]
Qiu 2016 0 100 39 100 16.5% 0.01 [0.00, 0.13]
Ruan 2017 0 35 10 35 4.4% 0.03 [0.00, 0.61]
Shen 2014 2 60 6 60 2.4% 0.31 [0.06, 1.60]
Wang 2016 0 185 25 185 10.7% 0.02 [0.00, 0.28]
Wang 2017 0 60 4 60 1.9% 0.10 [0.01, 1.97]
Wang H 2015 0 50 4 50 1.9% 0.10 [0.01, 1.95]
Xu 2014 (1) 0 40 3 40 1.5% 0.13 [0.01, 2.65]
Xu 2014 (2) 0 40 0 40 Not estimable
Zhao 2017 0 26 5 26 2.3% 0.07 [0.00, 1.41]
Zheng 2018 (1) 0 25 1 9 0.9% 0.11 [0.00, 2.99]
Zheng 2018 (2) 0 25 1 8 0.9% 0.10 [0.00, 2.66]
Zheng 2018 (3) 0 25 0 8 Not estimable
Zheng 2019 1 50 4 50 1.6% 0.23 [0.03, 2.18]
Zhou 2014 0 30 1 30 0.6% 0.32 [0.01, 8.24]
Subtotal (95% CI) 1096 966 60.2% 0.05 [0.03, 0.10]
Total events 3 134
Heterogeneity: Chi² = 11.70, df = 14 (P = 0.63); I² = 0%
Test for overall effect: Z = 8.67 (P < 0.00001)
Total (95% CI) 2066 1516 100.0% 0.08 [0.05, 0.12]
Total events 20 208
Heterogeneity: Chi² = 27.29, df = 34 (P = 0.79); I² = 0%
Test for overall effect: Z = 12.02 (P < 0.00001)
Test for subgroup differences: Chi² = 3.40, df = 1 (P = 0.07), I² = 70.6%
0.001
0.1
1
10
1000
Dezocine
Control


Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
6.1.1 Dezocine 0.025-0.05 mg/kg
Gao 2013 (1) 23 40 9 14 5.5% 0.75 [0.21, 2.65]
Qu 2012 0 60 16 60 2.5% 0.02 [0.00, 0.38]
Xu 2015 (1) 12 100 14 34 6.4% 0.19 [0.08, 0.48]
Xu 2015 (2) 4 100 13 33 5.6% 0.06 [0.02, 0.22]
Yuan 2015 (1) 17 30 6 10 5.0% 0.87 [0.20, 3.74]
Subtotal (95% CI) 330 151 25.2% 0.22 [0.07, 0.68]
Total events 56 58
Heterogeneity: Tau² = 1.11; Chi² = 13.75, df = 4 (P = 0.008); I² = 71%
Test for overall effect: Z = 2.61 (P = 0.009)
6.1.2 Dezocine 0.1 mg/kg
Gao 2013 (2) 7 40 7 13 5.3% 0.18 [0.05, 0.71]
Pi 2015 6 50 34 50 6.1% 0.06 [0.02, 0.18]
Qiu L 2016 1 60 15 60 3.7% 0.05 [0.01, 0.40]
Sun 2011 0 60 42 60 2.5% 0.00 [0.00, 0.06]
Wang 2015 1 70 53 70 3.7% 0.00 [0.00, 0.04]
Wang Z 2015 4 30 16 30 5.5% 0.13 [0.04, 0.48]
Wang Z 2016 5 30 15 30 5.7% 0.20 [0.06, 0.66]
Wu 2014 (1) 3 40 10 14 4.6% 0.03 [0.01, 0.17]
Xu 2015 (3) 0 100 13 33 2.5% 0.01 [0.00, 0.13]
Yuan 2015 (2) 4 30 6 10 4.6% 0.10 [0.02, 0.53]
Zhang 2013 1 40 18 40 3.7% 0.03 [0.00, 0.25]
Subtotal (95% CI) 550 410 47.9% 0.05 [0.02, 0.11]
Total events 32 229
Heterogeneity: Tau² = 0.88; Chi² = 23.21, df = 10 (P = 0.01); I² = 57%
Test for overall effect: Z = 7.59 (P < 0.00001)
6.1.3 Dezocine 0.15-0.3 mg/kg
Gao 2013 (3) 6 40 7 13 5.2% 0.15 [0.04, 0.61]
Wu 2014 (2) 1 40 8 13 3.3% 0.02 [0.00, 0.16]
Wu 2014 (3) 0 40 8 13 2.3% 0.01 [0.00, 0.16]
Yuan 2015 (3) 2 30 6 10 4.0% 0.05 [0.01, 0.32]
Subtotal (95% CI) 150 49 14.9% 0.04 [0.01, 0.16]
Total events 9 29
Heterogeneity: Tau² = 0.65; Chi² = 4.85, df = 3 (P = 0.18); I² = 38%
Test for overall effect: Z = 4.73 (P < 0.00001)
6.1.4 Dezocine 5mg
Bian 2012 1 50 15 50 3.7% 0.05 [0.01, 0.38]
Chen 2013 2 40 18 40 4.8% 0.06 [0.01, 0.30]
Zhang 2016 1 30 14 30 3.6% 0.04 [0.00, 0.33]
Subtotal (95% CI) 120 120 12.1% 0.05 [0.02, 0.15]
Total events 4 47
Heterogeneity: Tau² = 0.00; Chi² = 0.15, df = 2 (P = 0.93); I² = 0%
Test for overall effect: Z = 5.40 (P < 0.00001)
Total (95% CI) 1150 730 100.0% 0.07 [0.04, 0.12]
Total events 101 363
Heterogeneity: Tau² = 0.99; Chi² = 55.82, df = 22 (P < 0.0001); I² = 61%
Test for overall effect: Z = 9.45 (P < 0.00001)
Test for subgroup differences: Chi² = 5.24, df = 3 (P = 0.16), I² = 42.7%
0.001
0.1
1
10
1000
Dezocine
Control

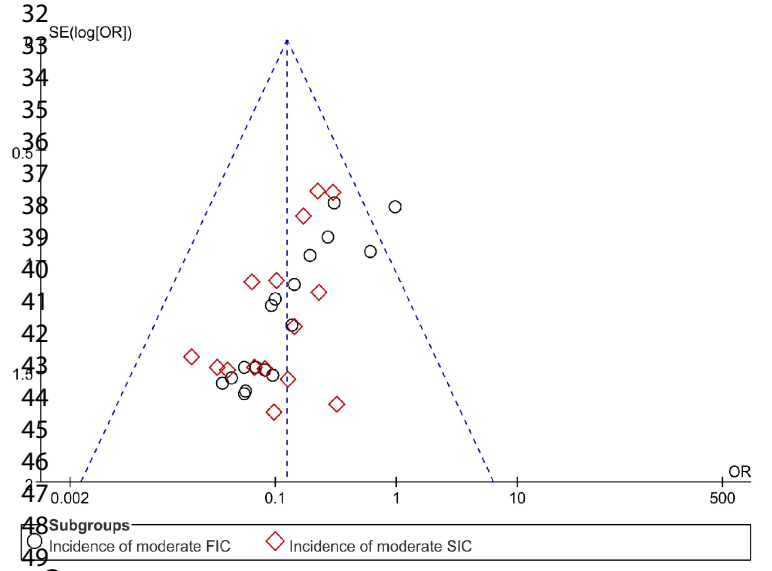

Dezocine
Control
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Fixed, 95% CI
M-H, Fixed, 95% CI
7.1.1 Dezocine 0.04-0.08 mg/kg
Li 2017 (1) 2 80 13 40 3.5% 0.05 [0.01, 0.25]
Shen 2014 5 60 17 60 3.2% 0.23 [0.08, 0.67]
Wang 2017 10 60 33 60 5.6% 0.16 [0.07, 0.38]
Wang L 2015 (1) 7 25 4 9 0.9% 0.49 [0.10, 2.36]
Wang L 2015 (2) 3 25 3 8 0.8% 0.23 [0.03, 1.48]
Subtotal (95% CI) 250 177 14.0% 0.18 [0.10, 0.30]
Total events 27 70
Heterogeneity: Chi² = 4.21, df = 4 (P = 0.38); I² = 5%
Test for overall effect: Z = 6.41 (P < 0.00001)
7.1.2 Dzocine 0.1 mg/kg
Fang 2016 6 62 39 62 7.2% 0.06 [0.02, 0.17]
Li 2017 (2) 1 80 14 40 3.8% 0.02 [0.00, 0.19]
Liu 2015 0 185 59 185 12.2% 0.01 [0.00, 0.09]
Meng 2013 0 40 29 40 6.0% 0.00 [0.00, 0.08]
Ruan 2017 2 35 20 35 3.9% 0.05 [0.01, 0.22]
Sheng 2017 0 48 31 48 6.4% 0.01 [0.00, 0.10]
Wang 2016 0 185 59 185 12.2% 0.01 [0.00, 0.09]
Wang H 2015 1 50 16 50 3.2% 0.04 [0.01, 0.34]
Wu 2014 (1) 3 50 40 50 7.7% 0.02 [0.00, 0.06]
Wu 2014 (2) 1 50 4 50 0.8% 0.23 [0.03, 2.18]
Zhao 2017 2 26 17 26 3.2% 0.04 [0.01, 0.23]
Zhou 2014 0 30 8 30 1.7% 0.04 [0.00, 0.79]
Zhu 2018 1 30 4 30 0.8% 0.22 [0.02, 2.14]
Subtotal (95% CI) 871 831 68.9% 0.03 [0.02, 0.04]
Total events 17 340
Heterogeneity: Chi² = 16.90, df = 12 (P = 0.15); I² = 29%
Test for overall effect: Z = 14.00 (P < 0.00001)
7.1.3 Dezocine 0.12 mg/kg
Wang L 2015 (3) 2 25 3 8 0.9% 0.14 [0.02, 1.11]
Subtotal (95% CI) 25 8 0.9% 0.14 [0.02, 1.11]
Total events 2 3
Heterogeneity: Not applicable
Test for overall effect: Z = 1.86 (P = 0.06)
7.1.4 Dezocine 2-5mg
Qiu 2016 5 100 45 100 8.8% 0.06 [0.02, 0.17]
Zheng 2018 (1) 7 25 6 9 1.3% 0.19 [0.04, 1.00]
Zheng 2018 (2) 1 25 4 8 1.2% 0.04 [0.00, 0.47]
Zheng 2018 (3) 1 25 4 8 1.2% 0.04 [0.00, 0.47]
Zheng 2019 8 50 22 50 3.8% 0.24 [0.09, 0.62]
Subtotal (95% CI) 225 175 16.2% 0.11 [0.06, 0.20]
Total events 22 81
Heterogeneity: Chi² = 5.51, df = 4 (P = 0.24); I² = 27%
Test for overall effect: Z = 7.44 (P < 0.00001)
Total (95% CI) 1371 1191 100.0% 0.06 [0.05, 0.08]
Total events 68 494
Heterogeneity: Chi² = 49.24, df = 23 (P = 0.001); I² = 53%
Test for overall effect: Z = 18.60 (P < 0.00001)
Test for subgroup differences: Chi² = 28.88, df = 3 (P < 0.00001), I² = 89.6%
0.001
0.1
1
10
1000
Dezocine
Control


SE(log[OR])
OR
0.001
0.1
1
10
1000
Subgroups
Fentanyl
Sufentanil

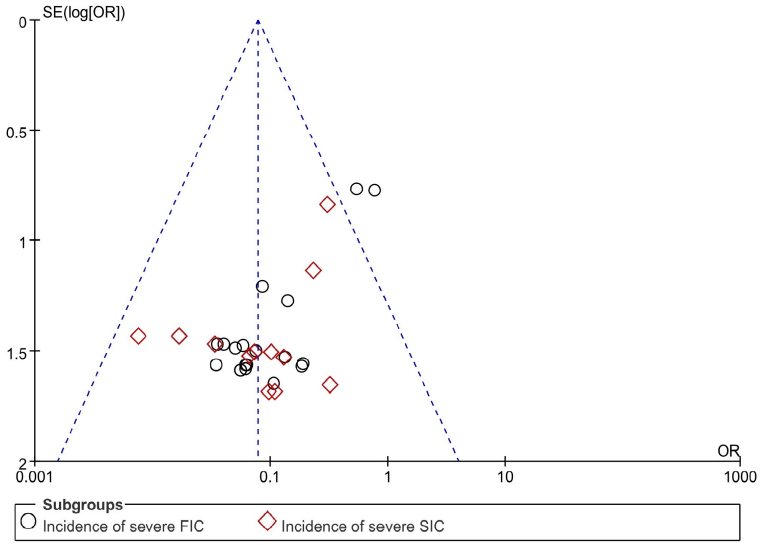
A


SE(log[OR])
0
0.5
1
1.5
2
OR
0.002
0.1
1
10
500
Subgroups
Incidence of mild FIC
Incidence of mild SIC

B


SE(log[OR])
OR
0.002
0.1
1
10
500
Subgroups
Incidence of moderate FIC
Incidence of moderate SIC

C


SE(log[OR])
0
0.5
1
1.5
2
OR
0.001
0.1
1
10
1000
Subgroups
Incidence of severe FIC
Incidence of severe SIC

D